Qwen-VLA: Unifying Vision-Language-Action Modeling across Tasks, Environments, and Robot Embodiments

Qwen Team

https://qwen.ai/blog?id=qwenvla
https://github.com/QwenLM/Qwen-VLA

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Abstract

Embodied intelligence is often studied through specialized models, each designed for a single scenario or task, such as manipulation and navigation, leading to fragmented capabilities and limited generalization across diverse tasks, environments, and robot embodiments. In this work, we investigate whether these heterogeneous embodied decision-making problems can be unified within a single vision-language-action model. We present Qwen-VLA, a unified embodied foundation model that extends Qwen's vision-language modeling stack from perception, understanding and reasoning to continuous action and trajectory generation through a DiT-based action decoder. Our approach adopts a large-scale joint pretraining recipe over diverse data sources, including robotics manipulation trajectories, human egocentric demonstrations, synthetic simulation data, vision-and-language navigation data, trajectory-centric supervision, and auxiliary vision-language data. To support multiple robot platforms within a shared model, we introduce embodiment-aware prompt conditioning, where robot-specific textual descriptions are prepended to specify the current embodiment and control convention. We further cast manipulation, navigation, and trajectory prediction into a unified action-and-trajectory prediction framework, enabling transferable visual grounding, spatial reasoning, and continuous action generation across robot morphologies, task families, and environments. Experiments on manipulation, navigation, and trajectory-centric benchmarks show that Qwen-VLA supports embodied control across task families and robot embodiments, with consistent multi-task performance and out-of-distribution generalization across variations in scene layout, background, lighting, object configuration, and robot embodiment. As a unified generalist policy, Qwen-VLA-Instruct simultaneously achieves 97.9% on LIBERO, 73.7% on Simpler-WidowX, 86.1/87.2% on RoboTwin-Easy/Hard, 69.0% OSR on R2R, and 59.6% SR on RxR, while further attaining 76.9% average OOD success in real-world ALOHA experiments and 26.6% zero-shot success rate on DOMINO dynamic manipulation.

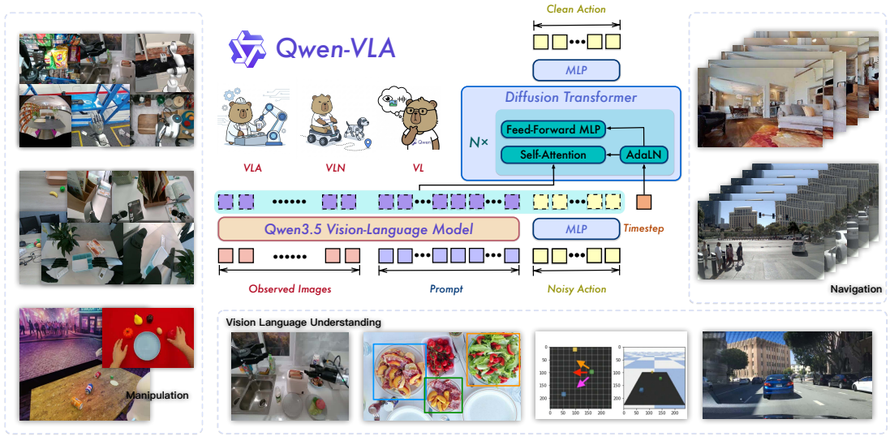

Figure 1: Overview of Qwen-VLA, a unified embodied model trained on mixed manipulation, navigation, and vision-language understanding data to generate both robot actions and textual responses.

Qwen-VLA：统一跨任务、环境与机器人本体的视觉-语言-动作建模

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摘要

具身智能通常通过专用模型来研究，每个模型只针对单一场景或任务（如操作与导航）设计，导致能力碎片化，并且在不同任务、环境和机器人具身形态之间的泛化能力有限。在本工作中，我们研究能否在单一视觉-语言-动作模型内统一这些异构的具身决策问题。我们提出 Qwen-VLA，一个统一的具身基础模型，它通过基于 DiT 的动作解码器，将 Qwen 的视觉-语言建模栈从感知、理解与推理扩展到连续动作和轨迹生成。我们的方法采用大规模联合预训练方案，覆盖多种数据源，包括机器人操作轨迹、人类第一视角演示、合成仿真数据、视觉-语言导航数据、以轨迹为中心的监督数据以及辅助视觉-语言数据。为在共享模型中支持多个机器人平台，我们引入具身形态感知的提示条件化，其中会在提示前附加机器人特定的文本描述，以指定当前具身形态和控制约定。我们进一步将操作、导航和轨迹预测统一为一个动作与轨迹预测框架，从而支持跨机器人形态、任务家族和环境的可迁移视觉定位、空间推理与连续动作生成。在操作、导航和以轨迹为中心的基准上的实验表明，Qwen-VLA 支持跨任务家族和机器人具身形态的具身控制，并在场景布局、背景、光照、物体配置和机器人具身形态变化下保持稳定的多任务性能与分布外泛化能力。作为统一的通用策略，Qwen-VLA-Instruct 同时在 LIBERO 上达到 97.9%，在 Simpler-WidowX 上达到 73.7%，在 RoboTwin-Easy/Hard 上达到 86.1/87.2%，在 R2R 上达到 69.0% OSR，在 RxR 上达到 59.6% SR，并且在真实世界 ALOHA 实验中进一步取得 76.9% 的平均 OOD 成功率，在 DOMINO 动态操作上取得 26.6% 的零样本成功率。

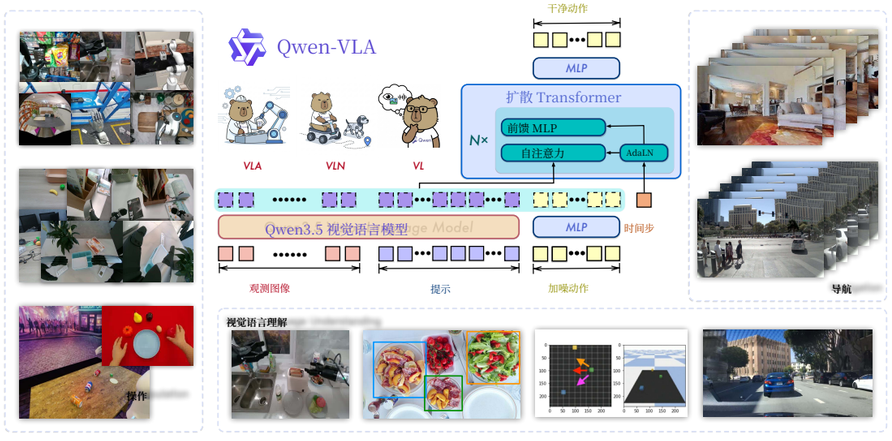

图 1：Qwen-VLA 概览，这是一个统一的具身模型，基于混合的操控、导航和视觉-语言理解数据进行训练，可生成机器人动作和文本响应。

1 Introduction

Embodied intelligence (Zitkovich et al., 2023; O'Neill et al., 2024) aims to build agents that can perceive the physical world, understand natural-language instructions, reason over spatial and temporal context, and execute actions to accomplish goals. Recent progress in vision-language models (VLMs) (Wang et al., 2024; Bai et al., 2025b; Team et al., 2025; Bai et al., 2025a; Team, 2026) has substantially improved open-world visual understanding and language grounding, while diffusion- or flow-based policies (Black et al., 2024; 2025; Intelligence et al., 2026) have shown strong potential for modeling continuous, high-dimensional robot actions. However, most existing embodied systems (NVIDIA et al., 2025; Luo et al., 2025; Yang et al., 2026) remain specialized to a narrow task family, robot embodiment, or evaluation setting. Manipulation models (Intelligence et al., 2026) are often trained for tabletop or dexterous control, and navigation models (Zhang et al., 2024; 2025b; Cheng et al., 2025) are designed around waypoint or action prediction in indoor environments. This fragmentation limits transfer across tasks, environments, and embodiments, and makes it difficult to scale embodied learning in the same way as general-purpose vision-language pretraining.

A central challenge is that embodied decision-making problems appear heterogeneous on the surface. Robot manipulation may require predicting end-effector poses (Black et al., 2024; Zheng et al., 2025; Zhang et al., 2026), joint positions (Fu et al., 2024; NVIDIA et al., 2025), gripper states, or dexterous-hand configurations (Luo et al., 2025); navigation may require predicting waypoints or discrete movement decisions (Zhang et al., 2025a; Wang et al., 2025a; Wei et al., 2025); egocentric human demonstrations may provide wrist and hand trajectories rather than robot control signals. These tasks also differ in observation format, control frequency, prediction horizon, action dimensionality, and evaluation protocol. Nevertheless, they share a common computational structure: an embodied agent must condition on visual observations, language instructions, and embodiment-specific constraints, then predict future actions or trajectories that are physically and semantically aligned with the task. This observation motivates a unified formulation for embodied modeling. We leverage this insight to design a joint pretraining framework that absorbs manipulation, navigation, egocentric human demonstrations, and vision-language reasoning into a single vision-language-action model. The resulting model generalizes across embodiments and task settings.

In this work, we present **Qwen-VLA**, a unified vision-language-action model for heterogeneous embodied decision-making. Qwen-VLA is built upon Qwen3.5-4B, a natively multimodal backbone from Qwen families that has demonstrated state-of-the-art performance across a wide range of vision-language benchmarks. The backbone brings strong fine-grained visual perception, robust referential grounding, multilingual instruction following, and structured reasoning capabilities, which are critical for embodied tasks that require interpreting spatial relations, identifying referred objects, and following complex multi-step instructions. On top of the backbone, we attach a DiT-based flow-matching action decoder that specializes in fine-grained continuous action generation. Instead of designing separate output heads or task-specific architectures for different embodiments, Qwen-VLA represents manipulation actions, navigation waypoints, and human egocentric motions in a shared action-and-trajectory prediction space. This design allows a single model to absorb supervision from diverse embodied datasets while preserving a unified inference interface.

A key component of Qwen-VLA is large-scale joint pretraining over diverse embodied and vision-language data. Our pretraining mixture includes robot manipulation trajectories, human egocentric demonstrations, synthetic simulation trajectories, vision-and-language navigation data, spatial grounding data, autonomous-driving VQA, fine-grained embodied action captions, and general vision-language data. This mixture is designed to cover both low-level motor priors and high-level semantic reasoning. Robot trajectories provide direct action supervision, egocentric human data supplies scalable real-world manipulation priors, navigation data introduces long-horizon instruction following and exploration, synthetic simulation improves controllability and out-of-distribution coverage, and vision-language data, i.e., selectively sampled for embodied-relevant capabilities such as spatial reasoning, referential grounding, and instruction following, preserves and strengthens the backbone's perception and reasoning foundations for downstream embodied tasks.

To enable cross-embodiment learning, we introduce **embodiment-aware prompt conditioning**. Each training example is prepended with a textual description of the current embodiment, including the robot platform, arm configuration, control convention, control frequency, and prediction horizon. The prompt serves as the sole interface through which the model is informed of embodiment-specific control semantics. Combined with a unified action representation, this allows the same action decoder to handle different control modes, action dimensions, and temporal horizons without changing model architecture. As a result, Qwen-VLA can learn shared visual grounding and spatial reasoning capabilities across heterogeneous embodiments while still respecting platform-specific action conventions.

1 引言

具身智能（Zitkovich et al., 2023; O'Neill et al., 2024）旨在构建能够感知物理世界、理解自然语言指令、对时空上下文进行推理并执行动作以完成目标的智能体。视觉-语言模型（VLMs）（Wang et al., 2024; Bai et al., 2025b; Team et al., 2025; Bai et al., 2025a; Team, 2026）的最新进展大幅提升了开放世界视觉理解和语言定位能力，而基于扩散或流的策略（Black et al., 2024; 2025; Intelligence et al., 2026）已展现出对连续、高维机器人动作建模的强大潜力。然而，大多数现有具身系统（NVIDIA et al., 2025; Luo et al., 2025; Yang et al., 2026）仍专门针对狭窄的任务家族、机器人具身形态或评估设定。操作模型（Intelligence et al., 2026）通常针对桌面或灵巧控制训练，导航模型（Zhang et al., 2024; 2025b; Cheng et al., 2025）则围绕室内环境中的路点或动作预测进行设计。这种碎片化限制了跨任务、环境和具身形态的迁移，也使具身学习难以像通用视觉-语言预训练那样实现规模化。

一个核心挑战在于，具身决策问题在表面上呈现出异质性。机器人操作可能需要预测末端执行器位姿（Black et al., 2024; Zheng et al., 2025; Zhang et al., 2026）、关节位置（Fu et al., 2024; NVIDIA et al., 2025）、夹爪状态或灵巧手构型（Luo et al., 2025）；导航可能需要预测路点或离散移动决策（Zhang et al., 2025a; Wang et al., 2025a; Wei et al., 2025）；以自我为中心的人类演示可能提供手腕和手部轨迹，而非机器人控制信号。这些任务在观测格式、控制频率、预测时域、动作维度和评估协议上也各不相同。然而，它们共享一种共同的计算结构：具身智能体必须基于视觉观测、语言指令以及特定具身约束进行条件建模，然后预测在物理和语义上与任务对齐的未来动作或轨迹。这一观察启发了具身建模的统一形式化。我们利用这一洞见设计了一个联合预训练框架，将操作、导航、以自我为中心的人类演示以及视觉-语言推理吸收到一个统一的视觉-语言-动作模型中。所得模型可泛化到不同具身形态和任务设置。

在本工作中，我们提出 Qwen-VLA，一种用于异构具身决策的统一视觉-语言-动作模型。Qwen-VLA 构建于 Qwen3.5-4B 之上，后者是 Qwen 系列中一个原生多模态骨干模型，已在广泛的视觉-语言基准上展现出最先进性能。该骨干模型带来强大的细粒度视觉感知、稳健的指代定位、多语言指令遵循以及结构化推理能力，这些能力对于具身任务至关重要，因为此类任务需要理解空间关系、识别被指代对象并遵循复杂的多步指令。在骨干模型之上，我们接入了一个基于 DiT 的流匹配动作解码器，专门用于细粒度连续动作生成。Qwen-VLA 没有为不同具身形态设计独立的输出头或任务特定架构，而是在共享的动作与轨迹预测空间中表示操作动作、导航路点和人类自我中心运动。这一设计使单一模型能够从多样化的具身数据集中吸收监督信号，同时保持统一的推理接口。

Qwen-VLA 的一个关键组成部分是在多样化的具身数据与视觉语言数据上进行大规模联合预训练。我们的预训练混合数据包括机器人操作轨迹、人类第一人称示范、合成仿真轨迹、视觉与语言导航数据、空间定位数据、自动驾驶视觉问答、细粒度具身动作描述以及通用视觉语言数据。该混合数据旨在同时覆盖底层运动先验和高层语义推理。机器人轨迹提供直接的动作监督，第一人称人类数据提供可扩展的真实世界操作先验，导航数据引入长时程指令遵循与探索能力，合成仿真提升可控性和分布外覆盖，而视觉语言数据——即针对空间推理、指代定位和指令遵循等具身相关能力进行选择性采样——则保持并增强主干模型在下游具身任务中的感知与推理基础。

为了实现跨本体学习，我们引入了本体感知的提示条件化机制。每个训练样本前都会附加对当前本体的文本描述，包括机器人平台、机械臂构型、控制约定、控制频率以及预测时域。该提示是模型获知特定本体控制语义的唯一接口。结合统一的动作表示，这使同一个动作解码器无需改变模型架构即可处理不同的控制模式、动作维度和时间范围。因此，Qwen-VLA 能够跨异构本体学习共享的视觉定位与空间推理能力，同时仍遵循平台特定的动作约定。

Training such a model is non-trivial because the VLM backbone and action decoder enter optimization in asymmetric states: the backbone already possesses general-purpose vision-language representations from large-scale pretraining, while the DiT action decoder is randomly initialized. Naively launching multimodal joint training from this point can waste computation on vision-independent decoder learning and destabilize the pretrained representations. We therefore adopt a staged training recipe grounded in a compression view of action learning. A language instruction such as "pick up the red cup" together with an embodiment prompt compactly encodes the task intent in a handful of tokens, yet the corresponding action trajectory may span hundreds of high-dimensional joint-position values. Bridging this dimensionality gap is a structured decompression problem, and our first training stage, i.e., **text-to-action DiT pretraining (T2A)**, teaches the decoder to serve as a language-conditioned action decompressor before any visual input is introduced. Once this compressed action prior is in place, continued pretraining grounds it in visual observations, supervised fine-tuning specializes it for downstream tasks, and reinforcement learning optimizes closed-loop task success. This recipe separates action-prior compression, visual grounding, task specialization, and success-driven refinement into distinct stages.

We evaluate Qwen-VLA across multiple embodied settings, including robot manipulation, vision-and-language navigation, out-of-distribution generalization, and cross-embodiment transfer. The evaluation is designed not only to measure in-domain task success, but also to test whether a unified model can generalize under changes in object layout, lighting, background, language instruction, sensor noise, and robot embodiment. Empirically, our results show that large-scale joint pretraining over heterogeneous embodied data yields strong multi-task performance and improves robustness beyond narrow specialist training. These findings support the view that manipulation, navigation, and trajectory-centric embodied tasks can be treated as different manifestations of a shared action-and-trajectory prediction problem.

Our contributions are summarized as follows:

- We introduce **Qwen-VLA**, a unified vision-language-action model that formulates manipulation, navigation, and egocentric action modeling within a shared action-and-trajectory space. Built on the Qwen3.5-4B vision-language backbone with a DiT-based flow-matching policy head, Qwen-VLA supports embodied control across multiple robot platforms and task families.
- We construct a large-scale joint pretraining mixture over heterogeneous data sources, including manipulation trajectories from multiple robots, egocentric human demonstrations, synthetic simulation, navigation, and curated vision-language data. We further propose embodiment-aware prompt conditioning that unifies diverse robot platforms, control conventions, and prediction horizons in one model without separate per-embodiment policies.
- We design a progressive training recipe that includes action pretraining, multimodal continued pretraining, supervised fine-tuning, and reinforcement learning to build a strong policy model that bridges the gap between discrete vision-language tokens and continuous action trajectories, improving both training stability and downstream transfer.
- We evaluate Qwen-VLA across manipulation, navigation, out-of-distribution robustness, and cross-embodiment generalization benchmarks. The results show that co-training and progressive learning improve multi-task performance under scene, object, lighting, and embodiment shifts.

2 Unified Embodied Model

2.1 Problem Formulation

We study a broad family of embodied decision-making tasks, including robot manipulation, vision-and-language navigation, trajectory prediction, and human egocentric action modeling. We propose to address these tasks within a unified embodied model, motivated by their shared computational structure: despite differences in output format and evaluation protocol, all tasks require an agent to ground language in visual observations, reason over spatial and temporal context, and predict future actions or trajectories.

We formulate all tasks in a unified conditional prediction framework. At time step t, the model receives a visual context o_t, a language instruction x, an embodiment description e, and an optional task identifier z. Here, o_t may consist of one or multiple image frames, video observations, or history windows; x specifies the task instruction; e is a textual prompt describing the current robot platform and control convention; and z identifies the task family when needed. The model is trained to predict a target sequence $y_{t:t+H-1}$ over a prediction horizon H:

$$p_\theta(y_{t:t+H-1} \mid o_t, x, e, z).$$

The target sequence $y_{t:t+H-1}$ is task-dependent but represented in a unified action-and-trajectory space.

训练这样的模型并非易事，因为 VLM 主干和动作解码器以不对称的状态进入优化：主干已通过大规模预训练具备通用视觉-语言表示，而 DiT 动作解码器则是随机初始化的。此时直接启动多模态联合训练，可能会将计算浪费在与视觉无关的解码器学习上，并破坏预训练表示的稳定性。因此，我们采用基于动作学习压缩视角的分阶段训练方案。诸如“拿起红色杯子”之类的语言指令连同本体提示，能在少量 token 中紧凑编码任务意图，而对应的动作轨迹可能包含数百个高维关节位置值。弥合这一维度差距是一个结构化解压缩问题，因此我们的第一个训练阶段，即文本到动作 DiT 预训练（T2A），会在引入任何视觉输入之前，教会解码器充当语言条件化的动作解压缩器。一旦这一压缩的动作先验就位，持续预训练会将其建立在视觉观测之上，监督微调会使其专门适配下游任务，强化学习则优化闭环任务成功率。该方案将动作先验压缩、视觉定位、任务专门化和成功驱动的精炼划分为不同阶段。

我们在多种具身环境中评估 Qwen-VLA，包括机器人操作、视觉-语言导航、分布外泛化和跨具身迁移。该评估不仅旨在衡量域内任务成功率，还用于检验统一模型能否在物体布局、光照、背景、语言指令、传感器噪声和机器人具身形态发生变化时实现泛化。从实证结果看，我们的结果表明，基于异构具身数据的大规模联合预训练能够带来强大的多任务性能，并相较狭窄的专家训练提升鲁棒性。这些发现支持这样一种观点：操作、导航以及以轨迹为中心的具身任务可以被视为共享的动作与轨迹预测问题的不同表现形式。

我们的贡献总结如下：

- 我们提出 Qwen-VLA，这是一种统一的视觉-语言-动作模型，它在共享的动作与轨迹空间中对操作、导航和自我中心动作建模进行统一表述。Qwen-VLA 构建于 Qwen3.5-4B 视觉-语言主干之上，并带有基于 DiT 的流匹配策略头，支持跨多个机器人平台和任务族的具身控制。
- 我们构建了一个大规模联合预训练混合数据，覆盖异构数据源，包括来自多台机器人的操作轨迹、第一人称人类演示、合成仿真、导航以及经过整理的视觉-语言数据。我们进一步提出了具身感知的提示条件化方法，将不同的机器人平台、控制规范和预测时域统一在一个模型中，而无需为每种具身分别制定策略。
- 我们设计了一套渐进式训练方案，包括动作预训练、多模态持续预训练、监督微调以及强化学习，以构建一个强大的策略模型，弥合离散视觉-语言 token 与连续动作轨迹之间的差距，从而同时提升训练稳定性和下游迁移能力。
- 我们在操作、导航、分布外鲁棒性以及跨具身泛化基准上评估了 Qwen-VLA。结果表明，在场景、物体、光照和具身变化下，协同训练与渐进式学习能够提升多任务性能。

2 统一具身模型

2.1 问题形式化

我们研究了一类广泛的具身决策任务，包括机器人操作、视觉与语言导航、轨迹预测以及人类第一人称动作建模。我们提出在一个统一的具身模型中处理这些任务，其动机在于它们共享的计算结构：尽管输出格式和评估协议存在差异，所有任务都要求智能体将语言建立在视觉观测之上，对空间与时间上下文进行推理，并预测未来的动作或轨迹。

我们在统一的条件预测框架中对所有任务进行建模。在时间步 t，模型接收视觉上下文 o_t、语言指令 x、具身描述 e，以及可选的任务标识符 z。其中，o_t 可以由一个或多个图像帧、视频观测或历史窗口组成；x 指定任务指令；e 是描述当前机器人平台和控制约定的文本提示；z 在需要时标识任务族。模型经过训练，以在预测时域 H 上预测目标序列 $y_{t:t+H-1}$：

$$p_\theta(y_{t:t+H-1} \mid o_t, x, e, z).$$

目标　　序列 $y_{t:t+H-1}$ 与任务相关，但在统一的动作与轨迹空间中表示　　e.

For manipulation tasks, it corresponds to future robot actions like end-effector positions; for navigation tasks, it represents navigation decisions or waypoints; for trajectory-centric tasks such as autonomous driving or motion forecasting, it denotes the precise future spatial trajectory of the agent or surrounding entities in continuous coordinate space; and for egocentric embodied data, it captures human body or hand motion trajectories represented in structured pose spaces such as MANO (Romero et al., 2017) or skeletal joint sequences. This unified formulation enables joint optimization across heterogeneous embodied datasets within a single model, facilitating transferable visual grounding, spatial reasoning, and action generation across task families. The formulation is also extensible along both the input and output axes: on the input side, augmenting the conditioning context o_t with episodic memory or persistent state would enable long-horizon planning and failure recovery; on the output side, co-predicting future visual states alongside actions would unify action generation with world modeling, allowing the agent to anticipate the consequences of its actions.

2.2 Model Architecture

Our model comprises a vision-language backbone for high-level understanding and reasoning and a flow-matching action expert for fine-grained action generation.

Vision-language backbone. We adopt Qwen3.5 (Team, 2026) as the backbone. Qwen3.5 is a natively multimodal model trained with early vision-language fusion: visual tokens produced by a ViT with spatial merging are interleaved directly into the text token stream, enabling unified processing of images, videos, and language within a single transformer. Its hybrid attention design combines gated linear attention in the majority of layers with grouped-query softmax attention at regular intervals, allowing efficient encoding of long multimodal sequences while retaining full-precision global reasoning where needed.

Action expert. We attach a single-stream DiT-style (Esser et al., 2024) flow matching policy as an action expert for predicting precise actions across both robot and human embodied data (Janner et al., 2022; Chi et al., 2023; Liang et al., 2023; Black et al., 2024). The action expert concatenates VLM hidden states with a noisy action chunk into one sequence and processes them through joint self-attention with AdaLN timestep conditioning (Peebles &Xie, 2023) and multi-section RoPE aligned with the backbone. This decoupled design lets the action expert specialize in fine-grained action generation and naturally handle the multimodality and high-frequency dynamics of embodied action distributions, while preserving the backbone's pretrained capabilities. The expert is trained with a flow-matching objective (Lipman et al., 2023) and produces action sequences via a few Euler integration steps at inference, enabling low-latency real-time control. In total, our action expert contains approximately 1.15B parameters: 16 DiT blocks account for the bulk (70.8M each, 1.13B combined), with the remaining parameters distributed among action projection MLPs that map between the raw action dimension and the DiT latent space (4.9M), a linear layer that transforms VLM hidden states to the DiT channel dimension (3.9M), timestep embedding (2.8M), and output AdaLN modulation (4.7M).

2.3 Embodiment-aware Prompt Conditioning

To support multiple robot embodiments within one shared model, we prepend a robot-specific textual prompt to each training example that describes the current platform, arm configuration, and control convention. The prompt follows the template:

```
The robot is {robot_tag} with {single arm / dual arms}[, waist][, and mobile
base]. The control frequency is {FPS} Hz. Please predict the next {chunk_size}
control actions to execute the following task: {ori_instruction}.
```

The robot tag and optional modifiers (*waist*, *mobile base*) are set per embodiment; `FPS` and `chunk_size` reflect the dataset's original control frequency and prediction horizon. Table 2 in Section 3.2 summarises the representative robot platforms covered by our pretraining corpus together with their arm configurations and action types.

2.4 Unified Action and Trajectory Representation

We unify the tensor interface and masking scheme, but do not force all embodiments into a single physical action semantic space. Each dataset preserves its native control convention, specified through the embodiment prompt and dataset-specific normalization. Concretely, each training sample contributes

对于操作任务，它对应于未来的机器人动作，例如末端执行器位置；对于导航任务，它表示导航决策或路点；对于以轨迹为中心的任务（如自动驾驶或运动预测），它表示智能体或周围实体在连续坐标空间中的精确未来空间轨迹；对于第一人称具身数据，它捕捉在 MANO（Romero et al., 2017）或骨骼关节序列等结构化姿态空间中表示的人体或手部运动轨迹。这种统一形式使得能够在单一模型中对异构具身数据集进行联合优化，促进跨任务家族的可迁移视觉定位、空间推理和动作生成。该形式在输入和输出两个方向上都具有可扩展性：在输入侧，用情景记忆或持久状态增强条件上下文 o_t，将能够实现长时程规划和故障恢复；在输出侧，在动作之外共同预测未来视觉状态，将使动作生成与世界建模相统一，使智能体能够预见其行动的后果。

2.2 模型架构

我们的模型由用于高层理解与推理的视觉-语言主干网络，以及用于细粒度动作生成的流匹配动作专家组成。

视觉-语言主干网络。我们采用 Qwen3.5（Team，2026）作为主干网络。Qwen3.5 是一个原生多模态模型，采用早期视觉-语言融合进行训练：由 ViT 生成并经空间合并的视觉 token 会直接交错插入文本 token 流中，从而能够在单个 Transformer 内统一处理图像、视频和语言。其混合注意力设计在大多数层中结合门控线性注意力，并以固定间隔使用分组查询 softmax 注意力，从而能够高效编码长多模态序列，同时在需要的地方保留全精度全局推理能力。

动作专家。我们附加一个单流 DiT 风格（Esser et al., 2024）的流匹配策略作为动作专家，用于跨机器人数据和人类具身数据预测精确动作（Janner et al., 2022; Chi et al., 2023; Liang et al., 2023; Black et al., 2024）。动作专家将 VLM 隐藏状态与带噪动作块拼接成一个序列，并通过联合自注意力进行处理，同时采用 AdaLN 时间步条件（Peebles &Xie, 2023）以及与主干网络对齐的多段 RoPE。这种解耦设计使动作专家能够专注于细粒度动作生成，并自然处理具身动作分布的多模态性和高频动态，同时保留主干网络的预训练能力。该专家使用流匹配目标（Lipman et al., 2023）进行训练，并在推理时通过少量欧拉积分步骤生成动作序列，从而实现低延迟的实时控制。总体而言，我们的动作专家包含约 1.15B 参数：其中 16 个 DiT 块占据主要部分（每个 70.8M，合计 1.13B），其余参数分布在以下部分：用于在原始动作维度与 DiT 潜空间之间进行映射的动作投影 MLP（4.9M）、将 VLM 隐藏状态转换为 DiT 通道维度的线性层（3.9M）、时间步嵌入（2.8M），以及输出 AdaLN 调制（4.7M）。

2.3 具身感知的提示条件化

为了在一个共享模型中支持多种机器人本体，我们在每个训练样本前添加一段机器人特定的文本提示，用于描述当前平台、机械臂配置和控制约定。该提示遵循以下模板：

```
T  该机器人是 {robot_tag}，带有 {single arm / dual arms}[，腰部][，和移动底座]。控制频率为
{FPS} Hz。请预测接下来的 {chunk_size} 个控制动作，以执行以下任务：{ori_instruction}。
```

机器人标签和可选修饰项（腰部、移动底座）按具身分别设置；FPS 和 chunk_size 反映数据集原始的控制频率和预测范围。第 3.2 节中的表 2 汇总了预训练语料覆盖的代表性机器人平台及其机械臂配置和动作类型。

2.4 统一的动作与轨迹表示

我们统一了张量接口和掩码方案，但不强制将所有具身形态归入单一的物理动作语义空间。每个数据集保留其原生控制约定，并通过具身提示和数据集特定的归一化加以指定。具体而言，每个训练样本贡献

a target tensor $\mathbf{Y} \in \mathbb{R}^{H \times K}$, where H is a fixed prediction horizon and K is a fixed channel dimension shared across all control modes.

Control signal types. We cover two families of continuous control signals. *Manipulation* signals include delta end-effector position $(\Delta x, \Delta y, \Delta z)$, end-effector rotation expressed as Euler angles or quaternions, absolute joint positions, gripper aperture, and dexterous-hand joint angles. *Navigation trajectory* signals follow the VLN convention and are represented as $(\Delta x, \Delta y, \Delta\theta)$ per waypoint, encoding relative displacement and heading change in the ground plane. Despite their different physical semantics, both families are sequences of real-valued vectors predicted over a horizon, and are therefore treated identically by the action expert.

Channel layout. A given control mode uses $c \leq K$ channels. These c task-relevant values are placed in the *leading* c dimensions of $\mathbf{Y}$, and the remaining $K - c$ dimensions are zero-padded. A per-channel binary mask $\mathbf{M} \in \{0,1\}^{H \times K}$ records which channels carry valid signals: $M_{h,k} = 1$ if and only if channel $k < c$ and time step h falls within the task's chunk length $H_{\text{task}} \leq H$. This scheme requires no embodiment-specific output heads; a single set of DiT parameters handles all control modes, with the mask preventing padded entries from influencing the gradient.

Task-aware conditioning. Each training sample is prefixed with the embodiment-aware prompt described in Section 2.3, which specifies the robot platform, arm configuration, control frequency, and prediction horizon. For VLN samples the prompt analogously states the navigation convention and waypoint horizon. These prompt tokens are processed by the VLM backbone and their hidden states are concatenated with the noisy action chunk as input to the DiT, so the action expert is always conditioned on the precise control specification of the current sample without any architectural changes.

2.5 Training Objectives

We train the entire model end-to-end with a weighted sum of two objectives that cover continuous action generation and vision-language understanding.

Flow-matching action loss. For all samples with continuous control targets (manipulation, VLN trajectory waypoints, and human egocentric data after action alignment), we supervise the action expert with a conditional flow-matching objective (Lipman et al., 2023). Given a clean target $\mathbf{Y}_0 \in \mathbb{R}^{H \times K}$ and noise $\mathbf{Y}_1 \sim \mathcal{N}(0, \mathbf{I})$, we form the linear interpolant $\mathbf{Y}_\tau = (1-\tau)\mathbf{Y}_0 + \tau\mathbf{Y}_1$ with $\tau \in [0,1]$, and train the expert v_θ to predict the conditional velocity field.

To avoid the gradient being dominated by padding, we apply a *per-channel, per-step* loss with two levels of averaging. For a sample whose control mode activates c channels and H_{task} time steps, let $\mathbf{M} \in \{0,1\}^{H \times K}$ be the validity mask defined in Section 2.4. We first compute the mean squared error for each active channel $k < c$:

$$\ell_k = \frac{\sum_{h=1}^{H} M_{h,k} \left\| \left(v_\theta(\mathbf{Y}_\tau, \tau \mid o_{1:t}, x, e, z) - (\mathbf{Y}_1 - \mathbf{Y}_0)\right)_{h,k} \right\|_2^2}{\sum_{h=1}^{H} M_{h,k}}, \tag{1}$$

and then average uniformly over the c active channels:

$$\mathcal{L}_{\text{act}} = \mathbb{E}_{\tau, \mathbf{Y}_0, \mathbf{Y}_1} \left[\frac{1}{c} \sum_{k=0}^{c-1} \ell_k \right]. \tag{2}$$

This two-level averaging ensures that each control dimension contributes equally to the gradient regardless of how many channels a given embodiment uses, and that padded positions are fully excluded. At inference, action chunks are produced by a few Euler integration steps from $\tau = 1$ to $\tau = 0$.

Vision-language loss. To preserve and strengthen the multimodal capabilities of the backbone, we keep a standard next-token prediction loss on auxiliary vision-language data, fine-grained embodied action captions, autonomous driving VQA, and general VL pretraining corpora:

$$\mathcal{L}_{\text{vl}} = -\sum_i \log p_\theta(w_i \mid w_{<i}, o_{1:t}), \tag{3}$$

where w_i are text tokens. This objective stabilizes language grounding under heavy embodied co-training and prevents catastrophic forgetting of perception and reasoning skills.

一个目标张量 $\mathrm{Y} \in \mathbb{R}^{H \times K}$，其中 H 是固定的预测时域，K 是在所有控制模式之间共享的固定通道维度。

控制信号类型。我们涵盖两类连续控制信号。操作信号包括末端执行器位置增量（Δx, Δy, Δz）、以欧拉角或四元数表示的末端执行器旋转、绝对关节位置、夹爪开度以及灵巧手关节角度。导航轨迹信号遵循 VLN 约定，并按每个路点表示为（Δx, Δy, $\Delta\theta$），编码地平面内的相对位移和航向变化。尽管二者的物理语义不同，但两类信号都是在某个预测时域上预测得到的实值向量序列，因此由动作专家以相同方式处理。

通道布局。给定控制模式使用 $\mathrm{c} \leq \mathrm{K}$ 个通道。这 c 个任务相关值被放置于 Y 的前 c 个维度中，其余 K − c 个维度用零填充。逐通道二值掩码 $\mathrm{M} \in \{0,1\}^{H \times K}$ 记录哪些通道携带有效信号：$\mathrm{M}_{h,k} = 1$ 当且仅当通道 $\mathrm{k} < \mathrm{c}$ 且时间步 h 落在任务的分块长度 $\mathrm{H}_{\text{task}} \leq \mathrm{H}$ 内。该方案无需具身特定的输出头；单组 DiT 参数即可处理所有控制模式，同时掩码防止填充项影响梯度。

任务感知的条件化。每个训练样本都会在前面加上第 2.3 节所述的具身感知提示，该提示指定机器人平台、机械臂配置、控制频率和预测时域。对于 VLN 样本，该提示类似地说明导航约定和航点时域。这些提示词元由 VLM 主干处理，其隐藏状态与含噪动作块拼接后作为 DiT 的输入，因此动作专家始终以当前样本的精确控制规格为条件，且无需任何架构改动。

2.5 训练目标

我们使用涵盖连续动作生成和视觉-语言理解的两个目标的加权和，对整个模型进行端到端训练。

流匹配动作损失。对于所有具有连续控制目标的样本（操作、VLN 轨迹航点，以及动作对齐后的人类第一视角数据），我们使用条件流匹配目标（Lipman et al., 2023）监督动作专家。给定干净目标 $\mathrm{Y}_0 \in \mathbb{R}^{H \times K}$ 和噪声 $\mathrm{Y}_1 \sim \mathcal{N}(0, \mathrm{I})$，我们构造线性插值 $\mathrm{Y}_\tau = (1-\tau)\mathrm{Y}_0 + \tau\mathrm{Y}_1$，其中 $\tau \in [0,1]$，并训练专家 v_θ 预测条件速度场。

为避免梯度被填充（padding）主导，我们采用逐通道、逐步骤的损失，并进行两级平均。对于控制模式激活 c 个通道和 H_{task} 个时间步的样本，令 $\mathrm{M} \in \{0,1\}^{H \times K}$ 为第 2.4 节中定义的有效性掩码。我们首先对每个激活通道 $\mathrm{k} < \mathrm{c}$ 计算均方误差：

$$\ell_k = \frac{\sum_{h=1}^{H} M_{h,k} \left\| \left(v_\theta(\mathbf{Y}_\tau, \tau \mid o_{1:t}, x, e, z) - (\mathbf{Y}_1 - \mathbf{Y}_0)\right)_{h,k} \right\|_2^2}{\sum_{h=1}^{H} M_{h,k}}, \tag{1}$$

然后对 c 个激活通道进行均匀平均：

$$\mathcal{L}_{\text{act}} = \mathbb{E}_{\tau, \mathbf{Y}_0, \mathbf{Y}_1} \left[\frac{1}{c} \sum_{k=0}^{c-1} \ell_k \right]. \tag{2}$$

这种两级平均确保每个控制维度对梯度的贡献相同，无论给定具身（embodiment）使用多少个通道；同时确保填充位置被完全排除。推理时，动作块（action chunks）通过少量欧拉积分步骤从 $\tau = 1$ 到 $\tau = 0$ 生成。

视觉-语言损失。为保持并增强主干网络的多模态能力，我们在辅助视觉-语言数据、细粒度具身动作描述、自动驾驶 VQA 以及通用 VL 预训练语料上保留标准的下一 token 预测损失：

$$\mathcal{L}_{\text{vl}} = -\sum_i \log p_\theta(w_i \mid w_{<i}, o_{1:t}), \tag{3}$$

其中 w_i a 是文本词元。该目标在大规模具身联合训练下稳定语言锚定 g
并防止 nts 对感知和推理技能的灾难性遗忘。

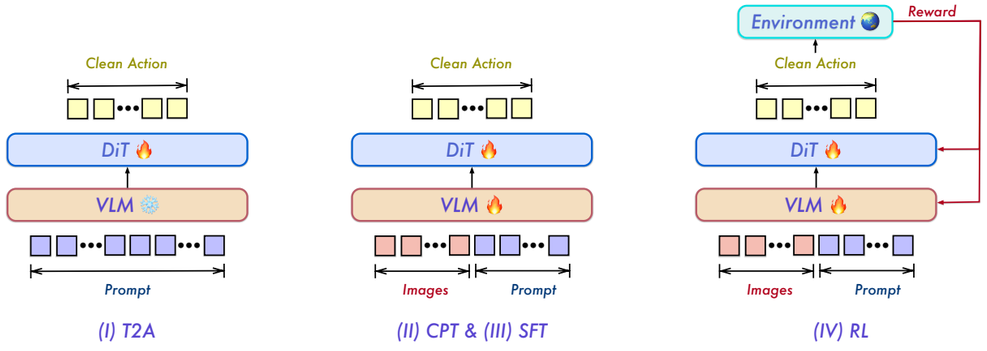

Figure 2: Training recipe of Qwen-VLA. Stage I (T2A) trains the DiT action decoder to reconstruct actions from text alone, building a structured action prior without visual input. Stage II (CPT) unfreezes both modules to ground this prior in visual observations. Stage III (SFT) branches into multi-task and real-robot tracks, and Stage IV (RL) optimizes closed-loop task success via environment rewards.

Joint objective. The overall training loss is a weighted combination

$$\mathcal{L} = \lambda_{\text{act}}\mathcal{L}_{\text{act}} + \lambda_{\text{vl}}\mathcal{L}_{\text{vl}}, \tag{4}$$

where the weights $\lambda_{\text{act}}, \lambda_{\text{vl}}$ are tuned to balance the gradient magnitudes of the two objectives. Within each mini-batch we mix samples from all task families according to a fixed sampling ratio, so that every optimization step jointly updates the backbone and the action expert with manipulation, VLN trajectory, and vision-language signals.

3 Large-Scale Joint Pretraining

3.1 Training Recipe

A usable VLA model requires co-training a cognitive backbone and a motor decoder, a division of labor analogous to the complementary roles of the cerebrum and cerebellum in biological motor control. In practice, however, these two modules enter training in deeply asymmetric states: the VLM backbone is already strongly pretrained, while the DiT action decoder is randomly initialized. Naive joint training from this starting point is inefficient and unstable. The decoder has to learn several things at once: the shape of the action distribution, how to condition on language and embodiment, the flow-matching dynamics of its own parameterization, and how to ground actions in vision. At the same time, every step pays the cost of encoding images. Meanwhile, noisy gradients from a fresh decoder may perturb the pretrained backbone before the decoder has learned useful action structure.

Our staged recipe is motivated by a *compression* perspective on action learning. Raw action trajectories are dense, high-frequency, and embodiment-dependent: a single manipulation episode may contain thousands of joint-position values spread across dozens of degrees of freedom. Yet the underlying task intent is compactly captured by a language instruction ("pick up the red cup") and an embodiment prompt specifying the robot platform and control convention. This description fits in a handful of tokens. There exists a vast dimensionality gap between this compressed task description and the full action signal; bridging it is a structured *decompression* problem.

We cast T2A as learning this decompression map. By withholding images and training the DiT solely on language-conditioned action prediction, we force the decoder to acquire a structured prior over the action space that is indexed entirely by language. This is more than a warm-start: the decoder learns how different linguistic descriptions select different regions of the action distribution, how embodiment prompts modulate the same task intent into platform-specific motor programs, and the temporal coherence and compositionality of full action trajectories at the sequence level, using only the compressed description. With this language-indexed action prior in place, subsequent multimodal training can focus its capacity on grounding the prior in concrete visual observations rather than learning action generation from scratch.

Based on this principle, we adopt a four-stage training recipe on top of a pretrained Qwen3.5 VLM backbone: (I) **text-to-action DiT pretraining (T2A)**, (II) **continued pretraining (CPT)**, (III) **supervised**

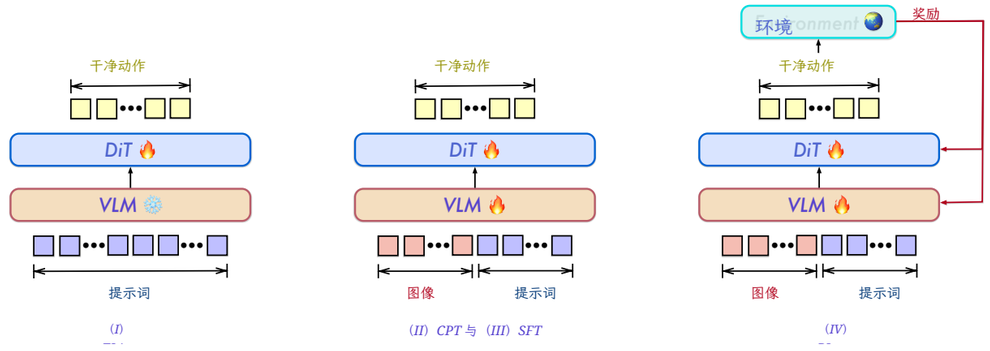

图 2：Qwen-VLA 的训练配方。阶段 I（T2A）训练 DiT 动作解码器，仅根据文本重建动作，在无视觉输入的情况下构建结构化动作先验。阶段 II（CPT）解冻两个模块，将这一先验建立在视觉观测之上。阶段 III（SFT）分支为多任务和真实机器人两条路线，阶段 IV（RL）通过环境奖励优化闭环任务成功率。

联合目标。总体训练损失是一个加权组合

$$\mathcal{L} = \lambda_{\text{act}}\mathcal{L}_{\text{act}} + \lambda_{\text{vl}}\mathcal{L}_{\text{vl}}, \tag{4}$$

其中权重 λ_{act}、λ_{vl} 被调校以平衡两个目标的梯度幅度。在每个小批次中，我们按照固定采样比例混合来自所有任务族的样本，使每次优化步骤都能利用操作、VLN 轨迹和视觉语言信号联合更新主干与动作专家。

3 大规模联合预训练

3.1 训练方案

一个可用的 VLA 模型需要联合训练认知骨干网络和运动解码器，这种分工类似于生物运动控制中大脑与小脑的互补作用。然而在实践中，这两个模块进入训练时处于深度不对称的状态：VLM 骨干网络已经过充分预训练，而 DiT 动作解码器是随机初始化的。从这一起点进行朴素联合训练既低效又不稳定。解码器必须同时学习多项内容：动作分布的形状、如何以语言和具身为条件、其自身参数化所涉及的流匹配动力学，以及如何将动作建立在视觉基础上。与此同时，每一步都要承担图像编码的开销。此外，来自全新解码器的噪声梯度可能会在解码器学到有用的动作结构之前扰动预训练骨干网络。

我们的分阶段训练方案源于从压缩视角看待动作学习。原始动作轨迹密集、高频且依赖具体具身：单个操作回合可能包含数千个关节位置值，分布在几十个自由度上。然而，底层任务意图可由一条语言指令（“拿起红色杯子”）和一个指定机器人平台与控制约定的具身提示紧凑地表达。这种描述只需少量 token 即可容纳。这种压缩后的任务描述与完整动作信号之间存在巨大的维度鸿沟；弥合这一鸿沟是一个结构化解压缩问题。

我们将 T2A 视为学习这种解压缩映射。通过不提供图像，并仅基于语言条件下的动作预测来训练 DiT，我们迫使解码器获得一个完全由语言索引的动作空间结构化先验。这不仅仅是热启动：解码器仅使用压缩描述，就能学习不同的语言描述如何选择动作分布中的不同区域、具身提示如何将同一任务意图调制为平台特定的运动程序，以及序列层面上完整动作轨迹的时间连贯性和组合性。有了这种由语言索引的动作先验，后续多模态训练便可将容量集中于将先验建立在具体视觉观察之上，而不是从头学习动作生成。

基于这一原则，我们在预训练的 Qwen3.5 VLM 主干网络之上采用四阶段训练方案：（I）文本到动作 DiT 预训练（T2A），（II）持续预训练（CPT），（III）监督

fine-tuning (SFT) in two parallel branches, and (IV) **reinforcement learning (RL)**. Each stage is defined by the gap it closes in the one before it.

Stage I: Text-to-action DiT pretraining (T2A). We freeze the VLM and train only the DiT, conditioning it on text and the embodiment prompt e (Section 2.3), but deliberately withholding images. The decoder thus operates as a pure language-to-action decompressor: it must reconstruct high-dimensional action distributions from the compact linguistic encoding alone, without any visual shortcut. In doing so, T2A installs a structured action prior in which language selects the region of action space, the embodiment prompt specifies the platform-specific motor parameterisation, and the flow-matching dynamics govern the generative process before any image is introduced.

Stage II: Continued pretraining (CPT). With the decoder warm-started, CPT focuses its multimodal capacity on exactly the problem that T2A could not address: grounding actions in visual observations and adapting the backbone to embodied perception. We unfreeze both modules and train on the heterogeneous mixture of Section 3.2, which deliberately combines simulation and real-robot trajectories so that the resulting checkpoint has seen both domains. This property is important for the post-training stages described below: SFT and RL can specialise toward either domain.

Stage III: Supervised fine-tuning (SFT). While CPT provides broad cross-embodiment and cross-task coverage, downstream deployment benefits from aligning the model with curated, high-quality demonstrations drawn from the target task distribution. We branch SFT into two parallel tracks from the CPT checkpoint. The first track performs multi-task SFT that jointly fine-tunes the model on heterogeneous tasks, including visual question answering, spatial grounding, manipulation, and navigation, under embodiment-balanced and task-balanced sampling. The second track fine-tunes from the CPT checkpoint on in-house teleoperation data for real-world robot deployment, testing whether CPT's cross-domain priors transfer to physical hardware.

Stage IV: Reinforcement learning (RL). SFT optimises a likelihood over demonstrations; what ultimately matters is closed-loop task success, a property of executed trajectories that no imitation objective can directly optimise. Starting from the multi-task SFT checkpoint, RL fine-tunes the policy with sparse binary success rewards collected exclusively in a single simulation environment (SimplerEnv), producing the final model Qwen-VLA-Instruct. This deliberately narrow RL setup tests whether task-success improvements obtained in one simulated environment can transfer to other out-of-distribution environments.

3.2 Pretraining Data

The quality and diversity of the pretraining corpus directly determine how well the cognitive backbone and motor decoder co-adapt across embodiments and task families. We construct a large and heterogeneous pretraining mixture to equip the model with broad embodied perception, spatial reasoning, and action generation capabilities. The mixture spans five data families: robot manipulation trajectories, human egocentric demonstrations, synthetic simulation data, navigation and trajectory-centric data, and auxiliary vision-language data. Table 1 summarizes the composition and sampling weights of each source.

Table 1: Pretraining data mixture composition.

Data Source	Proportion (%)
Robot Manipulation Trajectories	74.2
Human Egocentric Trajectories	6.0
Navigation Trajectories	7.5
Synthetic Simulation Trajectories (ours)	3.7
General Vision-Language Data	3.4
Spatial Grounding (2D)	2.5
Autonomous Driving VQA	2.4
Fine-Grained Embodied Action Caption	0.2
Total	**100.0**

在两个并行分支中进行微调（SFT），以及（IV）强化学习（RL）。每个阶段都由它在前一阶段中弥合的差距来定义。

阶段一：文本到动作的 DiT 预训练（T2A）。我们冻结 VLM，仅训练 DiT，使其以文本和具身提示 e（第 2.3 节）为条件，但有意不提供图像。因此，解码器充当一个纯语言到动作的解压器：它必须仅从紧凑的语言编码中重建高维动作分布，而没有任何视觉捷径。由此，T2A 在引入任何图像之前就建立了一种结构化的动作先验：语言选择动作空间的区域，具身提示指定平台特定的运动参数化，而流匹配动力学支配生成过程。

阶段二：持续预训练（CPT）。在解码器已热启动的情况下，CPT 将其多模态能力恰好集中于 T2A 无法解决的问题：将动作建立在视觉观测之上，并使主干网络适应具身感知。我们解冻两个模块，并在第 3.2 节的异构混合数据上训练；该混合数据有意结合了仿真轨迹和真实机器人轨迹，从而使得到的检查点已经见过这两个域。这一特性对于下文所述的后训练阶段很重要：SFT 和 RL 可以分别朝任一领域进行专门化。

阶段 III：监督微调（SFT）。虽然 CPT 提供了广泛的跨具身与跨任务覆盖，但下游部署仍受益于使模型与从目标任务分布中抽取的精选高质量演示对齐。我们将 SFT 从 CPT 检查点分支为两条并行路径。第一条路径执行多任务 SFT，在具身平衡和任务平衡采样下，对模型在异构任务（包括视觉问答、空间定位、操作和导航）上进行联合微调。第二条路径从 CPT 检查点出发，在内部遥操作数据上进行微调，以用于真实机器人部署，检验 CPT 的跨域先验能否迁移到物理硬件。

阶段 IV：强化学习（RL）。SFT 优化的是演示数据上的似然；最终真正重要的是闭环任务成功率，这是已执行轨迹的一种属性，任何模仿目标都无法直接优化。从多任务 SFT 检查点开始，RL 仅使用在单一仿真环境（SimplerEnv）中收集的稀疏二元成功奖励对策略进行微调，得到最终模型 Qwen-VLA-Instruct。这种刻意收窄的 RL 设置检验了在一个仿真环境中获得的任务成功率提升能否迁移到其他分布外环境。

3.2 预训练数据

预训练语料的质量和多样性直接决定了认知主干与运动解码器在不同具身形态和任务族之间协同适应的效果。我们构建了一个大规模、异构的预训练混合数据，以赋予模型广泛的具身感知、空间推理和动作生成能力。该混合数据涵盖五个数据族：机器人操作轨迹、人类第一视角演示、合成仿真数据、导航与轨迹中心数据，以及辅助视觉-语言数据。表1总结了各来源的组成和采样权重。

表1：预训练数据混合组成。

数据来源	比例（%）
机器人操作轨迹	74.2
人类第一人称视角轨迹	6.0
导航轨迹	7.5
合成仿真轨迹（本文）	3.7
通用视觉-语言数据	3.4
空间定位（2D）	2.5
自动驾驶视觉问答（VQA）	2.4
细粒度具身动作描述	0.2
总计	**100.0**

3.2.1 Robotics Manipulation Trajectories

Real and simulated robot manipulation trajectories form the core of our pretraining corpus, accounting for approximately 74.2% of the total pretraining mixture. The data spans tabletop manipulation, mobile manipulation, bimanual tasks, and dexterous hand control across a diverse set of robot embodiments.

Public datasets. To improve data scale, diversity, and cross-embodiment generalization, we train on a broad mixture of publicly available real-robot datasets, including RobotSet (Bharadhwaj et al., 2023), Galaxea (Jiang et al., 2025), AgiBot World (AgiBot-World-Contributors et al., 2025), RoboCOIN (Wu et al., 2025b), RoboMIND V1/V2 (Wu et al., 2025a; Hou et al., 2025), RDT-1B (Liu et al., 2025a), DROID (Khaz-atsky et al., 2024), BridgeData V2 (Walke et al., 2024), RH20T (Fang et al., 2023), RT-1 (Brohan et al., 2022), and BC-Z (Jang et al., 2022). These datasets cover diverse robotic settings, including tabletop manipulation, mobile manipulation, bimanual manipulation, dexterous hand control, and in-the-wild task execution, amounting to over 10,000 hours of interaction data across heterogeneous embodiments and task scenarios. This large-scale mixture provides complementary supervision over robot morphologies, camera viewpoints, object categories, backgrounds, language instructions, and action distributions, thereby reducing overfitting to any single embodiment or environment and improving robustness under distribution shifts.

In addition to real-robot data, we include simulation-based manipulation trajectories from InternData-A1 (Tian et al., 2025) and GR00T-X-Embodiment-Sim (NVIDIA et al., 2025), which are generated through motion planning in diverse virtual environments. These simulated trajectories augment scene and object diversity, especially for long-tail object configurations and task layouts. Before training, all real and simulated datasets are curated and converted into a unified observation-action format while preserving their original task semantics, enabling scalable training over heterogeneous robot experience.

Proprietary datasets. We supplement with over 1,000 hours of in-house collected real-robot trajectories covering a variety of manipulation tasks and robot platforms, together contributing approximately 20% of the total pretraining mixture. In addition, we generate over 8M synthetic simulation trajectories using our scalable simulation pipeline illustrated in Section 3.2.3, contributing a further 3.7% of the mixture and substantially expanding the diversity of scenes, objects, and task configurations beyond what real-robot teleoperation alone can cover.

Embodiment-aware prompt conditioning. As described in Section 2.3, each training example is prepended with a robot-specific prompt specifying the current platform, arm configuration, and control convention. Table 2 lists the representative embodiments in our pretraining corpus with their arm configurations and action types.

Table 2: Representative robot embodiments in the pretraining corpus. "EEF" = end-effector pose; "Joint" = joint angles; "Δ" = delta (relative) commands; "Abs" = absolute commands; "G" = gripper; "DH" = dexterous hand.

Robot	Arms	Action type
WidowX	Single	ΔEEF + G
Google Robot	Single	ΔEEF + G
Franka Panda	Single / Dual	ΔEEF + G; Abs Joint + G
ARX5	Dual	ΔEEF + G
Fourier GR-1	Dual	ΔEEF + G
Mobile ALOHA	Dual	ΔEEF + G; Abs Joint + G
AgiBot A2-D	Dual	Abs Joint + G; Abs Joint + DH
Galaxea R1	Dual	Abs Joint + G
AIRBOT MMK2	Dual	Abs Joint + DH
TienKung	Dual	Abs Joint + G; Abs Joint + DH
Real Human	Dual	ΔEEF (from MANO)

Action representation. Different datasets adopt different action conventions: some provide absolute end-effector poses in Cartesian space (Xie et al., 2026), others provide delta end-effector commands, and others use absolute or relative joint-space control signals. We preserve each dataset's original action format rather than converting to a shared representation, relying on embodiment-aware prompt conditioning to inform the model of the current control convention.

All action dimensions are normalized per dataset using quantile statistics. For each action dimension d

3.2.1 机器人操作轨迹

真实与仿真的机器人操作轨迹构成了我们预训练语料的核心，约占整个预训练混合数据的74.2%。这些数据涵盖桌面操作、移动操作、双臂任务和灵巧手控制，并覆盖多种不同的机器人具身形态。

公开数据集。为提升数据规模、多样性和跨具身泛化能力，我们在多种公开可用的真实机器人数据集上进行混合训练，包括 RobotSet（Bharadhwaj et al., 2023）、Galaxea（Jiang et al., 2025）、AgiBot World（AgiBot-World-Contributors et al., 2025）、RoboCOIN（Wu et al., 2025b）、RoboMIND V1/V2（Wu et al., 2025a; Hou et al., 2025）、RDT-1B（Liu et al., 2025a）、DROID（Khaz- atsky et al., 2024）、BridgeData V2（Walke et al., 2024）、RH20T（Fang et al., 2023）、RT-1（Brohan et al., 2022）和 BC-Z（Jang et al., 2022）。这些数据集覆盖多样化的机器人场景，包括桌面操作、移动操作、双臂操作、灵巧手控制以及野外任务执行，合计包含超过 10,000 小时的交互数据，涵盖异构具身形态和任务场景。这种大规模混合为机器人形态、相机视角、物体类别、背景、语言指令和动作分布提供了互补监督，从而减少对任一单一具身形态或环境的过拟合，并提升分布偏移下的鲁棒性。

除真实机器人数据外，我们还纳入来自 InternData-A1（Tian et al., 2025）和 GR00T-X-Embodiment-Sim（NVIDIA et al., 2025）的基于仿真的操作轨迹，这些轨迹通过在不同虚拟环境中进行运动规划生成。这些仿真轨迹增强了场景和物体的多样性，尤其适用于长尾物体配置和任务布局。在训练之前，所有真实和仿真数据集都会经过整理，并转换为统一的观测-动作格式，同时保留其原始任务语义，从而能够基于异构机器人经验进行可扩展训练。

专有数据集。我们补充了超过 1,000 小时的内部采集真实机器人轨迹，涵盖多种操作任务和机器人平台，共同约占总预训练混合数据的 20%。此外，我们使用第 3.2.3 节所示的可扩展仿真流程生成了超过 800 万条合成仿真轨迹，又为混合数据贡献了 3.7%，并显著扩展了场景、物体和任务配置的多样性，超出了仅靠真实机器人遥操作所能覆盖的范围。

具身感知的提示条件化。如第2.3节所述，每个训练样本前都会附加一个机器人特定提示，用于指定当前平台、机械臂配置和控制约定。表2列出了我们预训练语料库中具有代表性的具身形态及其机械臂配置和动作类型。

表2：预训练语料库中具有代表性的机器人具身形态。"EEF" = 末端执行器位姿；"Joint" = 关节角度；"Δ" = 增量（相对）指令；"Abs" = 绝对指令；"G" = 夹爪；"DH" =灵巧手。

机器人	Arms	动作类型
WidowX	单个	ΔEEF + G
Google 机器人	单个	ΔEEF + G
Franka Panda	单臂/双臂	ΔEEF + G；绝对关节 + G
ARX5	Dual	ΔEEF + G
Fourier GR-1	Dual	ΔEEF + G
Mobile ALOHA	Dual	ΔEEF + G；绝对关节 + G
AgiBot A2-D	Dual	绝对关节 + G；绝对关节 +DH
Galaxea R1	Dual	绝对关节 + G
AIRBOT MMK2	Dual	绝对关节 + DH
TienKung	Dual	绝对关节 + G；绝对关节 +DH
真人	Dual	Δ末端执行器（来自 MANO）

动作表示。不同数据集采用不同的动作约定：一些数据集提供笛卡尔空间中的绝对末端执行器位姿（Xie et al., 2026），另一些提供末端执行器增量命令，还有一些使用绝对或相对关节空间控制信号。我们保留每个数据集的原始动作格式，而不是统一转换为共享表示，并依赖具身感知提示条件来告知模型当前的控制约定。

所有动作 各维度使用分位数统计按数据集分别归一化。对于每个动作维度 d

in dataset k, we compute the 1st and 99th percentiles q_{01}^k and q_{99}^k over all trajectories and apply a linear mapping:

$$\tilde{a}_d = 2 \cdot \frac{a_d - q_{01}^k}{q_{99}^k - q_{01}^k} - 1, \tag{5}$$

clipping the result to $[-1, 1]$. This per-dataset quantile normalization removes scale differences across embodiments and action spaces while preserving relative motion structure within each source.

Language instructions. Language instructions are sourced from a combination of original dataset annotations and model-generated captions provided alongside the data. All instructions undergo a quality filtering and consistency checking pipeline: trajectories whose language annotations are inconsistent with the observed motion are discarded, ensuring reliable language–action alignment across diverse sources. Beyond action-labelled data, we incorporate fine-grained embodied video–caption pairs (described in Section 3.2.5) as auxiliary vision-language supervision. This dense language signal bridges the gap between coarse task labels and rich semantic descriptions of robot behaviour, enabling the model to generalise to a broader range of natural-language instructions at inference time.

Camera view representation. Many robot datasets provide observations from multiple camera viewpoints simultaneously, *e.g.* a head-mounted ego-centric camera alongside one or two wrist-mounted cameras. To make the view provenance explicit to the model, we wrap each image in the token stream with a pair of view-specific boundary tokens:

<|*tag*_start|> ⟨image⟩ <|*tag*_end|>

where *tag* identifies the camera source, *e.g.* `ego`, `cam_left_wrist`, or `cam_right_wrist`. The explicit view labels allow the VLM backbone to form view-aware representations and let the action expert attend selectively to each viewpoint when producing actions, without requiring any architectural changes or additional input channels.

Data cleaning. We apply a multi-stage filtering pipeline to remove trajectories with corrupted or missing frames, near-zero-variance action sequences indicative of static recordings, and episodes with anomalous lengths. For datasets without explicit action labels, pseudo-actions are recovered via finite differences on proprioceptive state sequences.

3.2.2 Egocentric Human Data

Compared with teleoperated robot trajectories, egocentric human demonstrations offer a more abundant and scalable source of real-world manipulation experience. Humans interact with diverse objects in open-world environments every day, naturally producing dexterous manipulation behaviours that span a much broader range of scenes, objects, and task semantics than robot teleoperation. Recent studies demonstrate that training on large-scale egocentric human videos endows vision-language-action models with richer manipulation priors and improves generalization to downstream robot tasks (Kareer et al., 2025; Luo et al., 2025; Li et al., 2026b; Luo et al., 2026; Zheng et al., 2026; Hu et al., 2026). Motivated by these findings, we incorporate a diverse collection of egocentric human manipulation datasets (6.0% of the total pretraining mixture) to provide the model with broad manipulation priors that complement the robot trajectory data.

Data sources. Our egocentric corpus is drawn from four publicly available datasets, each contributing complementary coverage in terms of task diversity, scene variety, and annotation granularity. **(1)** We adopt the Ego4D (Grauman et al., 2022) and EPIC-KITCHENS (Damen et al., 2022) subsets processed by VITRA (Li et al., 2026b), which applies an automated pipeline to segment egocentric human videos into atomic manipulation trajectories and generate fine-grained language annotations, together with frame-wise 3D hand and camera motion trajectories. **(2)** EgoDex (Hoque et al., 2026) is a large-scale dexterous manipulation dataset captured with Apple Vision Pro, comprising 829 hours of egocentric video with paired 3D hand and finger tracking across 194 diverse tabletop tasks. **(3)** EgoVerse (Punamiya et al., 2026) is a collaborative egocentric demonstration platform spanning over 1,300 hours across 1,965 tasks and 240 scenes, with standardized formats and manipulation-relevant annotations. **(4)** Xperience (Ropedia, 2026) is a large-scale egocentric multimodal dataset providing synchronized first-person recordings with depth, hand and body motion capture, and hierarchical language annotations.

Action representation. The model receives an egocentric image at an arbitrary time step together with a language instruction and predicts an action chunk comprising future bimanual wrist motion and hand

在数据集 k 中，我们在所有轨迹上计算第 1 百分位数和第 99 百分位数 q_{01}^k 和 q_{99}^k，并应用线性映射：

$$\tilde{a}_d = 2 \cdot \frac{a_d - q_{01}^k}{q_{99}^k - q_{01}^k} - 1, \tag{5}$$

将结果裁剪到 $[-1, 1]$。这种按数据集进行的分位数归一化消除了不同具身形态和动作空间之间的尺度差异，同时保留了每个来源内部相对运动结构。

语言指令。语言指令来自原始数据集标注与随数据提供的模型生成描述文本的组合。所有指令都会经过质量过滤与一致性检查流程：语言标注与观测运动不一致的轨迹会被丢弃，从而确保跨多样来源的可靠语言–动作对齐。除带动作标签的数据外，我们还引入细粒度具身视频–描述文本对（见第 3.2.5 节）作为辅助视觉-语言监督。这种密集语言信号弥合了粗粒度任务标签与机器人行为丰富语义描述之间的差距，使模型在推理时能够泛化到更广泛范围的的自然语言指令。

相机视角表示。许多机器人数据集会同时提供来自多个相机视角的观测，例如一个头戴式自我中心相机以及一个或两个腕部安装相机。为了让模型明确知道视角来源，我们在 token 流中用一对视角特定的边界 token 包裹每张图像：

<|*tag*_start|> ⟨image⟩ <|*tag*_end|>

其中标签用于标识相机来源，例如 ego、cam_left_wrist 或 cam_right_wrist。显式的视角标签使 VLM 主干能够形成视角感知表征，并让动作专家在生成动作时有选择地关注每个视点，而无需任何架构更改或额外输入通道。

数据清洗。我们采用多阶段过滤流水线，以移除帧损坏或缺失的轨迹、表明静态录制的近零方差动作序列，以及长度异常的片段。对于没有显式动作标签的数据集，通过本体感受状态序列上的有限差分恢复伪动作。

3.2.2 第一人称人类数据

与遥操作机器人轨迹相比，第一人称人类演示提供了更丰富且可扩展的真实世界操作经验来源。人类每天都在开放世界环境中与各种物体交互，自然产生灵巧操作行为，这些行为所覆盖的场景、物体和任务语义范围远超机器人遥操作。近期研究表明，在大规模第一人称人类视频上训练可使视觉-语言-动作模型获得更丰富的操作先验，并提升对下游机器人任务的泛化能力（Kareer et al., 2025; Luo et al., 2025; Li et al., 2026b; Luo et al., 2026; Zheng et al., 2026; Hu et al., 2026）。受这些发现启发，我们纳入多样化的第一人称人类操作数据集集合（占预训练混合总量的 6.0%），以为模型提供广泛的操作先验，补充机器人轨迹数据。

数据来源。我们的第一人称视角语料库来自四个公开可用数据集，每个数据集在任务多样性、场景多样性和标注粒度方面提供互补覆盖。（1）我们采用由 VITRA（Li et al., 2026b）处理的 Ego4D（Grauman et al., 2022）和 EPIC-KITCHENS（Damen et al., 2022）子集；VITRA 采用自动化流程将第一人称人类视频分割为原子级操作轨迹，并生成细粒度语言标注，以及逐帧三维手部与相机运动轨迹。（2）EgoDex（Hoque et al., 2026）是一个使用 Apple Vision Pro 采集的大规模灵巧操作数据集，包含 829 小时的第一人称视频，并配有成对的三维手部和手指跟踪数据，覆盖 194 项多样化桌面任务。（3）EgoVerse（Punamiya et al., 2026）是一个协作式第一人称演示平台，时长超过 1,300 小时，涵盖 1,965 项任务和 240 个场景，具有标准化格式和与操作相关的标注。（4）Xperience（Ropedia, 2026）是一个大规模第一人称多模态数据集，提供同步的第一人称录制数据，包含深度、手部和身体动作捕捉以及分层语言标注。

动作表示。模型在任意时间步接收一张自我中心图像和一条语言指令，并预测一个动作块，该动作块包含未来的双手腕运动和手部

articulation over a fixed prediction horizon. For each hand, the wrist motion is represented as the SE(3) transformation of the wrist coordinate frame at a future time step relative to the initial frame. During training, this relative action is represented with a translation vector and an axis-angle rotation, yielding 6 wrist action dimensions per hand. To represent hand articulation compactly, we perform principal component analysis (PCA) on the 45-dimensional axis-angle joint pose of the hand across all human datasets and retain the weights of the first 10 principal components. These low-dimensional coefficients, dubbed *eigengrasps* (Ciocarlie et al., 2007; Yuan et al., 2025), capture the dominant modes of human hand-pose variation while discarding per-joint redundancy. The model predicts relative wrist actions and 10 eigengrasp coefficients per hand, yielding a total of 32 action dimensions per time step for egocentric human data.

3.2.3 Synthetic Simulation Data

To improve the coverage, controllability, and robustness of embodied supervision, we construct a large-scale synthetic simulation data pipeline with two complementary components: (1) *vision-language-action data*, where the model predicts actions from both task instructions and image observations, and (2) *language-action data*, where the model predicts actions from language alone. These two sources of supervision play different but synergistic roles. The text-only component encourages the model to acquire high-level task abstractions and language-action regularities without depending on visual appearance, while the vision-conditioned component grounds those abstractions in realistic perceptual observations, scene variations, and embodied interaction dynamics. We developed our pipeline based on IsaacLab (Mittal et al., 2025) for our simulation environment and cuRobo (Sundaralingam et al., 2023) for collision-avoidance motion planning.

Vision-language-action data. For the vision-conditioned synthetic data, we use an internal early version of RoboInF (XLANG Lab, 2026) to generate standard VLA-style supervision, where the model receives a language instruction and image observations as input and predicts robot actions as output. RoboInF constructs manipulation scenes, generates scene-conditioned tasks, synthesizes executable success checks, produces robot motion programs through simulator feedback, and rolls out successful trajectories under domain randomization. Since the full technical details of the pipeline are provided in our RoboInF blog post, we only summarize the dataset composition and diversity here.

In this release, we focus on the random-placement scene generation setting for simplicity. We construct 20 tabletop scenes, each augmented with 10 different object initial-pose configurations, resulting in 200 base scene configurations. These scenes contain diverse everyday objects arranged in physically valid tabletop layouts, providing varied visual and spatial contexts for manipulation.

On top of these scenes, we generate 450 manipulation tasks spanning both short- and long-horizon behaviors. Short-horizon tasks typically involve a small number of primitive manipulation steps, such as *"Place the two green staplers side by side"*. Long-horizon tasks require multiple sequential interactions or more compositional reasoning, such as *"Group the drinks together and leave the cleaning sponge by itself"*, where the robot needs to manipulate several objects in sequence. As illustrated in Figure 3, this task mixture exposes the model to both atomic manipulation skills and higher-level instruction-following patterns over extended temporal horizons.

For each task, we generate 300 successful trajectories with different environment and execution augmentations. During rollout, we randomize visual, geometric, and control-related factors, including lighting, camera poses, backgrounds, table textures, robot initial states, object initial poses, and controller dynamics such as stiffness and damping. For visual randomization, we sample from a large pool of approximately 3K backgrounds and 1K table textures. For other factors, such as camera pose, robot state, lighting, and controller parameters, we define reasonable ranges manually and sample variations within those ranges. These augmentations are designed to improve robustness to distribution shifts in appearance, viewpoint, layout, and low-level execution dynamics.

A useful property of this generated data is that the trajectories are produced from motion-planning programs that are naturally decomposed into intermediate stages. We therefore additionally segment each full trajectory into several subtask trajectories, as shown in Figure 3. This provides supervision at multiple temporal granularities: full trajectories teach the model to follow high-level instructions over longer horizons, while subtask trajectories expose the model to shorter atomic behaviors and intermediate goals. We expect this mixture to help the model better connect high-level language instructions with the lower-level action segments needed to complete them.

Overall, the vision-conditioned synthetic data contains 359,848 full successful trajectories including subtask segments. The resulting dataset provides a large-scale source of diverse, controllable, and automatically verified VLA supervision, complementing real-world demonstrations and improving the

在固定预测时域内的关节运动。对于每只手，手腕运动被表示为未来时间步的手腕坐标系相对于初始坐标系的 SE(3) 变换。训练期间，这种相对动作用平移向量和轴角旋转表示，每只手产生 6 个手腕动作维度。为了紧凑地表示手部关节运动，我们在所有人类数据集上对手部的 45 维轴角关节姿态执行主成分分析（PCA），并保留前 10 个主成分的权重。这些低维系数被称为特征抓取（eigengrasps）（Ciocarlie et al., 2007; Yuan et al., 2025），它们捕捉人类手部姿态变化的主要模式，同时丢弃逐关节冗余。模型预测相对手腕动作以及每只手的 10 个特征抓取系数，对于自我中心人类数据，每个时间步共产生 32 个动作维度。

3.2.3 合成仿真数据

为了提高具身监督的覆盖范围、可控性和鲁棒性，我们构建了一个大规模合成仿真数据管线，包含两个互补组件：（1）视觉-语言-动作数据，其中模型根据任务指令和图像观测预测动作；（2）语言-动作数据，其中模型仅根据语言预测动作。这两类监督来源发挥着不同但协同的作用。纯文本组件促使模型在不依赖视觉外观的情况下习得高层任务抽象和语言-动作规律，而视觉条件组件则将这些抽象建立在真实感感知观测、场景变化和具身交互动态之上。我们基于 IsaacLab（Mittal et al., 2025）构建仿真环境，并使用 cuRobo（Sundaralingam et al., 2023）进行避障运动规划，从而开发了该管线。

视觉-语言-动作数据。对于视觉条件合成数据，我们使用 ROBOINF（XLANG Lab, 2026）的内部早期版本生成标准的 VLA 式监督，其中模型接收语言指令和图像观测作为输入，并预测机器人动作作为输出。ROBOINF 构建操作场景、生成以场景为条件的任务、合成可执行的成功检查、通过模拟器反馈生成机器人运动程序，并在域随机化下展开成功轨迹。由于该管线的完整技术细节已在我们的 ROBOINF 博客文章中提供，这里仅总结数据集的组成和多样性。

在本次发布中，为简单起见，我们聚焦于随机放置场景生成设置。我们构建了20个桌面场景，每个场景都增加了10种不同的物体初始位姿配置，从而得到200个基础场景配置。这些场景包含多样化的日常物体，并以符合物理有效性的桌面布局进行排列，为操作任务提供多样的视觉和空间上下文。

在这些场景的基础上，我们生成了450个操作任务，涵盖短时程和长时程行为。短时程任务通常涉及少量基础操作步骤，例如“将两个绿色订书机并排放置”。长时程任务需要多次顺序交互或更强的组合推理，例如“将饮料归到一起，并将清洁海绵单独放置”，其中机器人需要按顺序操作多个物体。如图3所示，这种任务混合使模型在更长的时间范围内同时接触原子操作技能和更高层级的指令遵循模式。

对于每个任务，我们生成300条成功轨迹，并采用不同的环境和执行增强。在轨迹展开过程中，我们对视觉、几何和控制相关因素进行随机化，包括光照、相机位姿、背景、桌面纹理、机器人初始状态、物体初始位姿，以及刚度和阻尼等控制器动态。对于视觉随机化，我们从约3000个背景和1000种桌面纹理的大型池中采样。对于其他因素，如相机位姿、机器人状态、光照和控制器参数，我们手动定义合理范围，并在这些范围内采样变化。这些增强旨在提升模型对外观、视角、布局以及底层执行动态中分布偏移的鲁棒性。

这种生成数据的一个有用特性是，轨迹由运动规划程序生成，而这些程序天然可分解为若干中间阶段。因此，我们还额外将每条完整轨迹分割成若干子任务轨迹，如图 3 所示。这提供了多种时间粒度上的监督：完整轨迹教会模型在更长时间跨度上遵循高层指令，而子任务轨迹则让模型接触更短的原子行为和中间目标。我们期望这种混合能帮助模型更好地将高层语言指令与完成这些指令所需的低层动作片段联系起来。

总体而言，视觉条件合成数据包含 359,848 条含子任务片段的完整成功轨迹。所得数据集提供了大规模、多样、可控且可自动验证的 VLA 监督来源，补充了真实世界示范，并提升

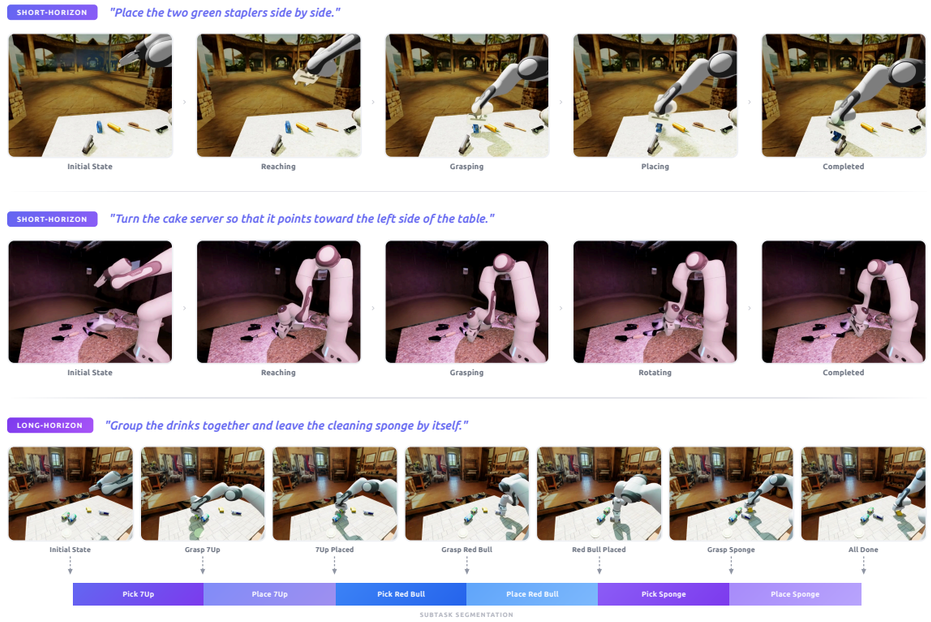

Figure 3: Examples of data generated through ROBOINF. The top row shows a short-horizon task, *"Place the two green staplers side by side,"* which consists of a compact sequence of reaching, grasping, transporting, and placing. The bottom row shows a long-horizon task, *"Group the drinks together and leave the cleaning sponge by itself,"* which requires multiple object manipulations and can be decomposed into subtask segments such as picking and placing each drink.

model's generalization across task types, visual conditions, and execution variations.

Language-action data. We additionally construct a text-only action dataset as a complementary source of supervision. The goal of this component is to expose the model to a broad space of manipulation intents and action patterns before introducing visual observations, thereby encouraging generalization in the language-action joint space. Concretely, we brainstorm a diverse set of atomic and compositional manipulation tasks that can be executed by the robot in simulation, and generate corresponding trajectories using robot-state-centric pipelines without image inputs. This data primarily serves as a form of semantic and behavioral pretraining, helping the model capture common task structures such as moving, grouping, separating, aligning, rotating, stacking, and reordering objects from language instructions alone.

Specifically, we define six task template families that span the core single-arm manipulation primitives: pick-and-place, linear pushing, linear pulling, rotation with repositioning, rotation toward a viewpoint direction, and positional swapping of two objects. Each template encodes a canonical motion structure while randomizing the spatial parameters and natural language description across instances. Instructions are procedurally composed from template-specific vocabularies of verbs, prepositions, object names, and spatial references to produce varied descriptions of the same underlying behavior (e.g., "Place the apple into the container," "Move the sponge toward the target location"). This design provides precise control over task coverage and difficulty while remaining simple enough to scale to millions of trajectories.

To ensure broad coverage of embodiment-specific kinematics and workspace geometry, we instantiate all six task templates across six single-arm robot configurations (Franka Panda, UR10e, UR5e, Kinova Gen3, TM12, and xArm7) spanning different degrees of freedom, arm morphologies, and reachable workspaces. For each robot–task pair, we generate approximately 200k trajectories, yielding roughly 7.2M trajectories and over 14,000 hours of simulated robot trajectory data in total. Trajectory synthesis proceeds without physics simulation or scene rendering: for each instance, we sample end-effector goal poses

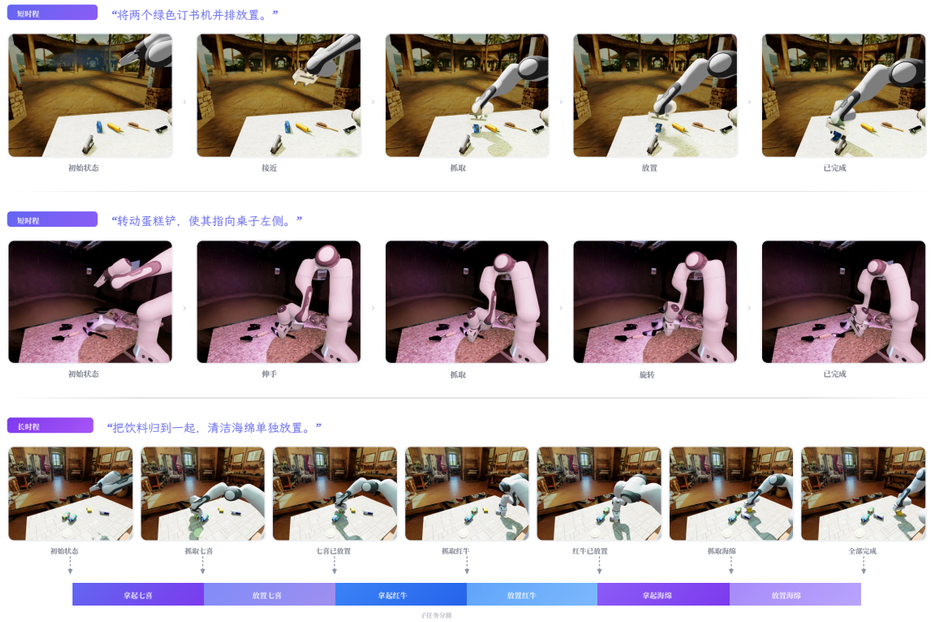

图 3：通过 ROBOINF 生成的数据示例。顶行展示了一个短时程任务“将两个绿色订书机并排放置”，该任务由一段紧凑的伸取、抓取、运输和放置序列组成。底行展示了一个长时程任务“将饮料归拢在一起，并将清洁海绵单独放置”，该任务需要多次物体操作，并可分解为诸如逐一拾取和放置每件饮料等子任务片段。

模型在任务类型、视觉条件和执行变化方面的泛化能力。

语言-动作数据。我们额外构建了一个纯文本动作数据集，作为补充监督来源。该组件的目标是在引入视觉观测之前，让模型接触广泛的操作意图和动作模式，从而促进其在语言-动作联合空间中的泛化能力。具体而言，我们构思了一组可由机器人在仿真中执行的多样化原子式和组合式操作任务，并使用以机器人状态为中心的流程在无图像输入的情况下生成相应轨迹。该数据主要作为一种语义与行为预训练形式，帮助模型仅从语言指令中捕获常见任务结构，例如移动、分组、分离、对齐、旋转、堆叠和重排物体。

具体而言，我们定义了六个任务模板族，涵盖核心的单臂操作基元：拾取放置、直线推动、直线拉动、重新定位并旋转、朝视角方向旋转，以及两个物体的位置交换。每个模板都编码了一种规范的运动结构，同时在各个实例之间随机化空间参数和自然语言描述。指令由模板特定的动词、介词、物体名称和空间参照词汇按程序组合而成，从而为同一底层行为生成多样的描述（例如，“将苹果放入容器中”，“将海绵移向目标位置”）。该设计可精确控制任务覆盖范围和难度，同时保持足够简单，以便扩展到数百万条轨迹。

为确保广泛覆盖各具身本体的运动学与工作空间几何特性，我们在六种单臂机器人配置（Franka Panda、UR10e、UR5e、Kinova Gen3、TM12 和 xArm7）上实例化全部六种任务模板，这些配置涵盖不同的自由度、机械臂形态和可达工作空间。对于每个机器人–任务对，我们生成约 20 万条轨迹，总计产生约 720 万条轨迹以及超过 14,000 小时的模拟机器人轨迹数据。轨迹合成过程无需物理仿真或场景渲染：对于每个实例，我们采样末端执行器的目标位姿

within the robot's reachable workspace bounds, randomizing positions, orientations, and inter-pose distances, and compute collision-free joint-space paths using a GPU-accelerated batch motion planner. The generation pipeline is parallelized across multiple GPUs, with each worker independently sampling target configurations, planning motions, and writing successful trajectories. Only kinematically valid trajectories for which the planner produces feasible, collision-free solutions across all required motion segments are retained.

Each trajectory records joint positions, joint velocities, end-effector poses (position and orientation), and gripper states at 50 Hz, providing a rich and flexible supervision signal that can support a variety of action representations and control modes during training.

During training, this dataset serves as the primary corpus for Stage I T2A pretraining (Section 3.1), where the DiT action decoder is initialized on language–action correspondences before visual grounding is introduced. By exposing the decoder to a large and diverse set of language–action pairs without visual conditioning, this component pre-empts visual shortcuts and establishes a strong action prior that transfers to and complements the downstream vision-conditioned training stages.

3.2.4 Navigation data.

We incorporate Navigation data (7.5%), which stands out for its long-horizon trajectories coupled with rich visual information. The mobile robot is assumed to have 3 degrees of freedom: translation in the plane and rotation about the vertical axis (heading angle). Navigation data comprises navigation videos (sampled at 2 FPS), instructions, and trajectory information, jointly equipping the model with mobile capabilities such as instruction following, object searching, and target tracking.

Instruction following. Instruction-following navigation data (4.3%) covers a diverse range of navigation horizons and instruction granularities. In these episodes, the mobile robot must execute commands that intermix motion primitives (e.g., move forward, turn right/left, stop) with textual landmarks. This data encourages the model to align navigation history with the given instruction and predict the subsequent trajectory.

Object searching. Object-searching data (2.3%) derives from navigation histories where the robot seeks a text-described object within large-scale indoor environments. The robot is challenged to remember previously navigated areas while exploring efficiently to locate the target. Such episodes foster the model's long-horizon exploration and identification capabilities.

Target tracking. Target-tracking data (1.0%) consists of episodes in which the robot follows a moving target (e.g., a described person) through indoor spaces, maintaining a safe following distance and keeping the target in view. The resulting trajectories blend reactive motion adjustments with longer-horizon path decisions triggered by turns or occlusions.

3.2.5 Vision-language data.

We further incorporate auxiliary vision-language supervision (8.5% combined) to strengthen semantic grounding, fine-grained instruction following, and general visual reasoning.

Fine-grained embodied action caption. Most existing robot datasets provide only coarse task-level labels (e.g., "pick up the ceramic bowl"), which are insufficient for precise action prediction: the same label may correspond to vastly different execution strategies (e.g., grasping from the left vs. the right, rotating clockwise vs. counter-clockwise), creating ambiguity during policy learning. Drawing inspiration from fine-grained text control in video generation, we construct dense action descriptions that tightly couple language to each individual episode. To build a diverse yet manageable annotation set, we draw episodes from major open-source manipulation datasets across all task categories and apply a clustering-based selection strategy: episodes within each task are grouped by their action information, and representative ones are sampled to ensure broad coverage. Each selected episode is annotated along 13 dimensions: action primitive, actor identity, object recognition and disambiguation, contact region, source and target location, trajectory and orientation, gripper state, and body motion. We use a two-stage pipeline for annotation. In the first stage, a vision-language model (Qwen3.6-plus) watches the video and extracts a coarse action sequence along with the manipulated object. In the second stage, the video is densely sampled into frames to preserve subtle motion details; the model then re-watches these frames conditioned on the first-stage output and produces fine-grained, step-by-step descriptions enriched with contact points, spatial relationships, and motion trajectories. For multi-view datasets, an additional refinement pass incorporates wrist- or side-camera footage to correct details only visible from close-up or

在机器人可达工作空间范围内，随机化位置、朝向和位姿间距离，并使用 GPU 加速的批量运动规划器计算无碰撞的关节空间路径。生成流水线跨多个 GPU 并行化，每个工作进程独立采样目标配置、规划运动并写入成功轨迹。仅保留那些规划器在所有所需运动段均能生成可行、无碰撞解的、在运动学上有效的轨迹。

每条轨迹以 50Hz 记录关节位置、关节速度、末端执行器位姿（位置和朝向）以及夹爪状态，提供丰富而灵活的监督信号，可在训练期间支持多种动作表示和控制模式。

在训练期间，该数据集作为第一阶段 T2A 预训练（第 3.1 节）的主要语料库；在该阶段中，DiT 动作解码器先基于语言–动作对应关系进行初始化，随后再引入视觉接地。通过让该解码器在无视觉条件的情况下接触大量且多样化的语言–动作对，这一组件预先规避了视觉捷径，并建立了强大的动作先验，该先验可迁移至下游的视觉条件训练阶段并对其形成补充。

3.2.4 导航数据。

我们引入导航数据（7.5%），其突出之处在于具有长时程轨迹并伴随丰富的视觉信息。假定移动机器人具有 3 个自由度：平面内的平移以及绕垂直轴的旋转（航向角）。导航数据包括导航视频（以 2 FPS 采样）、指令和轨迹信息，共同赋予模型指令跟随、物体搜索和目标跟踪等移动能力。

指令跟随。指令跟随导航数据（4.3%）涵盖多样化的导航范围和指令粒度。在这些回合中，移动机器人必须执行将运动基元（例如前进、右转/左转、停止）与文本地标混合在一起的命令。该数据促使模型将导航历史与给定指令对齐，并预测后续轨迹。

物体搜索。物体搜索数据（2.3%）源自机器人在大规模室内环境中寻找由文本描述的物体的导航历史。机器人面临挑战：既要记住此前导航过的区域，又要高效探索以定位目标。此类回合培养模型的长时程探索与识别能力。

目标跟踪。目标跟踪数据（1.0%）由机器人在室内空间中跟随移动目标（例如所描述的人）的回合组成，同时保持安全跟随距离并使目标保持在视野内。由此生成的轨迹将反应式运动调整与由转弯或遮挡触发的更长时程路径决策相结合。

3.2.5 视觉-语言数据。

我们进一步引入辅助视觉-语言监督（合计 8.5%），以增强语义锚定、细粒度指令遵循和通用视觉推理能力。

细粒度具身动作描述。大多数现有机器人数据集仅提供粗粒度的任务级标签（例如“拿起陶瓷碗”），这些标签不足以支持精确的动作预测：同一标签可能对应差异极大的执行策略（例如从左侧还是右侧抓取、顺时针还是逆时针旋转），从而在策略学习过程中造成歧义。受视频生成中细粒度文本控制的启发，我们构建了将语言与每个独立片段紧密耦合的稠密动作描述。为了构建多样且规模可控的标注集，我们从涵盖所有任务类别的主要开源操作数据集中抽取片段，并采用基于聚类的选择策略：每个任务内的片段按其动作信息进行分组，并抽样选取代表性片段以确保广泛覆盖。每个选中的片段均沿13个维度进行标注：动作基元、执行者身份、物体识别与消歧、接触区域、源位置与目标位置、轨迹与朝向、夹爪状态以及身体运动。我们采用两阶段流程进行标注。在第一阶段，视觉语言模型（Qwen3.6-plus）观看视频，并提取粗粒度动作序列以及被操作物体。在第二阶段，将视频密集采样为帧，以保留细微运动细节；随后，模型以第一阶段输出为条件重新观看这些帧，并生成细粒度的逐步描述，其中包含接触点、空间关系和运动轨迹。对于多视角数据集，还会增加一次细化处理，纳入腕部或侧视相机画面，以修正细节，这些细节仅在近景或

alternative perspectives. All generated captions are finally reviewed and corrected by human annotators. This pipeline yields approximately 48,000 fine-grained video–caption pairs (0.2% of the pretraining mixture). Table 3 contrasts a coarse label with our fine-grained caption for the same episode.

Table 3: Coarse task label vs. fine-grained action caption for the same episode.

Coarse label	"Pick up, rotate, and place the ceramic bowl."
Fine-grained	*Step 1:* Pick up the ceramic bowl from the right far edge. *Step 2:* Rotate the bowl clockwise for two full circles. *Step 3:* Place the bowl at the center of the table.

Building on the annotated data, we further fine-tune a dedicated VLM for embodied action captioning. We also design a companion evaluation benchmark consisting of QA pairs derived from the ground-truth annotations together with a scoring rubric for assessing caption quality.

Autonomous driving VQA. We incorporate autonomous-driving visual question-answering data (2.4%) as a complementary source of visually grounded, decision-oriented supervision. This mixture focuses on four transferable capabilities: (1) *Temporal scene understanding*: driving scenes evolve under ego-motion, dynamic agents, and changing hazards. We include LingoQA (Marcu et al., 2024), DriveAction (Hao et al., 2025), and MMAU (Fang et al., 2024) to strengthen event understanding and future-state prediction. (2) *Surround-view spatial reasoning*: multi-camera observations provide wide-field geometric context. We include Impromptu-VLA (Chi et al., 2025), nuScenes-QA (Qian et al., 2024), nuScenes-MQA (Inoue et al., 2024), MapLM (Cao et al., 2024), and WaymoQA (Yu et al., 2025b) to improve viewpoint-robust localization and spatial-relation understanding across diverse vehicles, sensors, and road layouts. (3) *Language-grounded localization*: embodied agents must bind language to objects, regions, and spatial relations before acting. We include CODA-LM (Chen et al., 2025a), Talk2Car (Deruyttere et al., 2019), DrivingVQA Corbiere et al. (2025), DriveLM (Sima et al., 2024), W3DA (Zhou et al., 2025), and GRAID (El-maaroufi et al., 2025) for object-level grounding and region-aware question answering. (4) *Planning-aware reasoning*: safe decision making requires linking perception to route, lane, and trajectory constraints. We include Bench2Drive-VL (Jia et al., 2026), DriveGPT4 (Xu et al., 2024), OmniDrive (Wang et al., 2025b), Senna (Jiang et al., 2024), NAVSIM-RecogDrive (Li et al., 2026c), and NAVSIM-Traj to provide supervision for maneuver selection and future trajectory prediction. Prior to pretraining, all data sources undergo standardization into a unified conversational format. We convert multiple-choice annotations into free-form QA supervision and normalize bounding boxes to a 1000-scale coordinate system. Additionally, multi-frame samples receive frame tags to maintain temporal order, while surround-view samples are assigned view tags to explicitly denote camera identity. Collectively, these structured sources offer transferable supervision for object-centric grounding, temporal state estimation, intent prediction, and goal-conditioned decision-making. This comprehensively complements the manipulation and navigation data utilized in our embodied pretraining.

Spatial grounding. We include 2D bounding box grounding data (2.5%) to reinforce object-level spatial understanding. Accurate spatial grounding is a prerequisite for language-conditioned manipulation, where the model must localize task-relevant objects from instruction-specified descriptions.

General vision-language data. To preserve robust visual perception and language grounding during embodied VLA pretraining, we supplement the primary action-centric dataset with a curated mixture of general vision-language data (3.4% of total tokens). This auxiliary mixture explicitly comprises: (1) captioning data for image–text alignment; (2) knowledge-oriented VQA and visual reasoning for world knowledge retention; (3) OCR and text-grounding samples for fine-grained textual perception; (4) interleaved instruction-following data for stable cross-modal alignment; and (5) common grounding tasks (e.g., referring expression, spatial relation prediction) for object-level language grounding. Critically, to better support embodied reasoning, we strategically up-weight video-centric, spatial-relation, and 3D-aware supervision within this mixture. Collectively, these general vision-language objectives mitigate catastrophic forgetting, retain foundational world-knowledge priors, and enhance robustness in language-conditioned embodied perception.

4 Post-Training

The large-scale pretraining of T2A and CPT produces **Qwen-VLA-Base**, a generalist vision-language-action model that exhibits broad cross-task and cross-embodiment generalization. Although this broad coverage equips the model with versatile knowledge, it does not yet yield the precision needed for

其他视角下可见。所有生成的描述最终均由人工标注员审核并修正。该流程产出约48,000个细粒度视频–描述对（占预训练混合数据的0.2%）。表3对比了同一片段的粗粒度标签与我们的细粒度描述。

表3：同一回合中粗粒度任务标签与细粒度动作描述的对比。

粗粒度标签	"拿起、旋转并放置陶瓷碗。"
细粒度	步骤 1：从右侧远端边缘拿起陶瓷碗。 步骤 2：将碗顺时针旋转整整两圈。 步骤 3：将碗放在桌子中央。

在标注数据的基础上，我们进一步微调一个专用的 VLM，用于具身动作描述生成。我们还设计了一个配套的评估基准，该基准由从真实标注中衍生的 QA 对以及用于评估描述质量的评分细则组成。

自动驾驶 VQA。我们将自动驾驶视觉问答数据（2.4%）作为具有视觉依据且面向决策的监督信号的补充来源。该混合数据聚焦于四项可迁移能力：（1）时序场景理解：驾驶场景会随自车运动、动态主体和不断变化的风险而演变。我们纳入 LingoQA (Marcu et al., 2024)、DriveAction (Hao et al., 2025) 和 MMAU (Fang et al., 2024)，以增强事件理解和未来状态预测。（2）环视空间推理：多相机观测提供广域几何上下文。我们纳入 Impromptu-VLA (Chi et al., 2025)、nuScenes-QA (Qian et al., 2024)、nuScenes-MQA (Inoue et al., 2024)、MapLM (Cao et al., 2024) 和 WaymoQA (Yu et al., 2025b)，以提升在多样化车辆、传感器和道路布局下的视角鲁棒定位与空间关系理解。（3）语言接地的定位：具身智能体在行动前必须将语言与物体、区域和空间关系绑定。我们纳入 CODA-LM (Chen et al., 2025a)、Talk2Car (Deruyttere et al., 2019)、DrivingVQA Corbiere et al. (2025)、DriveLM (Sima et al., 2024)、W3DA (Zhou et al., 2025) 和 GRAID (El- maaroufi et al., 2025)，用于对象级接地和区域感知问答。（4）规划感知推理：安全决策需要将感知与路线、车道和轨迹约束关联起来。我们纳入 Bench2Drive-VL (Jia et al., 2026)、DriveGPT4 (Xu et al., 2024)、OmniDrive (Wang et al., 2025b)、Senna (Jiang et al., 2024)、NAVSIM-RecogDrive (Li et al., 2026c) 和 NAVSIM-Traj，为机动选择与未来轨迹预测提供监督。在预训练之前，所有数据源都会被标准化为统一的对话格式。我们将多项选择标注转换为自由形式 QA 监督，并将边界框归一化到 1000 尺度的坐标系中。此外，多帧样本会获得帧标签以保持时序顺序，而环视样本会被分配视图标签，以显式标明相机身份。总体而言，这些结构化数据源为以对象为中心的接地、时序状态估计、意图预测和目标条件决策提供可迁移的监督。这全面补充了我们在具身预训练中使用的操作与导航数据。

空间定位。我们加入 2D 边界框定位数据（2.5%），以强化对象级空间理解。准确的空间定位是语言条件化操控的前提，模型必须根据指令指定的描述定位与任务相关的对象。

通用视觉语言数据。为在具身 VLA 预训练期间保持稳健的视觉感知和语言定位能力，我们用精选的通用视觉语言数据混合来补充以动作为主的数据集（占总 token 的 3.4%）。该辅助混合数据明确包含：(1) 用于图文对齐的图像描述数据；(2) 面向知识保留的 VQA 与视觉推理，用于保留世界知识；(3) OCR 和文本定位样本，用于细粒度文本感知；(4) 交错式指令遵循数据，用于稳定的跨模态对齐；(5) 常见定位任务（例如指称表达、空间关系预测），用于对象级语言定位。关键的是，为更好地支持具身推理，我们在该混合数据中策略性地提高以视频为中心、空间关系和 3D 感知监督的权重。总体而言，这些通用视觉语言目标缓解灾难性遗忘，保留基础世界知识先验，并增强语言条件化具身感知的鲁棒性。

4 后训练

T2A 与 CPT 的大规模预训练产生了 Qwen-VLA-Base，这是一个通用的视觉-语言-动作模型，展现出广泛的跨任务与跨具身泛化能力。尽管这种广泛覆盖为模型赋予了多样化的知识，但它尚未达到所需的精度，以满足

reliable closed-loop control on specific downstream tasks. To bridge this gap, we introduce a two-stage post-training procedure that specializes base model for accurate task execution:

(i) A multi-task *supervised fine-tuning* (SFT) stage that jointly fine-tunes both the VLM backbone and the action expert on heterogeneous tasks, including visual question answering, spatial grounding, manipulation, and navigation, under embodiment-balanced and task-balanced sampling.
(ii) A *reinforcement learning* (RL) stage, initialized from the SFT checkpoint, that further refines the policy by optimizing directly against task-success-driven rewards obtained from on-policy rollouts in simulation, producing the final model **Qwen-VLA-Instruct**.

Across both stages, we use cosine-decayed learning rates with separate group-wise schedules for the vision-language backbone and the action decoder, along with gradient clipping consistent with pretraining. The specific data and methods for each stage are detailed in the subsequent sections.

4.1 Multi-Task Supervised Fine-Tuning

Data. The SFT data mixture comprises three categories: general vision-language samples, vision-language navigation episodes, and robotic manipulation demonstrations. First, we curate a set of vision-language samples—covering visual question answering, spatial grounding, and action captioning—to maintain the backbone's visual understanding throughout fine-tuning. Second, we collect robotic manipulation demonstrations from simulation, including trajectories across a diverse set of platforms spanning single-arm, dual-arm, and humanoid robots, all using a prediction horizon of 16 action steps per chunk (Community, 2026). Following the same embodiment-aware prompt design used in pretraining (Section 2.3), every training sample is prefixed with a textual prompt that specifies the robot platform, arm configuration, control frequency, and prediction horizon. Third, we collect vision-language navigation episodes from continuous control environments, covering varied indoor scenes and instruction styles. We retain only successfully completed episodes, with a prediction horizon of 8 waypoints per chunk.

Objective. The SFT stage jointly optimizes two losses: next-token prediction on vision-language tokens and flow matching on action tokens. The loss weights are set to 0.1 for vision-language next-token prediction and 1.0 for both manipulation and navigation action prediction, ensuring that the model retains its language and visual understanding while focusing gradient capacity on action generation.

4.2 Reinforcement Learning

Likelihood-based SFT optimizes a surrogate that measures how well the policy *imitates* demonstrations, but the quantity that ultimately matters in embodied control is closed-loop *task success*: whether the executed trajectory achieves the goal. These two objectives can diverge: a policy may assign high likelihood to plausible actions yet fail when its own outputs shift the state distribution away from the demonstration support. The RL stage closes this gap by directly optimizing a task-success-driven reward on trajectories rolled out by the policy itself, producing the final model. We implement this stage using the RLinf framework (Yu et al., 2025a).

Policy optimization objective. We adopt Proximal Policy Optimization (PPO) (Schulman et al., 2017) with Generalized Advantage Estimation (GAE) (Schulman et al., 2015) as the RL algorithm. Let π_θ denote the current policy and $\pi_{\theta_{\text{old}}}$ the policy that generated the rollout batch. The clipped surrogate objective is

$$\mathcal{L}^{\text{actor}}(\theta) = -\,\mathbb{E}_t\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\ \text{clip}\big(r_t(\theta),\,1{-}\epsilon,\,1{+}\epsilon\big)\,\hat{A}_t\Big)\Big], \tag{6}$$

where $r_t(\theta) = \pi_\theta(a_t \mid s_t)\ /\ \pi_{\theta_{\text{old}}}(a_t \mid s_t)$ is the importance ratio, with $s_t = (o_t, x, e)$ the current state (visual observation, task instruction, and embodiment prompt following Section 2.3) and a_t the predicted action chunk of horizon H, $\hat{A}_t$ is the GAE advantage estimate with discount γ=0.99 and trace-decay λ=0.95, and ϵ=0.2 is the clipping threshold applied symmetrically. The total loss combines the policy surrogate with a value-function regression term:

$$\mathcal{L}(\theta) = \mathcal{L}^{\text{actor}}(\theta) + c_v\,\mathcal{L}^{\text{value}}(\theta), \tag{7}$$

where $\mathcal{L}^{\text{value}}$ is a loss on value predictions and c_v=1. We perform four optimization epochs per rollout batch.

Value estimation. Rather than training a separate critic network, we attach a lightweight value head directly to the vision-language backbone. The value head mean-pools all VLM hidden states and maps them to a scalar value estimate through a linear projection. We apply stop-gradient on the VLM hidden

在特定下游任务上进行可靠的闭环控制。为弥合这一差距，我们引入一种两阶段后训练流程，使基础模型专门用于准确执行任务：

(i) 一个多任务监督微调（SFT）阶段，在具身均衡和任务均衡采样下，对异构任务（包括视觉问答、空间定位、操作和导航）联合微调 VLM 主干与动作专家。(ii) 一个强化学习（RL）阶段，从 SFT 检查点初始化，通过在仿真中基于在策略 rollout 获得的任务成功驱动奖励进行直接优化，进一步精炼策略，最终生成模型 Qwen-VLA-Instruct。

在两个阶段中，我们对视觉-语言主干和动作解码器采用余弦衰减学习率，并使用各自的分组调度，同时采用与预训练一致的梯度裁剪。各阶段的具体数据和方法将在后续章节中详述。

4.1 多任务监督微调

数据。SFT 数据混合包含三类：通用视觉-语言样本、视觉-语言导航片段和机器人操作演示。首先，我们整理一组视觉-语言样本——涵盖视觉问答、空间定位和动作描述——以在微调过程中保持主干的视觉理解能力。其次，我们从仿真中收集机器人操作演示，包括跨多种平台（涵盖单臂、双臂和人形机器人）的轨迹，且每个块均使用 16 个动作步的预测时域（Community, 2026）。遵循预训练中使用的相同具身感知提示设计（第 2.3 节），每个训练样本都以文本提示为前缀，用于指定机器人平台、机械臂配置、控制频率和预测时域。第三，我们从连续控制环境中收集视觉-语言导航片段，涵盖多样化的室内场景和指令风格。我们仅保留成功完成的片段，每个块使用 8 个路点的预测时域。

目标。SFT 阶段联合优化两个损失：视觉-语言 token 上的下一 token 预测，以及动作 token 上的流匹配。损失权重设置为：视觉-语言下一 token 预测为 0.1，操作与导航动作预测均为 1.0，从而确保模型保留其语言和视觉理解能力，同时将梯度容量集中于动作生成。

4.2 强化学习

基于似然的 SFT 优化的是一个替代目标，用于衡量策略对示范的模仿程度，但在具身控制中最终重要的是闭环任务成功率：即所执行的轨迹是否达成目标。这两个目标可能会偏离：策略可能为看似合理的动作赋予高似然，但当其自身输出使状态分布偏离示范支撑区域时便会失败。RL 阶段通过在策略自身 rollout 的轨迹上直接优化由任务成功驱动的奖励来弥合这一差距，从而生成最终模型。我们使用 RLinf 框架（Yu et al., 2025a）实现该阶段。

策略优化目标。我们采用近端策略优化（PPO）（Schulman et al., 2017）与广义优势估计（GAE）（Schulman et al., 2015）作为 RL 算法。令 π_θ 表示当前策略，$\pi_{\theta_{\text{old}}}$ 表示生成 rollout 批次的策略。裁剪后的替代目标为

$$\mathcal{L}^{\text{actor}}(\theta) = -\,\mathbb{E}_t\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\ \text{clip}\big(r_t(\theta),\,1{-}\epsilon,\,1{+}\epsilon\big)\,\hat{A}_t\Big)\Big], \tag{6}$$

其中 $r_t(\theta) = \pi_\theta(a_t \mid s_t)\ /\ \pi_{\theta_{\text{old}}}(a_t \mid s_t)$ 是重要性比率，$s_t = (o_t, x, e)$ 是当前状态（视觉观测、任务指令以及遵循第2.3节的具身提示），a_t 是时域为 H 的预测动作块，$\hat{A}_t$ 是 GAE 优势估计，折扣为 γ=0.99、迹衰减为 λ=0.95，ϵ=0.2 是对称应用的裁剪阈值。总损失将策略代理项与价值函数回归项结合起来：

$$\mathcal{L}(\theta) = \mathcal{L}^{\text{actor}}(\theta) + c_v\,\mathcal{L}^{\text{value}}(\theta), \tag{7}$$

其中 $\mathcal{L}^{\text{value}}$ 是价值预测损失，且 c_v=1。每个 rollout 批次执行四个优化轮次。

价值估计。我们没有训练单独的评论家网络，而是将轻量级价值头直接附加到视觉-语言主干上。价值头对所有 VLM 隐藏状态进行均值池化，并通过线性投影将它们映射为标量价值估计。我们在 VLM 隐藏

states before they enter the value head, so that value-function gradients do not propagate back through the pretrained backbone. The value head is trained with a clipped MSE loss and a separate learning rate (10^{-4}, approximately 20× the actor learning rate of 5×10^{-6}), allowing it to converge rapidly while the policy updates remain conservative.

Log-probability estimation under flow matching. A key challenge in applying PPO to our flow-matching action decoder is computing the log-probability $\log \pi_\theta(a_t \mid s_t)$ required for the importance ratio $r_t(\theta)$. Unlike autoregressive token-based policies where log-probabilities are directly available from the softmax output, flow-matching models define an implicit density through a learned velocity field and iterative denoising. We convert the deterministic probability-flow ODE into a corresponding SDE by injecting controlled noise at each Euler denoising step (Song et al., 2021), so that each transition becomes an explicit Gaussian whose log-probability can be computed analytically without numerical ODE integration. During rollout we store the intermediate denoising states; at the PPO update we re-evaluate the velocity field under the current parameters and recompute the Gaussian log-probability, yielding the importance ratio at negligible additional cost. By default we randomly select a single denoising step per rollout for the log-probability estimate, requiring only one extra DiT forward pass during recomputation. Both log-probabilities and advantages are computed at the *action-chunk* level: each chunk of H=16 action steps receives a single scalar reward and a single advantage estimate, matching the temporal granularity of the flow-matching decoder's output.

Reward design. We use a sparse binary reward from the simulator: R=1 if the task goal is achieved at episode end and R=0 otherwise. Credit assignment across action chunks is handled by GAE (eq. (6)), which propagates the episode-level signal back through the value baseline. No learned reward model is used; all signals come from the simulator's ground-truth task-completion semantics.

Rollout infrastructure. On-policy rollouts are collected in simulation environments that mirror the downstream evaluation benchmarks. We adopt a decoupled client-server architecture in which a remote benchmark server hosts the simulation environments and the training process communicates with it via a network interface. This separation allows the simulation workload to scale independently of the GPU-intensive policy optimization. We instantiate N=128 parallel environment instances distributed across multiple manipulation task suites with non-uniform allocation proportional to task difficulty. Each training iteration collects 8 rollout epochs of 128 environment steps each, yielding $N \times \lfloor 128/H \rfloor \times 8 = 128 \times 8 \times 8 = 8{,}192$ transition chunks per iteration (where H=16 is the action chunk length). During rollout, actions are sampled from the policy with temperature τ=1.0; at evaluation time, we reduce the temperature to τ=0.6 to sharpen the action distribution. The embodiment prompts used during rollout are identical to those in SFT, ensuring that the RL stage does not reintroduce a distribution shift in the prompt conditioning.

Out-of-domain generalization. Because the embodiment prompt is the sole platform-specific interface, the RL-refined policy transfers to out-of-domain environments, tasks, and embodiments without an additional adaptation head or domain-specific fine-tuning. At deployment, we simply replace the embodiment prompt with the textual description of the physical robot platform, including its arm configuration, control frequency, and action space, while keeping the VLM backbone and DiT action decoder unchanged. This zero-shot generalization is enabled by two properties of our training recipe: (1) the CPT stage has already exposed the model to both simulated and real-world visual distributions (Section 3.1), so the backbone's visual representations are not specific to the simulation renderer; and (2) the RL stage optimizes task-success metrics that are invariant to the visual domain, encouraging the policy to rely on semantically meaningful visual features rather than renderer-specific textures. We quantify the cumulative effect of each post-training stage in Section 5.2.3.

5 Experiments

We conduct extensive experiments to evaluate the performance of Qwen-VLA in both simulation and real-world settings, covering two core embodied AI domains: robotic manipulation and visual navigation. We evaluate two model variants throughout: **Qwen-VLA-Base**, trained with large-scale pretraining on diverse embodied data, and **Qwen-VLA-Instruct**, further fine-tuned with instruction-following data collected across diverse simulated environments and tasks, capable of executing fine-grained manipulation and navigation with greater precision and reliability in one unified model.

状态进入价值头之前，对其施加停止梯度，使价值函数的梯度不会通过预训练主干反向传播。价值头使用裁剪后的 MSE 损失和单独的学习率（10^{-4}，约为 20× 倍 5×10^{-6} 的 actor 学习率）进行训练，使其能够快速收敛，同时策略更新保持保守。

流匹配下的对数概率估计。将 PPO 应用于我们的流匹配动作解码器时，一个关键挑战是计算重要性比率 $r_t(\theta)$ 所需的对数概率 $\log \pi_\theta(a_t \mid s_t)$。与可直接从 softmax 输出获得对数概率的自回归 token 策略不同，流匹配模型通过学到的速度场和迭代去噪来定义隐式密度。我们在每个 Euler 去噪步骤注入受控噪声（Song et al., 2021），将确定性的概率流 ODE 转换为相应的 SDE，从而使每次转移都成为一个显式高斯分布，其对数概率可以解析计算，而无需数值 ODE 积分。在 rollout 期间，我们存储中间去噪状态；在 PPO 更新时，我们在当前参数下重新评估速度场并重新计算高斯对数概率，从而以可忽略的额外成本得到重要性比率。默认情况下，我们为对数概率估计在每次 rollout 中随机选择一个去噪步骤，这仅需要在重新计算时额外进行一次 DiT 前向传播。对数概率和优势均在动作块级别计算：每个包含 H=16 个动作步骤的块获得单个标量奖励和单个优势估计，这与流匹配解码器输出的时间粒度相匹配。

奖励设计。我们使用来自模拟器的稀疏二值奖励：若在回合结束时达成任务目标，则 R=1，否则 R=0。跨动作块的信用分配由 GAE（式 (6)）处理，它将回合级信号通过价值基线进行反向传播。未使用学习得到的奖励模型；所有信号均来自模拟器对任务完成情况的真实语义。

rollout 基础设施。On-policy rollout 在模拟环境中采集，这些环境与下游评估基准保持一致。我们采用解耦的客户端-服务器架构：远程基准服务器托管模拟环境，训练进程通过网络接口与其通信。这种分离使模拟负载能够独立于 GPU 密集型的策略优化进行扩展。我们实例化 N=128 个并行环境实例，并将其分布到多个操作任务套件中，按任务难度成比例进行非均匀分配。每次训练迭代收集 8 个 rollout epoch，每个 epoch 包含 128 个环境步，从而每次迭代产生 $N \times \lfloor 128/H \rfloor \times 8 = 128 \times 8 \times 8 = 8{,}192$ 个过渡块（其中 H=16 为动作块长度）。在 rollout 期间，动作从策略中按温度 τ=1.0 采样；在评估时，我们将温度降至 τ=0.6，以锐化动作分布。rollout 期间使用的具身提示与 SFT 中的相同，确保 RL 阶段不会在提示条件中重新引入分布偏移。

域外泛化。由于具身提示是唯一的平台特定接口，经 RL 精调后的策略无需额外的适配头或领域特定微调，即可迁移到域外环境、任务和具身形态中。部署时，我们只需将具身提示替换为对物理机器人平台的文本描述，包括其机械臂配置、控制频率和动作空间，同时保持 VLM 主干和 DiT 动作解码器不变。这种零样本泛化能力得益于我们训练方案的两个特性：(1) CPT 阶段已使模型同时接触模拟和真实世界的视觉分布（第 3.1 节），因此主干的视觉表征并不特定于模拟渲染器；(2) RL 阶段优化的是对视觉域不变的任务成功指标，从而鼓励策略依赖具有语义意义的视觉特征，而不是渲染器特定的纹理。我们在第 5.2.3 节量化了每个后训练阶段的累积效果。

5 实验

我们开展了大量实验，以评估 Qwen-VLA 在仿真和真实环境中的性能，涵盖具身 AI 的两个核心领域：机器人操作与视觉导航。我们在整个评估过程中考察了两个模型变体：Qwen-VLA-Base，基于多样化的具身数据进行大规模预训练；以及 Qwen-VLA-Instruct，使用从多样化的仿真环境和任务中收集的指令跟随数据进一步微调，能够在单一统一模型中更精确、更可靠地执行细粒度操作与导航。

5.1 Main Results

5.1.1 Manipulation Results in Simulation

To comprehensively evaluate the manipulation capabilities of Qwen-VLA, we benchmark on four simulation environments covering single-arm and dual-arm settings: LIBERO (Liu et al., 2023), Simpler (Li et al., 2024), RoboCasa-GR1 Tabletop Tasks (Nasiriany et al., 2024), and RoboTwin 2.0 (Mu et al., 2024).

- **LIBERO** is a single-arm tabletop benchmark comprising diverse tasks across four splits: Libero-Spatial, Libero-Object, Libero-Goal, and Libero-Long.
- **Simpler-WidowX** is a real-to-sim evaluation suite using the WidowX robot arm.
- **RoboCasa-GR1** is a bimanual humanoid benchmark with 24 atomic kitchen tasks.
- **RoboTwin 2.0** is a dual-arm benchmark with 50 bimanual tasks divided into Easy and Hard difficulty tiers.

Together, these benchmarks test a wide range of manipulation skills from tabletop pick-and-place to long-horizon bimanual kitchen tasks. For all benchmarks, we set the action chunk length $H = 16$ and report the average success rate following the standard evaluation protocol in StarVLA (Community, 2026).

Table 4: **Robot manipulation results across benchmarks: specialists vs. a single generalist.** Specialist models are fine-tuned separately on each benchmark, whereas our generalist (Qwen-VLA) is trained once on all embodiments jointly and evaluated across all platforms without per-benchmark adaptation. **Bold**: best; underline: second best.

Method	Type	Benchmark				
		LIBERO	RoboCasa-GR1	Simpler-WidowX	RoboTwin-Easy	RoboTwin-Hard
π_0 (Black et al., 2024)	Specialist	94.4	–	–	65.9	58.4
StarVLA-OFT (Community, 2026)		96.6	48.8	64.6	50.4	–
GR00T N1.6 (NVIDIA et al., 2025)		97.2	49.9	63.2	47.6	–
$\pi_{0.5}$ (Black et al., 2025)		97.6	37.0	46.9	82.7	76.8
ABot-M0 (Yang et al., 2026)		**98.6**	**58.3**	–	86.0	85.0
Being-H0.5 (Luo et al., 2026)		97.6	53.3	–	–	–
Qwen-VLA-Base	Generalist	90.8	40.4	64.3	64.3	66.4
Qwen-VLA-Instruct		97.9	56.7	**73.7**	**86.1**	**87.2**

Table 4 compares our generalist model against state-of-the-art specialist policies. Specialist models are fine-tuned separately on each benchmark and remain embodiment-specific, while our single generalist is trained jointly on diverse, large-scale data spanning multiple scenes and embodiments, enabling deployment across all four platforms through embodiment-aware prompting alone.

A single generalist outperforms most specialists. Despite being an all-in-one model, Qwen-VLA-Instruct surpasses the majority of specialist baselines. On LIBERO, it achieves 97.9%, on par with the best specialists. On RoboCasa-GR1, it reaches 56.7%, outperforming $\pi_{0.5}$ (37.0%), GR00T N1.6 (49.9%), and Being-H0.5 (53.3%). On Simpler-WidowX, it attains 73.7%, surpassing specialist baselines including StarVLA-OFT (64.6%). On RoboTwin-Easy/Hard, it scores 86.1%/87.2%, **exceeding** the previous best specialist ABot-M0 (86.0%/85.0%). These results confirm that joint multi-embodiment training does not sacrifice task-specific performance; in several benchmarks, the generalist even outperforms dedicated models.

Pretraining provides a strong foundation, and instruction tuning yields substantial gains. Qwen-VLA-Base already achieves reasonable performance (e.g., 90.8% on LIBERO, 64.3% on Simpler-WidowX), indicating that large-scale pretraining learns transferable manipulation primitives across embodiments. With instruction tuning, Qwen-VLA-Instruct improves consistently across all benchmarks (+7.1% on LIBERO, +16.3% on RoboCasa-GR1, +9.4% on Simpler-WidowX, +21.8% on RoboTwin-Easy, +20.8% on RoboTwin-Hard), confirming that a modest amount of task-specific alignment is sufficient to convert general pretrained representations into precise, deployable policies.

5.1.2 Manipulation Results in the Real World

We collect the real-world dataset on ALOHA, a representative bimanual robot embodiment. The platform consists of two 6-DoF robotic arms equipped with parallel-jaw grippers, forming a bimanual manipulation system. It is instrumented with three RGB cameras, including two wrist-mounted cameras and one first-person-view camera. Unlike Qwen-VLA-Instruct, which is fine-tuned on simulated

5.1 主要结果

5.1.1 仿真中的操作结果

为了全面评估 Qwen-VLA 的操作能力，我们在四个仿真环境中进行基准测试，涵盖单臂和双臂设置：LIBERO（Liu 等，2023）、Simpler（Li 等，2024）、RoboCasa-GR1 Tabletop Tasks（Nasiriany 等，2024）以及 RoboTwin 2.0（Mu 等，2024）。

- LIBERO 是一个单臂桌面基准，包含四个子集中的多样化任务：Libero-Spatial、Libero-Object、Libero-Goal 和 Libero-Long。
- Simpler-WidowX 是一个使用 WidowX 机械臂的真实到仿真评估套件。
- RoboCasa-GR1 是一个双臂人形机器人基准，包含 24 个原子级厨房任务。
- RoboTwin 2.0 是一个双臂基准，包含 50 个双臂任务，分为简单和困难两个难度层级。

这些基准共同测试了广泛的操作技能，从桌面拾放到长时程双臂厨房任务。对于所有基准，我们将动作块长度设为 H ＝16，并按照 StarVLA（Community, 2026）中的标准评估协议报告平均成功率。

表 4：各基准上的机器人操作结果：专家模型 vs. 单一通用模型。专家模型分别在各基准上微调，而我们的通用模型（Qwen-VLA）在所有具身形态上联合训练一次，并在所有平台上评估，无需针对每个基准进行适配。粗体：最佳；下划线：次佳。

方法	Type	基准				
		LIBERO	RoboCasa-GR1	Simpler-WidowX	RoboTwin-Easy	RoboTwin-Hard
π_0 (Black 等，2024)	专用模型	94.4	–	–	65.9	58.4
StarVLA-OFT（社区，2026）		96.6	48.8	64.6	50.4	–
GR00T N1.6（NVIDIA 等，2025）		97.2	49.9	63.2	47.6	–
$\pi_{0.5}$ （Black 等，2025）		97.6	37.0	46.9	82.7	76.8
ABot-M0（Yang 等，2026）		**98.6**	**58.3**	–	86.0	85.0
Being-H0.5（Luo 等，2026）		97.6	53.3	–	–	–
Qwen-VLA-Base	通用型	90.8	40.4	64.3	64.3	66.4
Qwen-VLA-Instruct		97.9	56.7	**73.7**	**86.1**	**87.2**

表 4 将我们的通用模型与最先进的专用策略进行了比较。专用模型在每个基准上分别微调，并且仍特定于具体具身；而我们的单一通用模型则是在涵盖多个场景和具身的多样化大规模数据上联合训练的，仅通过具身感知提示即可部署到全部四个平台。

单一通用模型优于大多数专用模型。尽管 Qwen-VLA-Instruct 是一个一体化模型，但它仍超越了大多数专用基线。在 LIBERO 上，它达到 97.9%，与最佳专用模型持平。在 RoboCasa-GR1 上，它达到 56.7%，优于 $\pi_{0.5}$（37.0%）、GR00T N1.6（49.9%）和 Being-H0.5（53.3%）。在 Simpler-WidowX 上，它达到 73.7%，超过包括 StarVLA-OFT（64.6%）在内的专用基线。在 RoboTwin-Easy/Hard 上，它取得 86.1%/87.2% 的成绩，超过此前最佳专用模型 ABot-M0（86.0%/85.0%）。这些结果证实，联合多具身训练不会牺牲任务特定性能；在若干基准上，通用模型甚至优于专用模型。

预训练提供了坚实的基础，而指令微调带来了显著的提升。Qwen-VLA-Base 已经达到了合理的性能（例如，在 LIBERO 上达到 90.8%，在 Simpler-WidowX 上达到 64.3%），这表明大规模预训练学习到了可跨实施例迁移的操作原语。通过指令微调，Qwen-VLA-Instruct 在所有基准测试中均持续提升（在 LIBERO 上提升 +7.1%，在 RoboCasa-GR1 上提升 +16.3%，在 Simpler-WidowX 上提升 +9.4%，在 RoboTwin-Easy 上提升 +21.8%，在 RoboTwin-Hard 上提升 +20.8%），这证实了适度的任务特定对齐足以将通用的预训练表示转化为精确、可部署的策略。

5.1.2 真实世界中的操作结果

我们在 ALOHA 上采集真实世界数据集，ALOHA 是一种具有代表性的双臂机器人本体。该平台由两条配备平行夹爪的 6 自由度机械臂组成，构成双臂操作系统。它配备三个 RGB 相机，包括两个腕部安装相机和一个第一人称视角相机。不同于在模拟中微调的 Qwen-VLA-Instruct

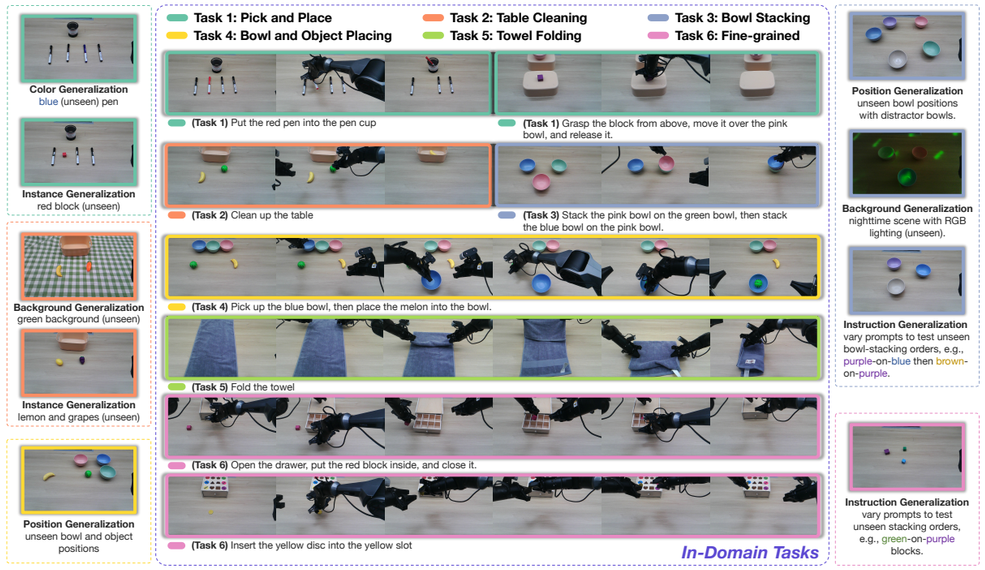

Figure 4: Overview of real-world evaluation tasks on the ALOHA bimanual platform.

environments, we further validate Qwen-VLA on real-world tasks by fine-tuning on demonstrations collected through ALOHA across multiple tasks. We prepare two variants sharing the same architecture: Qwen-VLA-aloha$_{\text{w/o pretrain}}$ trained from scratch, and Qwen-VLA-aloha$_{\text{w/ pretrain}}$ fine-tuned from Qwen-VLA-Base, to evaluate the transferability of large-scale embodied pretraining to real-world settings.

We evaluate the model on a diverse set of real-world manipulation tasks, covering both in-domain and out-of-distribution (OOD) scenarios. The in-domain evaluation contains six task categories: pick and place, table cleaning, bowl stacking, bowl picking and object placement, towel folding, fine-grained manipulation. The OOD evaluation further tests generalization to unseen colors, instances, positions, background, language instructions.

In-domain tasks. The in-domain evaluation covers six task categories of increasing complexity:

- **Task 1: Pick and Place.** The robot picks up basic objects, including a block and a pen, and places them at specified target locations.
- **Task 2: Table Cleaning.** The robot cleans up objects from the table and places them into a target container.
- **Task 3: Bowl Stacking.** The robot stacks bowls in the color order specified by the language instruction.
- **Task 4: Bowl Pick & Place.** The robot picks up a bowl and places target objects into it according to the language instruction.
- **Task 5: Towel Folding.** The robot folds a towel on the table, testing its ability to manipulate deformable objects.
- **Task 6: Fine-grained Manipulation.** This category includes several precise manipulation tasks, such as opening a drawer to place an object inside and then closing it, inserting an object into a slot, and stacking two blocks.

OOD tasks. The OOD evaluation includes five settings that assess different types of generalization:

- **Color Generalization.** This setting evaluates color generalization on Task 1 and Task 4, where the robot needs to handle unseen color specifications, such as picking up a bowl with a specified color (unseen) and placing an object into it, or putting a pen with a specified color (unseen) into the target container.
- **Instance Generalization.** This setting evaluates instance generalization on Task 1 and Task 2, where the robot encounters novel objects, such as grapes (unseen) and a lemon (unseen) during table cleaning,

图 4：ALOHA 双臂平台上的真实世界评估任务概览。

环境中，我们进一步通过在 ALOHA 上采集的跨多个任务的演示数据进行微调，在真实世界任务上验证 Qwen-VLA。我们准备了两个共享相同架构的变体：从头训练的 Qwen-VLA-aloha$_{\text{w/o}}$ pretrain，以及从 Qwen-VLA-Base 微调得到的 Qwen-VLA-aloha$_{\text{w/ pretrain}}$ fine-tuned，以评估大规模具身预训练向真实世界场景的可迁移性。

我们在多样化的真实世界操作任务上评估该模型，涵盖域内和分布外（OOD）场景。域内评估包含六个任务类别：拾取与放置、桌面清理、碗堆叠、碗拾取与物体放置、毛巾折叠、细粒度操作。OOD 评估进一步测试对未见过的颜色、实例、位置、背景和语言指令的泛化能力。

In-d领域任务。域内评估涵盖六个复杂度递增的任务类别 y:

- 任务 1：拾取与放置。机器人拾取包括积木和笔在内的基本物体，并将它们放到指定目标位置。
- 任务 2：桌面清理。机器人清理桌面上的物体，并将其放入目标容器中。
- 任务 3：叠碗。机器人按照语言指令指定的颜色顺序叠放碗。
- 任务 4：取放碗。机器人根据语言指令拿起一个碗，并将目标物体放入其中。
- 任务 5：叠毛巾。机器人在桌面上折叠毛巾，测试其操作可变形物体的能力。
- 任务 6：细粒度操作。该类别包括若干精确操作任务，例如打开抽屉并将物体放入其中后再关闭抽屉、将物体插入槽内，以及堆叠两个积木。

OOD 任务 s. OOD 评估包含五种设置，分别评估不同类型的泛化能力：

- 颜色泛化。该设置评估任务 1 和任务 4 上的颜色泛化能力，其中机器人需要处理未见过的颜色指定，例如拿起一个指定颜色（未见）的碗并将物体放入其中，或将一支指定颜色（未见）的笔放入目标容器中。
- 实例泛化。该设置评估任务 1 和任务 2 上的实例泛化，其中机器人在清理桌面时会遇到新的物体，例如葡萄（未见过的）和一个柠檬（未见过的），

or a red block (unseen) during object placement.

- **Position Generalization.** This setting evaluates position generalization on Task 3 and Task 4, where bowls and target objects appear at spatial configurations (unseen).
- **Background Generalization.** This setting evaluates background and lighting generalization on Task 2 and Task 3, including lighting conditions (unseen) for bowl stacking and a green background (unseen) for table cleaning.
- **Instruction Generalization.** This setting evaluates instruction generalization on Task 3 and Task 6, where the robot is required to execute different actions according to the given language instruction. Examples include stacking bowls in a specified order (unseen) or stacking blocks according to color relations, e.g., placing the green block (unseen) on the purple block (unseen).

Table 5: **In-domain performance across short-horizon and long-horizon task categories.** Qwen-VLA-aloha$_{\text{w/o pretrain}}$ and Qwen-VLA-aloha$_{\text{w/ pretrain}}$ share the same Qwen-VLA model architecture; the former is trained from scratch, and the latter is fine-tuned from Qwen-VLA-Base.

Model	Short-horizon Tasks			Long-horizon Tasks			Avg.
	Pick and Place	Table Cleaning	Bowl Stacking	Bowl Pick & Place	Towel Folding	Fine-grained Manipulation	
GR00T N1.6 (NVIDIA et al., 2025)	30.8	38.5	53.8	19.2	19.2	10.3	28.6
$\pi_{0.5}$ (Black et al., 2025)	73.1	84.6	88.5	69.2	**80.8**	33.3	71.6
Qwen-VLA-aloha$_{\text{w/o pretrain}}$	30.8	53.8	61.5	64.1	50.0	30.8	48.5
Qwen-VLA-aloha$_{\text{w/ pretrain}}$	**96.2**	**92.3**	**98.7**	**87.2**	65.4	**61.5**	**83.6**

Results on in-domain tasks. Table 5 summarises in-domain results. Qwen-VLA-aloha$_{\text{w/ pretrain}}$ achieves the best average success rate of 83.6%, outperforming strong baselines including GR00T N1.6 (NVIDIA et al., 2025) and $\pi_{0.5}$ (Black et al., 2025) on most task categories. The benefit of large-scale pretraining is evident from the comparison with Qwen-VLA-aloha$_{\text{w/o pretrain}}$: fine-tuning from Qwen-VLA-Base raises the average success rate from 48.5% to 83.6%, with particularly large gains on Pick and Place, Table Cleaning, Bowl Stacking, Bowl Pick & Place, and Fine-grained Manipulation. These results confirm that pretraining provides a strong foundation for real-world manipulation that transfers effectively across diverse in-domain tasks.

Results on OOD tasks. Table 6 reports the OOD generalization performance. Qwen-VLA-aloha$_{\text{w/ pretrain}}$ consistently achieves the best results across all five generalization settings, including color, instance, position, background, and instruction generalization. Its average OOD success rate reaches 76.9%, substantially outperforming $\pi_{0.5}$ (Black et al., 2025) by 35.4 percentage points and Qwen-VLA-aloha$_{\text{w/o pretrain}}$ by 40.7 percentage points. The improvement is particularly significant under background and instruction generalization, where Qwen-VLA-aloha$_{\text{w/ pretrain}}$ achieves 80.8% and 84.6%, respectively. These results suggest that pretraining not only improves in-domain performance but also provides stronger robustness to unseen visual conditions, object instances, positions, and instruction variations.

Overall analysis. Overall, the results demonstrate that pretraining plays a critical role in real-world manipulation learning. While Qwen-VLA-aloha$_{\text{w/o pretrain}}$ and Qwen-VLA-aloha$_{\text{w/ pretrain}}$ share the same architecture, training from scratch leads to much lower performance, especially under OOD settings. This comparison indicates that the performance gain does not simply come from model architecture, but mainly from the pretrained Qwen-VLA-Base model. Moreover, the strong OOD performance of Qwen-VLA-aloha$_{\text{w/ pretrain}}$ shows that pretrained vision-language-action representations can effectively support generalization across both visual variations and instruction variations.

5.1.3 Navigation Results

We evaluate the navigation capabilities of Qwen-VLA on the vision-and-language navigation in continuous environments (VLN-CE (Krantz et al., 2020; Ku et al., 2020)). Specifically, we test Qwen-VLA Base and Instruct versions on the Val-Unseen split of the R2R and RxR benchmarks, and compare them with the open-source baselines. We use the default setting of VLN-CE and, in particular, implement a `sliding-window` waypoint action compatible with the predicted trajectory of Qwen-VLA.

As shown in Table 7, Qwen-VLA-Instruct achieves the best performance on most metrics across both R2R and RxR benchmarks. On R2R Val-Unseen, it attains the highest Oracle Success rate (69.0) and Success Rate (57.5), surpassing StreamVLN by 4.8 and 0.6 points respectively, while maintaining competitive SPL. On the more challenging RxR Val-Unseen split, Qwen-VLA-Instruct leads in both SR (59.6) and SPL (47.8), outperforming all baselines by a notable margin. Joint training on VLA and VLN data maintains reasonable performance on both benchmarks.

或在物体放置过程中遇到一个红色方块（未见过的）。

- 位置泛化。该设置评估任务 3 和任务 4 中的位置泛化能力，其中碗和目标物体出现在未见过的空间配置中。
- 背景泛化。该设置评估任务 2 和任务 3 中的背景与光照泛化能力，包括叠碗任务中未见过的光照条件，以及桌面清洁任务中未见过的绿色背景。
- 指令泛化。该设置评估任务 3 和任务 6 中的指令泛化能力，其中机器人需要根据给定的语言指令执行不同的动作。示例包括按指定顺序叠碗（该顺序未见过），或根据颜色关系堆叠积木，例如将绿色积木（未见过）放在紫色积木（未见过）上。

表 5：短时域与长时域任务类别上的域内性能。Qwen-VLA-aloha$_{\text{w/o pretrain}}$ 与 Qwen-VLA-aloha$_{\text{w/ pretrain}}$ 共享相同的 Qwen-VLA 模型架构；前者从头训练，后者基于 Qwen-VLA-Base 微调。

模型	短时程任务			长时程任务			Avg.
	抓取与放置	桌面清理	叠碗	碗抓取与放置	毛巾折叠	细粒度操作	
GR00T N1.6（NVIDIA 等，2025）	30.8	38.5	53.8	19.2	19.2	10.3	28.6
$\pi_{0.5}$（Black 等，2025）	73.1	84.6	88.5	69.2	**80.8**	33.3	71.6
Qwen-VLA-aloha$_{\text{w/o 预训练}}$	30.8	53.8	61.5	64.1	50.0	30.8	48.5
Qwen-VLA-aloha$_{\text{w/ 预训练}}$	**96.2**	**92.3**	**98.7**	**87.2**	65.4	**61.5**	**83.6**

域内任务结果。表 5 汇总了域内结果。Qwen-VLA-aloha$^{\text{w/ pretrain}}$ 取得了 83.6% 的最佳平均成功率，在大多数任务类别上优于包括 GR00T N1.6（NVIDIA 等，2025）和 $\pi_{0.5}$（Black 等，2025）在内的强基线。与 Qwen-VLA-aloha$^{\text{w/o pretrain}}$ 的对比清楚体现了大规模预训练的优势：从 Qwen-VLA-Base 进行微调将平均成功率从 48.5% 提升至 83.6%，其中在抓取与放置、清理桌面、叠碗、碗抓取与放置和细粒度操作上提升尤为显著。这些结果证实，预训练为真实世界操作提供了坚实基础，并能够有效迁移到多样的域内任务中。

OOD 任务结果。表 6 报告了 OOD 泛化性能。Qwen-VLA-aloha$^{\text{w/ pretrain}}$ 在包括颜色、实例、位置、背景和指令泛化在内的全部五种泛化设置中始终取得最佳结果。其平均 OOD 成功率达到 76.9%，大幅优于 $\pi_{0.5}$（Black 等，2025）35.4 个百分点，并优于 Qwen-VLA-aloha$^{\text{w/o pretrain}}$ 40.7 个百分点。在背景和指令泛化下提升尤为显著，Qwen-VLA-aloha$^{\text{w/ pretrain}}$ 分别达到 80.8% 和 84.6%。这些结果表明，预训练不仅提升了域内性能，还对未见过的视觉条件、物体实例、位置和指令变化提供了更强的鲁棒性。

总体分析。总体而言，结果表明预训练在真实世界操作学习中起着关键作用。虽然 Qwen-VLA-aloha$^{\text{w/o pretrain}}$ 和 Qwen-VLA-aloha$^{\text{w/ pretrain}}$ 具有相同的架构，但从头训练会导致性能低得多，尤其是在分布外（OOD）设置下。这一比较表明，性能提升并非仅仅来自模型架构，而主要来自预训练的 Qwen-VLA-Base 模型。此外，Qwen-VLA-aloha$^{\text{w/ pretrain}}$ 在 OOD 下的强劲表现表明，预训练的视觉-语言-动作表征能够有效支持对视觉变化和指令变化的泛化。

5.1.3 导航结果

我们在连续环境中的视觉-语言导航（VLN-CE（Krantz et al., 2020; Ku et al., 2020））任务上评估 Qwen-VLA 的导航能力。具体而言，我们在 R2R 和 RxR 基准的 Val-Unseen 划分上测试 Qwen-VLA Base 和 Instruct 版本，并将它们与开源基线进行比较。我们使用 VLN-CE 的默认设置，并且特别实现了一种与 Qwen-VLA 的预测轨迹兼容的滑动窗口路点动作。

如表7所示，Qwen-VLA-Instruct 在 R2R 和 RxR 两个基准的大多数指标上均取得最佳性能。在 R2R Val-Unseen 上，它取得了最高的 Oracle 成功率（69.0）和成功率（57.5），分别比 StreamVLN 高出 4.8 和 0.6 个百分点，同时保持了具有竞争力的 SPL。在更具挑战性的 RxR Val-Unseen 划分上，Qwen-VLA-Instruct 在 SR（59.6）和 SPL（47.8）两项指标上均领先，并以显著优势优于所有基线。在 VLA 与 VLN 数据上进行联合训练，可在两个基准上保持合理性能。

Table 6: **OOD performance across generalization categories.** Qwen-VLA-aloha$_{\text{w/o pretrain}}$ and Qwen-VLA-aloha$_{\text{w/ pretrain}}$ share the same Qwen-VLA model architecture; the former is trained from scratch, and the latter is fine-tuned from pretrained Qwen-VLA-Base.

Model	Color	Instance	Position	Background	Instruction	Avg.
GR00T N1.6 (NVIDIA et al., 2025)	46.2	38.5	3.8	19.2	19.2	25.4
$\pi_{0.5}$ (Black et al., 2025)	57.7	61.5	19.2	26.9	42.3	41.5
Qwen-VLA-aloha$_{\text{w/o pretrain}}$	42.3	30.8	34.6	30.8	42.3	36.2
Qwen-VLA-aloha$_{\text{w/ pretrain}}$	**88.5**	**76.9**	**53.8**	**80.8**	**84.6**	**76.9**

Table 7: **Comparison with open-source baselines on VLN-CE.** We compare the Qwen-VLA Base and Instruct versions with widely regarded open-source baselines on the Val-Unseen split of the R2R and RxR benchmarks. **Bold**: best; underline: second best.

Method	**R2R Val-Unseen**				**RxR Val-Unseen**			
	NE↓	OS↑	SR↑	SPL↑	NE↓	SR↑	SPL↑	nDTW↑
NaVid (Zhang et al., 2024)	5.7	49.2	41.9	36.5	**5.7**	45.7	38.2	–
Uni-NaVid (Zhang et al., 2025b)	5.6	53.3	47.0	42.7	6.2	48.7	40.9	–
NaVILA (Cheng et al., 2025)	5.2	62.5	54.0	49.0	6.8	49.3	44.0	58.8
StreamVLN (Wei et al., 2025)	**5.0**	64.2	56.9	**51.9**	6.2	52.9	46.0	**61.9**
Qwen-VLA-Base	5.2	61.7	53.8	49.4	6.4	55.1	45.8	56.2
Qwen-VLA-Instruct	5.1	**69.0**	**57.5**	51.2	5.8	**59.6**	**47.8**	57.1

5.1.4 Out-of-Distribution Evaluation on Static Manipulation

To test whether pre-training priors generalise beyond the fine-tuning distribution, we construct a controlled OOD benchmark where fine-tuning covers only simple pick-and-place yet evaluation demands unseen task types. We develop SimplerEnv-OOD, a suite of 6 out-of-distribution tasks spanning 3 tabletop scenes built upon the SimplerEnv framework with a WidowX robotic arm. All models are fine-tuned solely on the Bridge training split, which contains only simple pick-and-place demonstrations with fixed object pairings. None of the OOD task instructions, spatial relations, or manipulation primitives appear in the training data. The six tasks are:

- **MoveAway:** Move the spoon farther from the cloth such that the final distance exceeds the initial distance. This tests *positional generalization* through distance-increasing placement.
- **MoveRight:** Move the spoon to the right side of the cloth. This tests *positional generalization* through directional placement with orthogonal-axis constraints.
- **PlaceNear:** Move the carrot to a position adjacent to but not on top of the plate. This tests *positional generalization* through proximity-based placement.
- **PlaceRight:** Place the spoon onto the right half region of the cloth. This tests *positional generalization* with region-constrained placement, requiring the policy to interpret a sub-region specifier ("right half") and achieve both correct target contact and spatial precision within the designated area.
- **PutFront:** Place the carrot in front of the plate. This tests *positional generalization* along the front/back directional axis.
- **StackYellow:** Pick up the yellow block and stack it on the green block. The training data only contains stacking green on yellow; this reversed order tests *visual generalization* by requiring color-based reasoning and generalization to novel object-color bindings not seen during training.

As shown in Table 8, Qwen-VLA-Instruct achieves the highest average success rate of **32.0%**, substantially outperforming $\pi_{0.5}$ (12.6%) across most tasks. The advantage is strongest on positional generalization: $\pi_{0.5}$ fails completely on MoveRight and PlaceNear, while Qwen-VLA-Instruct reaches 33.3% and 39.6%, respectively. Consistent gains also appear on MoveAway (43.8% vs. 26.1%) and PlaceRight (47.9% vs. 32.1%), indicating stronger generalization to unseen spatial instructions. On StackYellow (22.9% vs. 4.2%), where the training data only contains stacking green on yellow but the task requires the reversed order, Qwen-VLA-Instruct demonstrates stronger generalization to novel color-object bindings.

5.1.5 Out-of-Distribution Evaluation on Dynamic Manipulation

Deploying robots in complex environments requires mastering dynamic manipulation and necessitating adaptation to independent object motion (Liang et al., 2026). To systematically evaluate this critical

表 6：各泛化类别下的 OOD 性能。Qwen-VLA-aloha$^{\text{w/o pretrain}}$ 与 Qwen-VLA-aloha$^{\text{w/ pretrain}}$ 共享相同的 Qwen-VLA 模型架构；前者从头训练，后者由预训练的 Qwen-VLA-Base 微调而来。

模型	颜色	实例	位置	背景	指令	Avg.
GR00T N1.6（NVIDIA 等，2025）	46.2	38.5	3.8	19.2	19.2	25.4
$\pi_{0.5}$（Black 等，2025）	57.7	61.5	19.2	26.9	42.3	41.5
Qwen-VLA-alohaw/o 预训练	42.3	30.8	34.6	30.8	42.3	36.2
Qwen-VLA-alohaw/ 预训练	**88.5**	**76.9**	**53.8**	**80.8**	**84.6**	**76.9**

表 7：在 VLN-CE 上与开源基线方法的比较。我们在 R2R 和 RxR 基准的 Val-Unseen 划分上，将 Qwen-VLA Base 和 Instruct 版本与广受认可的开源基线方法进行比较。粗体：最佳；下划线：次佳。

方法	**R2R Val-Unseen**				**RxR Val-Unseen**			
	NE↓	OS↑	SR↑	SPL↑	NE↓	SR↑	SPL↑	nDTW↑
NaVid（Zhang 等，2024）	5.7	49.2	41.9	36.5	**5.7**	45.7	38.2	–
Uni-NaVid（Zhang 等，2025b）	5.6	53.3	47.0	42.7	6.2	48.7	40.9	–
NaVILA（Cheng 等，2025）	5.2	62.5	54.0	49.0	6.8	49.3	44.0	58.8
StreamVLN（Wei 等，2025）	**5.0**	64.2	56.9	**51.9**	6.2	52.9	46.0	**61.9**
Qwen-VLA-Base	5.2	61.7	53.8	49.4	6.4	55.1	45.8	56.2
Qwen-VLA-Instruct	5.1	**69.0**	**57.5**	51.2	5.8	**59.6**	**47.8**	57.1

5.1.4 静态操作上的分布外评估

为了检验预训练先验能否泛化到微调分布之外，我们构建了一个受控的 OOD 基准，其中微调仅涵盖简单的抓取放置任务，而评估则要求处理未见过的任务类型。我们开发了 SimplerEnv-OOD，这是一套包含 6 个分布外任务的基准，涵盖 3 个桌面场景，基于 SimplerEnv 框架并使用 WidowX 机械臂构建。所有模型仅在 Bridge 训练划分上进行微调，该划分仅包含具有固定物体配对的简单抓取放置演示。OOD 任务指令、空间关系或操作原语均未出现在训练数据中。这六项任务分别是：

- MoveAway：将勺子移得离布更远，使最终距离超过初始距离。这通过增大距离的放置方式来测试位置泛化能力。
- MoveRight：将勺子移到布的右侧。该任务通过具有正交轴约束的方向性放置来测试位置泛化能力。
- PlaceNear：将胡萝卜移到与盘子相邻但不在盘子正上方的位置。该任务通过基于邻近关系的放置来测试位置泛化能力。
- PlaceRight：将勺子放到布的右半部分区域上。该任务通过区域约束放置来测试位置泛化能力，要求策略解释子区域指定语（"右半部分"），并在指定区域内同时实现正确的目标接触和空间精度。
- PutFront：将胡萝卜放在盘子前方。该任务测试沿前后方向轴的位置泛化能力。
- StackYellow：拿起黄色积木并将其叠放到绿色积木上。训练数据只包含将绿色叠放到黄色上；这种相反顺序要求基于颜色的推理，并泛化到训练中未见过的全新物体—颜色绑定，从而测试视觉泛化能力。

如表8所示，Qwen-VLA-Instruct取得了最高的平均成功率32.0%，在大多数任务上大幅优于 $\pi_{0.5}$（12.6%）。其优势在位置泛化方面最为显著：$\pi_{0.5}$ 在MoveRight和PlaceNear上完全失败，而Qwen-VLA-Instruct分别达到33.3%和39.6%。在MoveAway（43.8% 对 26.1%）和PlaceRight（47.9% 对 32.1%）上也呈现出稳定提升，表明其对未见空间指令具有更强的泛化能力。在StackYellow（22.9% 对 4.2%）上，训练数据仅包含将绿色叠放到黄色上，而该任务要求相反顺序，Qwen-VLA-Instruct表现出对全新颜色—物体绑定更强的泛化能力。

5.1.5 动态操作中的分布外评估

在复杂环境中部署机器人需要掌握动态操作，并适应独立物体的运动（Liang et al., 2026）。为了系统地评估这一关键

Table 8: **OOD generalization results on SimplerEnv-OOD. Bold**: best; underline: second best.

Method	OOD Task						Avg.
	MoveAway	MoveRight	PlaceNear	PlaceRight	PutFront	StackYellow	
$\pi_{0.5}$ (Black et al., 2025)	26.1	0.0	0.0	32.1	**13.0**	4.2	12.6
Qwen-VLA-Base	31.3	31.6	16.7	47.1	6.3	18.8	25.3
Qwen-VLA-Instruct	**43.8**	**33.3**	**39.6**	**47.9**	4.2	**22.9**	**32.0**

Table 9: **Comparison with state-of-the-art methods on DOMINO. Bold**: best; underline: second best.

Method	SR (%)↑	MS↑
Fine-tuned on dynamic manipulation data		
OpenVLA (Kim et al., 2024)	1.5	6.1
RDT-1B (Liu et al., 2025b)	5.3	17.7
π_0 (Black et al., 2024)	8.2	24.0
$\pi_{0.5}$ (Black et al., 2025)	9.6	26.2
InternVLA-M1 (Chen et al., 2025b)	5.4	27.6
VLA-Adapter (Wang et al., 2026)	4.4	24.3
π_0-FAST (Pertsch et al., 2025)	3.5	20.9
OpenVLA-OFT (Kim et al., 2025)	9.1	24.1
StarVLA-OFT (Community, 2026)	10.9	30.5
PUMA (Fang et al., 2026)	17.2	35.0
Zero-shot to dynamic manipulation		
OpenVLA-OFT (Kim et al., 2025)	6.7	20.0
$\pi_{0.5}$ (Black et al., 2025)	7.5	20.4
LingBot-VLA w/ depth (Wu et al., 2026)	11.8	26.7
LingBot-VA (Li et al., 2026a)	24.1	36.1
Qwen-VLA-Base	21.1	37.4
Qwen-VLA-Instruct	**26.6**	**39.5**

capability, the recently introduced DOMINO benchmark (Fang et al., 2026) pioneers a comprehensive evaluation paradigm. It innovatively introduces a hierarchical motion design and a continuous Manipulation Score to capture execution quality under unpredictable constraints. These unique features make DOMINO an exceptional stress test for determining whether a vision-language-action model possesses generalizable spatial-to-kinematic mapping capabilities rather than merely executing memorized static routines. We evaluated Qwen-VLA in a zero-shot setting across all 35 DOMINO suites and report the Success Rate (SR) alongside the Manipulation Score (MS).

As shown in Table 9, Qwen-VLA establishes the best overall performance on the DOMINO benchmark. Qwen-VLA-Instruct achieves the highest SR (26.6%) and MS (39.5). In the zero-shot category, our model demonstrates a substantial advantage over standard vision-language-action architectures by surpassing both OpenVLA-OFT (Kim et al., 2025) and $\pi_{0.5}$ (Black et al., 2025) by over 19 percentage points in SR. Furthermore, it edges out the representative WAM-style method LingBot-VA (Li et al., 2026a) while demonstrating stronger continuous execution quality under moving-object dynamics. Crucially, Qwen-VLA outperforms the entire suite of baselines explicitly fine-tuned for dynamic manipulation. For example, the advanced method PUMA (Fang et al., 2026) relies on DOMINO-specific fine-tuning and temporal motion inputs to maintain dynamic awareness. Despite lacking these tailored adaptations and relying solely on current-frame observations, Qwen-VLA-Instruct surpasses PUMA by 9.4 percentage points in SR and 4.5 points in MS. Our model achieves these impressive gains at inference using only current-frame observations, without any dynamic-manipulation fine-tuning. This result highlights the advantage of our unified action-and-trajectory pretraining strategy in learning transferable spatial-to-kinematic priors.

We attribute this out-of-distribution generalization to the combination of decisive action generation and broad embodied pretraining. The flow-matching action decoder produces coherent action chunks. This mechanism reduces hesitation and helps the policy act precisely within narrow temporal windows. Additionally, large-scale joint pretraining across manipulation, navigation, trajectory prediction, and vision-language data provides transferable priors over visual grounding, spatial reasoning, and continuous control. These robust priors enable Qwen-VLA to generalize effectively to moving-object manipulation despite the complete absence of dynamic training data.

Tab表 8：在 SimplerEnv-OOD 上的 OOD 泛化结果。粗体：最佳；下划线：次佳 .

方法	OOD 任务						Avg.
	移开	向右移动	放置到附近	向右放置	放到前面	堆叠黄色	
$\pi_{0.5}$（Black 等，2025）	26.1	0.0	0.0	32.1	**13.0**	4.2	12.6
Qwen-VLA-Base	31.3	31.6	16.7	47.1	6.3	18.8	25.3
Qwen-VLA-Instruct	**43.8**	**33.3**	**39.6**	**47.9**	4.2	**22.9**	**32.0**

表 9：在 DOMINO 上与最先进方法的比较。粗体：最佳；下划线：次佳。

方法	成功率 (%)↑	MS↑
在动态操作数据上微调		
OpenVLA（Kim 等，2024）	1.5	6.1
RDT-1B（Liu 等，2025b）	5.3	17.7
π_0（Black 等，2024）	8.2	24.0
$\pi_{0.5}$（Black 等，2025）	9.6	26.2
InternVLA-M1（Chen 等，2025b）	5.4	27.6
VLA-Adapter（Wang 等，2026）	4.4	24.3
π_0-FAST（Pertsch 等，2025）	3.5	20.9
OpenVLA-OFT（Kim 等，2025）	9.1	24.1
StarVLA-OFT（社区，2026）	10.9	30.5
PUMA（Fang 等，2026）	17.2	35.0
零样本迁移至动态操作		
OpenVLA-OFT（Kim 等，2025）	6.7	20.0
$\pi_{0.5}$（Black 等，2025）	7.5	20.4
LingBot-VLA（带深度）（Wu 等，2026）	11.8	26.7
LingBot-VA（Li 等，2026a）	24.1	36.1
Qwen-VLA-Base	21.1	37.4
Qwen-VLA-Instruct	**26.6**	**39.5**

能力，最近提出的 DOMINObenchmark（Fang et al., 2026）开创了一种全面的评估范式。它创新性地引入了分层运动设计和连续的操作评分（Manipulation Score），以捕捉不可预测约束下的执行质量。这些独特特性使 DOMINO 成为一项出色的压力测试，用于判断视觉-语言-动作模型是否具备可泛化的空间到运动学映射能力，而非仅仅执行记忆中的静态例程。我们在零样本设置下对所有 35 个 DOMINO 套件评估了 Qwen-VLA，并报告成功率（SR）以及操作评分（MS）。

如表 9 所示，Qwen-VLA 在 DOMINO 基准上取得了最佳整体性能。Qwen-VLA-Instruct 实现了最高的 SR（26.6%）和 MS（39.5）。在零样本类别中，我们的模型相较标准视觉-语言-动作架构展现出显著优势，在 SR 上比 OpenVLA-OFT（Kim 等，2025）和 $\pi_{0.5}$（Black 等，2025）均高出超过 19 个百分点。此外，它还略胜具有代表性的 WAM 风格方法 LingBot-VA（Li 等，2026a），同时在移动物体动态场景下展现出更强的连续执行质量。关键在于，Qwen-VLA 优于整套专门针对动态操作显式微调的基线。例如，先进方法 PUMA（Fang 等，2026）依赖针对 DOMINO 的特定微调与时间运动输入来维持动态感知。尽管缺乏这些定制适配且仅依赖当前帧观测，Qwen-VLA-Instruct 在 SR 上仍超过 PUMA 9.4 个百分点，在 MS 上高出 4.5 分。我们的模型在推理时仅使用当前帧观测、且未进行任何动态操作微调，就取得了这些显著提升。这一结果凸显了我们统一的动作与轨迹预训练策略在学习可迁移的空间到运动学先验方面的优势。

我们将这种分布外泛化归因于果断动作生成与广泛具身预训练的结合。流匹配动作解码器会生成连贯的动作片段。该机制减少了犹豫，并帮助策略在狭窄的时间窗口内精确行动。此外，跨操作、导航、轨迹预测和视觉语言数据的大规模联合预训练，提供了关于视觉定位、空间推理和连续控制的可迁移先验。这些稳健的先验使 Qwen-VLA 能够在完全缺乏动态训练数据的情况下，仍有效泛化到移动物体操作任务。

5.1.6 Out-of-Distribution Generalization in Real World

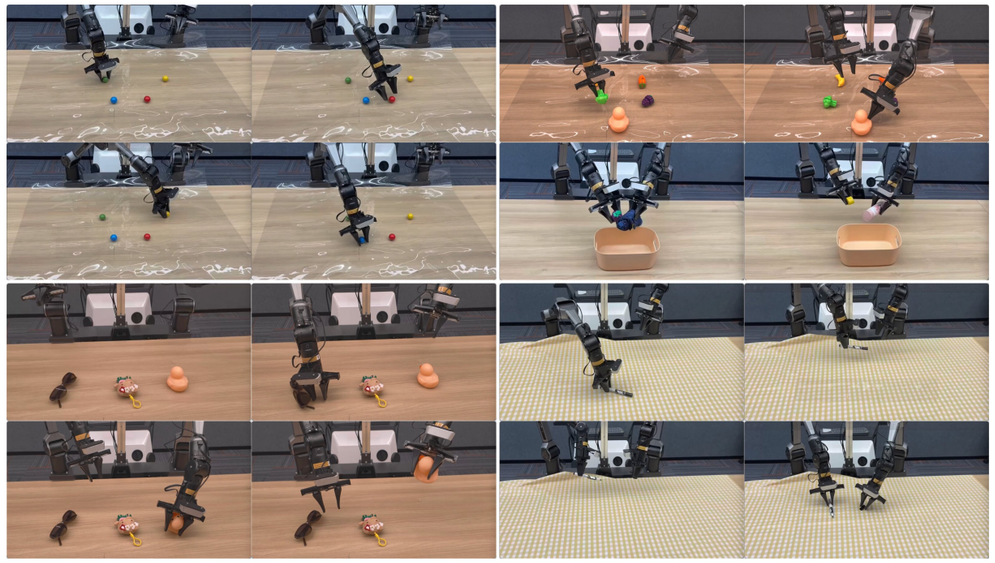

Figure 5: **Qualitative out-of-distribution generalization of Qwen-VLA-Base on the ALOHA dual-arm robot. Top-left:** color-conditioned grasping of green, blue, red, and yellow balls. **Top-right:** upper two panels show grasping of novel objects (green broccoli, toy duck); lower two panels show a compositional "clean up the table" task with sequential pick-and-place into a bin (blue umbrella, toy duck, bottled yogurt). **Bottom-left:** interaction with completely unseen objects (sunglasses, plush doll, toy duck). **Bottom-right:** atomic action transfer through uncapping a pen and placing the cap, robust to unseen yellow backgrounds.

To qualitatively examine continual pre-training, we evaluate **Qwen-VLA-Base** on the ALOHA dual-arm robot under four out-of-distribution settings: color grounding, novel object manipulation, unseen object recognition, and background robustness. Representative rollouts are shown in Figure 5.

Color grounding (top-left). We test whether Qwen-VLA-Base can discriminate between visually similar objects that differ only in color. Across four trials the robot is prompted with "grasp {color} ball" for green, blue, red, and yellow, respectively. In all cases the model correctly identifies the target color and produces accurate reaching-and-grasping trajectories, indicating color-conditioned semantic grounding after continual pre-training.

Novel object grasping and compositional clean-up (top-right). We next test whether the model can *zero-shot* grasp objects never seen in manipulation pretraining data. The top two panels show successful grasping prompted by "grasp {object}" for a green broccoli and a toy duck, both absent from the manipulation training set. The model successfully localizes and grasps the broccoli; for the toy duck, it correctly identifies the object and reaches toward it, though the irregular shape prevents a stable grasp. The bottom two panels further demonstrate a compositional task prompted by "clean up the table," where the robot sequentially picks up a blue umbrella, a toy duck, and a bottled yogurt and places them into a bin. All three are everyday objects absent from the manipulation set. Qwen-VLA-Base clears all objects in sequence, showing zero-shot visual recognition of novel objects and re-planning after each pick-and-place cycle. This behavior is consistent with the mixed VL–VLA–VLN training setup: visual-language data exposes the model to a broader object vocabulary, which can transfer to manipulation even without in-domain demonstrations.

Unseen object interaction (bottom-left). We further probe zero-shot generalization by issuing "approach {object}" instructions for object categories entirely absent from pretraining: sunglasses, a plush doll, and a toy duck. Notably, the "approach" action itself is almost never present in our manipulation data, yet the model correctly interprets the instruction and navigates the end-effector toward each target object. For the toy duck, the model even succeeds in grasping it by the head. For sunglasses, the model

5.1.6 真实世界中的分布外泛化

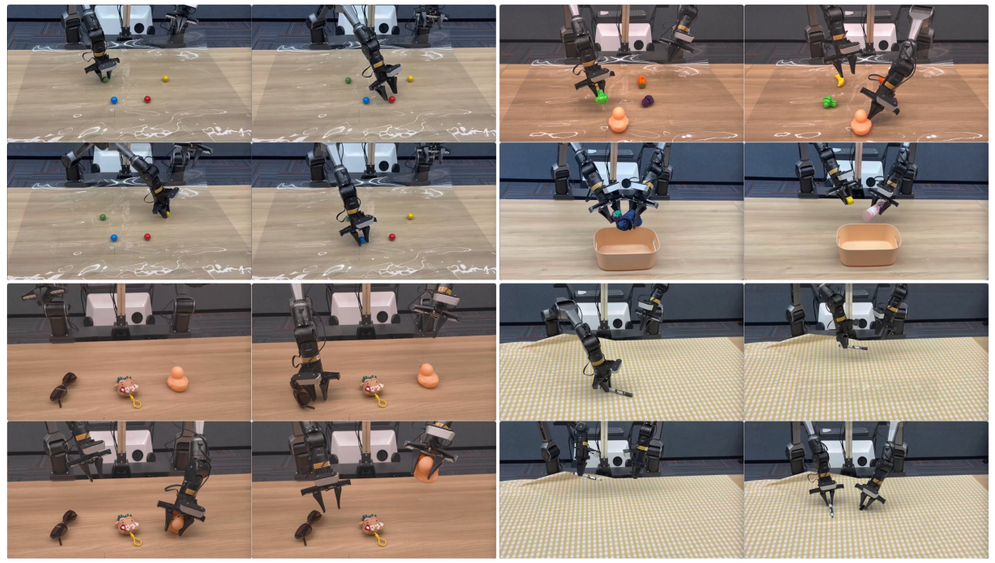

图 5：Qwen-VLA-Base 在 ALOHA 双臂机器人上的定性分布外泛化。左上：对绿色、蓝色、红色和黄色球进行颜色条件抓取。右上：上方两幅图展示对新物体的抓取（绿色西兰花、玩具鸭）；下方两幅图展示组合式"清理桌子"任务，依次拾取物体并放入收纳箱（蓝色雨伞、玩具鸭、瓶装酸奶）。左下：与完全未见过的物体交互（太阳镜、毛绒玩偶、玩具鸭）。右下：通过打开笔帽并放置笔帽实现原子动作迁移，并对未见过的黄色背景具有鲁棒性。

为了定性考察持续预训练，我们在四种分布外设置下于 ALOHA 双臂机器人上评估 Qwen-VLA-Base：颜色定位、新物体操作、未见物体识别和背景鲁棒性。代表性执行轨迹如图 5 所示。

颜色定位（左上）。我们测试 Qwen-VLA-Base 能否区分仅在颜色上不同的视觉相似物体。在四次试验中，机器人分别以"抓取 {color} 球"为提示，对应绿色、蓝色、红色和黄色。在所有情况下，模型都正确识别目标颜色，并生成准确的接近与抓取轨迹，表明持续预训练后具备颜色条件语义定位能力。

新颖物体抓取与组合式清理（右上）。接下来，我们测试模型能否零样本抓取在操作预训练数据中从未见过的物体。上方两幅图展示了在"抓取 {object}"提示下成功抓取一颗绿色西兰花和一个玩具鸭，二者均未出现在操作训练集中。模型成功定位并抓取了西兰花；对于玩具鸭，模型正确识别出该物体并朝其伸手，尽管其不规则形状导致无法稳定抓取。下方两幅图进一步展示了一个由"清理桌子"提示的组合式任务，机器人依次拿起一把蓝色雨伞、一个玩具鸭和一瓶酸奶，并将它们放入箱子。这三者都是日常物体，且未出现在操作集中。Qwen-VLA-Base 依次清除了所有物体，展现出对新颖物体的零样本视觉识别能力，以及在每次抓取-放置循环后重新规划的能力。这一行为与 VL–VLA–VLN 混合训练设置相一致：视觉-语言数据使模型接触到更广泛的物体词汇，即使没有域内演示，这些词汇也能迁移到操作任务中。

未见物体交互（左下）。我们通过针对完全未出现在预训练中的物体类别——太阳镜、毛绒玩偶和玩具鸭——发出"接近 {object}"指令，进一步考察零样本泛化能力。值得注意的是，"接近"动作本身几乎从未出现在我们的操作数据中，但模型仍能正确理解该指令，并将末端执行器导航至每个目标物体。对于玩具鸭，模型甚至成功抓住了它的头部。对于太阳镜，模型

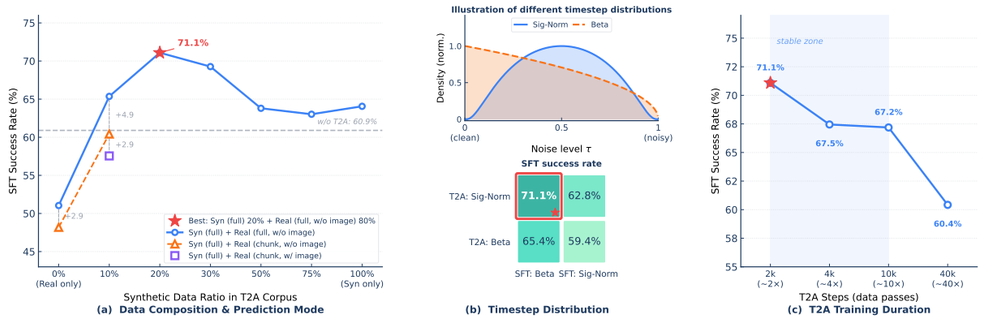

Figure 6: T2A pretraining ablations. **(a) Data composition and prediction mode.** SFT success rate (%) vs. synthetic data ratio in the T2A corpus. The dashed line marks the baseline without T2A (60.9%). Full-sequence prediction with ∼20% synthetic + 80% real data achieves the best result (71.1%, +10.2 pp over no T2A). Chunk prediction consistently underperforms full-sequence (e.g. +4.9 pp at 10% synthetic), and including image tokens during T2A further hurts chunk-mode performance (−2.9 pp), confirming that images should be fully suppressed at T2A. **(b) Flow-matching timestep distribution.** The combination of Sigmoid-Normal $p(\tau)$ at T2A and Beta $p(\tau)$ at SFT achieves the highest success rate (71.1%); reversing the choice at either stage degrades performance. **(c) T2A training duration.** Performance peaks at 2,000 steps (71.1%) and plateaus through 10,000 steps; 40,000 steps causes a slight drop due to overfitting to the T2A corpus.

accurately localizes the object and approaches it, though the thin and flat geometry prevents a stable grasp. This reflects the physical boundary of shape generalization rather than a failure in visual recognition or instruction following. This result suggests instruction-following transfer from mixed VL–VLA–VLN continual pre-training: the model can follow novel action verbs and recognize unseen objects using linguistic and visual coverage from the general-purpose pre-training data.

Background robustness (bottom-right). Finally, we probe robustness to visual domain shift. Qwen-VLA-Base successfully uncaps a pen and places the cap on the table, which is a dexterous two-stage action requiring precise force control. While this task may have been encountered during pre-training, the yellow background used here was never seen during training. Despite this unseen visual context, the policy completes the task without any degradation. This invariance to background perturbation is consistent with the visual diversity of the general-purpose VL understanding data used during continual pre-training. Exposure to natural images with varied scenes, lighting, and backgrounds may encourage the visual encoder to focus on task-relevant objects rather than spurious background cues.

5.2 Ablation Studies

5.2.1 Text-to-Action Pretraining Ablations

As established in Section 3.1, T2A trains the decoder to reconstruct action distributions from language and embodiment prompts alone, building a structured action prior before visual grounding begins. We now ablate five design choices that shape this prior: data composition, sequence prediction mode, visual input, flow-matching timestep distribution, and training duration. All ablations share the same downstream training recipe; results are task-success rate (%) after SFT from the T2A checkpoint on Simpler-WidowX.

Data composition. T2A acquires action priors from language alone, so its data composition directly shapes what the decoder learns. We consider two sources. *Syn* (language-only synthetic trajectories from our simulation pipeline) is cheap to scale and broad in task coverage, but its trajectories are kinematically idealised. *Real* (real-robot teleoperation data, vision dropped) captures authentic physical dynamics but is expensive to collect and narrower in task scope.

As shown in Fig. 6(a), purely real T2A data yields 51.0%, while purely synthetic data reaches 64.1%—better, but still suboptimal. Mixing ∼20% synthetic with 80% real achieves the best result (**71.1%**), a +10.2 pp gain over the no-T2A baseline (60.9%). Syn broadens the coverage of language–action correspondences that the decoder can learn, while Real anchors the prior in physically plausible dynamics.

图 6：T2A 预训练消融实验。(a) 数据组成与预测模式。SFT 成功率（%）与 T2A 语料中合成数据比例的关系。虚线表示无 T2A 的基线（60.9%）。使用 ∼20% 合成数据和 + 80% 真实数据的全序列预测取得了最佳结果（71.1%，相较无 T2A 提升 +10.2 个百分点）。分块预测始终不如全序列预测（例如在 10% 合成数据下低 +4.9 个百分点），并且在 T2A 中加入图像 token 会使分块模式性能进一步下降（−2.9 个百分点），这证实了在 T2A 时应完全抑制图像。(b) 流匹配时间步分布。在 T2A 阶段使用 Sigmoid-Normal p(τ) 与在 SFT 阶段使用 Beta p(τ) 的组合取得了最高成功率（71.1%）；在任一阶段反转该选择都会降低性能。(c) T2A 训练时长。性能在 2,000 步时达到峰值（71.1%），并一直保持平稳至 10,000 步；40,000 步会因对 T2A 语料过拟合而略有下降。

能够准确定位该物体并靠近它，尽管其薄而平的几何形状妨碍了稳定抓取。这反映的是形状泛化的物理边界，而非视觉识别或指令遵循方面的失败。这一结果表明，混合 VL–VLA–VLN 持续预训练带来了指令遵循迁移：模型能够遵循新的动作动词，并利用通用预训练数据中的语言与视觉覆盖来识别未见物体。

背景鲁棒性（右下）。最后，我们考察对视觉域偏移的鲁棒性。Qwen-VLA-Base 成功拔下笔帽并将笔帽放在桌上，这是一个需要精确力控的灵巧两阶段动作。尽管这项任务可能在预训练期间遇到过，但此处使用的黄色背景在训练期间从未出现过。尽管存在这种未见过的视觉上下文，策略仍能完成该任务，且没有任何性能下降。这种对背景扰动的不变性与持续预训练期间使用的通用视觉语言理解数据的视觉多样性相一致。接触具有多样场景、光照和背景的自然图像，可能会促使视觉编码器关注任务相关物体，而不是虚假的背景线索。

5.2 消融研究

5.2.1 文本到动作预训练消融实验

如第 3.1 节所述，T2A 训练解码器仅根据语言和具身提示重建动作分布，从而在视觉定位开始之前构建结构化的动作先验。我们现在对塑造该先验的五个设计选择进行消融：数据组成、序列预测模式、视觉输入、流匹配时间步分布和训练时长。所有消融实验均采用相同的下游训练方案；结果是在 Simpler-WidowX 上从 T2A 检查点进行 SFT 后的任务成功率（%）。

数据组成。T2A 仅从语言中获取动作先验，因此其数据组成直接塑造了解码器所学到的内容。我们考虑两种来源。Syn（来自我们仿真流程的仅语言合成轨迹）易于扩展且任务覆盖范围广，但其轨迹在运动学上是理想化的。Real（真实机器人遥操作数据，丢弃视觉）捕捉真实的物理动态，但采集成本高且任务范围较窄。

如图 6(a) 所示，纯真实 T2A 数据可达到 51.0%，而纯合成数据可达到 64.1%——虽然更好，但仍非最优。将 ∼20% 合成数据与 80% 真实数据混合可取得最佳结果（71.1%），相比无 T2A 基线（60.9%）提升 +10.2 个百分点。Syn 拓宽了解码器可学习的语言–动作对应关系的覆盖范围，而 Real 则将先验锚定在物理上合理的动力学中。

Sequence prediction mode: full-length vs. chunk. Full-sequence prediction supervises the entire trajectory in one forward pass, exposing the decoder to long-horizon temporal structure and natural start/termination patterns. Chunk prediction instead splits each trajectory into fixed-length windows and trains the decoder to produce only the next window from a task instruction, which truncates long-horizon dependencies and creates positional ambiguity at chunk boundaries.

Fig. 6(a) shows that full-sequence prediction consistently outperforms chunk prediction across data mixes. At 10% synthetic data, the gap is +4.9 pp (65.4% vs. 60.4%); at 0% synthetic (real-only) the gap is +2.9 pp (51.0% vs. 48.2%). Full trajectories let the decoder learn how language maps to complete action sequences, including trajectory-level coherence and compositionality, whereas chunks provide only local fragments that leave this mapping incomplete. Importantly, the action representation is identical between T2A and downstream stages: actions are encoded as delta end-effector displacements relative to the first frame of each chunk, and the embodiment prompt specifies the robot platform, normalisation convention, and prediction horizon, allowing the decoder to handle heterogeneous embodiments within the same T2A objective.

Visual input during T2A. T2A is deliberately a *vision-free* stage. Including image observations would raise per-step cost by roughly an order of magnitude and, more importantly, would let the decoder shortcut through visual cues instead of learning to ground actions in language.

Fig. 6(a) confirms this: at 10% synthetic data, chunk prediction with images scores 57.6%, while the same setting without images reaches 60.4%—a −2.9 pp penalty for including vision. When images are present, the decoder can rely on visual correlations rather than building a reliable language–action mapping, diluting exactly the prior that T2A is designed to establish. Image tokens should be fully suppressed during T2A; visual grounding is deferred to CPT, by which time the decoder already possesses a well-formed action prior.

Flow-matching timestep distribution. Flow-matching policies condition on a scalar timestep $\tau \in [0, 1]$ that interpolates between the clean target (τ=0) and pure noise (τ=1). The sampling distribution $p(\tau)$ determines which noise levels receive the most gradient signal during training. The standard practice is to use a *Beta* distribution, which concentrates density toward the clean end; we follow this convention at the CPT and SFT stages. However, since T2A operates without visual conditioning, the decoder has no backbone features to guide denoising, and we hypothesise that shifting gradient toward intermediate timesteps—where the signal-to-noise ratio is most informative for learning the language–action mapping—may be more effective. We therefore compare Beta against *Sigmoid-Normal*, which peaks at intermediate noise levels (Fig. 6(b)).

As shown in Fig. 6(b), Sigmoid-Normal $p(\tau)$ at T2A combined with Beta $p(\tau)$ at SFT achieves the highest success rate (**71.1%**). Swapping to Beta at T2A drops performance to 65.4% (−5.7 pp), while using Sigmoid-Normal at SFT instead of Beta yields 62.8% (−8.3 pp). The worst combination—Beta at both stages—reaches only 59.4%. Without visual conditioning to guide denoising, intermediate timesteps carry the most learning signal for the language–action prior; once the VLM backbone supplies rich conditioning at CPT/SFT, a Beta-shaped $p(\tau)$ that spreads gradient more uniformly becomes more sample-efficient. We therefore use Sigmoid-Normal at T2A and Beta at CPT/SFT throughout our final pipeline.

T2A training duration. Longer T2A training lets the decoder more fully explore the action distribution, but excessive training on a fixed T2A corpus risks overfitting to its idiosyncrasies and reducing the plasticity available to CPT. We sweep T2A duration over a wide range to locate the sweet spot.

As shown in Fig. 6(c), performance peaks at 2,000 steps (**71.1%**) and remains comparable at 4,000 steps (67.5%) and 10,000 steps (67.2%). At 40,000 steps we observe a notable degradation to 60.4%. The rapid convergence is expected: the language–action mapping that T2A learns is structurally lower-dimensional than the full vision–language–action space, so a modest number of steps suffices to capture it; prolonged training begins to memorise specific trajectory instances rather than refining the structural prior. We adopt **2,000 T2A steps** as the default in our final pipeline, offering a favourable cost–quality tradeoff.

5.2.2 Multi-Embodiment Co-training

We study two design axes that are critical for multi-embodiment co-training: the role of vision-language data during action learning and the projection policy for heterogeneous action spaces.

Impact of vision-language data on action learning. Although vision-language supervision operates in discrete token space while action prediction targets continuous trajectories, the two objectives share a common perceptual and linguistic backbone. A natural question is whether retaining VL data during

序列预测模式：全长预测与分块预测。全序列预测在一次前向传播中监督整个轨迹，使解码器接触到长时程时间结构以及自然的起始/终止模式。分块预测则将每条轨迹切分为固定长度窗口，并训练解码器仅根据任务指令生成下一个窗口，这会截断长时程依赖，并在块边界处造成位置歧义。

图 6(a) 表明，在不同数据混合下，全序列预测始终优于分块预测。在使用 10% 合成数据时，差距为 +4.9 个百分点（65.4% 对 60.4%）；在使用 0% 合成数据（仅真实数据）时，差距为 +2.9 个百分点（51.0% 对 48.2%）。完整轨迹使解码器能够学习语言如何映射到完整的动作序列，包括轨迹级的一致性和组合性，而分块仅提供局部片段，使这种映射不完整。重要的是，T2A 与下游阶段的动作表示相同：动作被编码为相对于每个分块首帧的末端执行器增量位移，并且具身提示指定了机器人平台、归一化约定和预测时域，从而使解码器能够在同一 T2A 目标下处理异构具身。

T2A 期间的视觉输入。T2A 被有意设计为无视觉阶段。加入图像观测会使每步成本提高约一个数量级，更重要的是，会让解码器通过视觉线索走捷径，而不是学习将动作与语言对应起来。

图 6(a) 证实了这一点：在 10% 合成数据下，带图像的分块预测得分为 57.6%，而相同设置下不带图像则达到 60.4%——因引入视觉而带来 −2.9 个百分点的损失。当存在图像时，解码器可能依赖视觉相关性，而不是构建可靠的语言–动作映射，从而恰好稀释了 T2A 旨在建立的先验。在 T2A 期间应完全抑制图像 token；视觉对齐则推迟到 CPT，届时解码器已经具备成型良好的动作先验。

流匹配时间步分布。流匹配策略以标量时间步 $\tau \in [0, 1]$ 为条件，该时间步在干净目标（τ=0）与纯噪声（τ=1）之间插值。采样分布 $p(\tau)$ 决定了训练期间哪些噪声水平获得最多的梯度信号。标准做法是使用 Beta 分布，该分布将密度集中于干净端；我们在 CPT 和 SFT 阶段遵循这一惯例。然而，由于 T2A 在无视觉条件下运行，解码器没有可引导去噪的骨干特征，我们假设将梯度转向中间时间步——在这些时间步上，信噪比对于学习语言–动作映射最具信息量——可能更有效。因此，我们将 Beta 与 Sigmoid-Normal 进行比较，后者在中间噪声水平处达到峰值（图 6(b)）。

如图 6(b) 所示，在 T2A 阶段采用 Sigmoid-Normal $p(\tau)$ 并结合 SFT 阶段的 Beta $p(\tau)$，取得了最高成功率（71.1%）。若在 T2A 阶段改用 Beta，性能降至 65.4%（−5.7 个百分点）；而在 SFT 阶段用 Sigmoid-Normal 替代 Beta，则得到 62.8%（−8.3 个百分点）。最差组合——两个阶段都使用 Beta——仅达到 59.4%。在没有视觉条件引导去噪的情况下，中间时间步为语言–动作先验承载了最多的学习信号；一旦 VLM 主干在 CPT/SFT 阶段提供丰富条件，采用 Beta 形状的 $p(\tau)$ 能更均匀地分配梯度，从而更具样本效率。因此，我们在整个最终流程中于 T2A 阶段使用 Sigmoid-Normal，并在 CPT/SFT 阶段使用 Beta。

T2A 训练时长。更长的 T2A 训练能让解码器更充分地探索动作分布，但在固定的 T2A 语料上过度训练，可能导致对其特有模式过拟合，并削弱可供 CPT 使用的可塑性。我们在较大范围内扫描 T2A 时长，以找到最佳平衡点。

如图 6(c) 所示，性能在 2,000 步时达到峰值（71.1%），在 4,000 步（67.5%）和 10,000 步（67.2%）时仍保持相当水平。在 40,000 步时，我们观察到性能显著下降至 60.4%。这种快速收敛在预期之内：T2A 学习到的语言—动作映射在结构上比完整的视觉—语言—动作空间维度更低，因此只需适中的步数即可捕捉到它；而延长训练会开始记忆特定的轨迹实例，而不是细化结构先验。我们在最终流程中采用 2,000 个 T2A 步作为默认设置，提供了有利的成本—质量权衡。

5.2.2 多具身协同训练

我们研究了多具身协同训练至关重要的两个设计维度：动作学习过程中视觉—语言数据的作用，以及面向异构动作空间的投影策略。

视觉-语言数据对动作学习的影响。尽管视觉-语言监督在离散 token 空间中运作，而动作预测面向连续轨迹，但这两个目标共享一个共同的感知与语言主干。一个自然的问题是，在……期间保留 VL 数据是否

embodied fine-tuning helps or hurts action quality. We compare three configurations in Fig. 7: **(a)** *Action-only*: the backbone is fine-tuned exclusively on embodied trajectories with no VL supervision; **(b)** *Action + VL data*: a fraction of VL data (VQA, captioning, spatial grounding) is mixed into the training corpus.

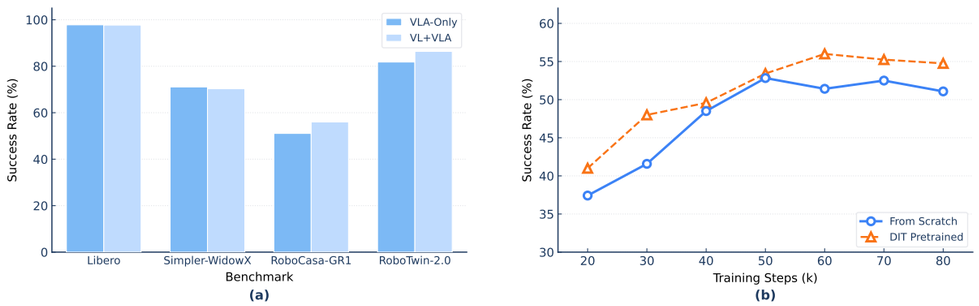

Figure 7: Vision-language co-training ablations. **(a) Impact of VL data on action learning.** Task-average success rate (%) across four benchmarks for VLA-Only (action data only) vs. VL+VLA (action + vision-language data co-training). On simpler benchmarks (Libero, Simpler-WidowX) the two configurations perform comparably, confirming that VL co-training introduces no interference. On benchmarks that demand fine-grained object recognition and compositional instruction parsing, VL+VLA yields clear gains: +4.9 pp on RoboCasa-GR1 (51.1% $\rightarrow$ 56.0%) and +4.6 pp on RoboTwin-2.0 (81.8% $\rightarrow$ 86.4%). **(b) Transferability of pretrained DiT.** SFT success rate (%) on RoboCasa-GR1 vs. training steps (k) when pairing the original Qwen3.5 VLM backbone with a DiT decoder initialised from scratch vs. a pretrained DiT. The pretrained DiT converges faster and reaches a higher peak.

As shown in Fig. 7 (a), mixing VL data into training yields clear benefits on manipulation benchmarks. On Libero and Simpler-WidowX, VLA-Only and VL+VLA achieve nearly identical success rates. On RoboCasa-GR1, which requires whole-body control in cluttered kitchen scenes with diverse objects and spatial configurations, VL+VLA improves over VLA-Only by **+4.9%** absolute (51.1% $\rightarrow$ 56.0%). Moreover, we attach the pretrained DiT to a fresh Qwen3.5-4B VLM and compare against a from-scratch baseline where the DiT is randomly initialised. As shown in Fig. 7 (b), the pretrained DiT consistently outperforms the from-scratch baseline throughout training, converging faster in early stages and reaching a higher peak success.

Projection design for heterogeneous embodiments. Different robot platforms have different action dimensionalities and semantics (e.g. 7-DoF arm vs. 29-DoF whole-body). To learn a shared representation space across heterogeneous action formats, the projection between the DiT latent space and per-embodiment action vectors must reconcile this mismatch. To this end, we compare three representative designs.

Multi-MLP. Each embodiment owns a private encoder MLP that maps its native action dimension to the DiT hidden size, and a private decoder MLP that maps back. Let d_i denote the action dimensionality of embodiment i, h the DiT hidden size, and N the number of embodiments. Each embodiment owns a private encoder $f_i^{\text{enc}}: \mathbb{R}^{d_i} \rightarrow \mathbb{R}^h$ and decoder $f_i^{\text{dec}}: \mathbb{R}^h \rightarrow \mathbb{R}^{d_i}$, totalling $2h\sum_{i=1}^{N} d_i$ parameters.

Concatenation. Actions from all N embodiments are concatenated into a single vector of dimensionality $D = \sum_{i=1}^{N} d_i$; each embodiment occupies a dedicated, non-overlapping segment. A shared encoder $f^{\text{enc}}: \mathbb{R}^D \rightarrow \mathbb{R}^h$ and decoder $f^{\text{dec}}: \mathbb{R}^h \rightarrow \mathbb{R}^D$ process the full vector ($2h\sum_{i=1}^{N} d_i$ parameters), and each embodiment retrieves its predictions from the corresponding segment.

Zero-Padding. A single shared MLP encodes all embodiments; shorter action vectors are right-padded with zeros to $d_{\max} = \max_i d_i$ before encoding. A shared encoder $f^{\text{enc}}: \mathbb{R}^{d_{\max}} \rightarrow \mathbb{R}^h$ and decoder $f^{\text{dec}}: \mathbb{R}^h \rightarrow \mathbb{R}^{d_{\max}}$ process the padded vector ($2h\, d_{\max}$ parameters); each embodiment reads only its d_i-dimensional prefix. Since $d_{\max} \leq \sum_i d_i$, *Zero-Padding* uses fewer projection parameters than both *Multi-MLP* and *Concatenation*.

As shown in Table 10, all co-training projection designs achieve performance comparable to the single-embodiment baselines on both benchmarks (Bridge single-train 62.8% vs. co-train ~63%; Robocasa single-train 53.4% vs. co-train 52–53%), confirming that multi-embodiment co-training introduces no significant penalty relative to training on each dataset independently. Among the three designs, performance differences are marginal (<1.2% absolute on both benchmarks), indicating that the choice of projection

具身微调有助于还是损害动作质量。我们在图 7 中比较了三种配置：(a) 仅动作：主干仅在具身轨迹上微调，没有任何 VL 监督；(b) Action+ VL 数据：将一部分 VL 数据（VQA、图像描述、空间定位）混入训练语料。

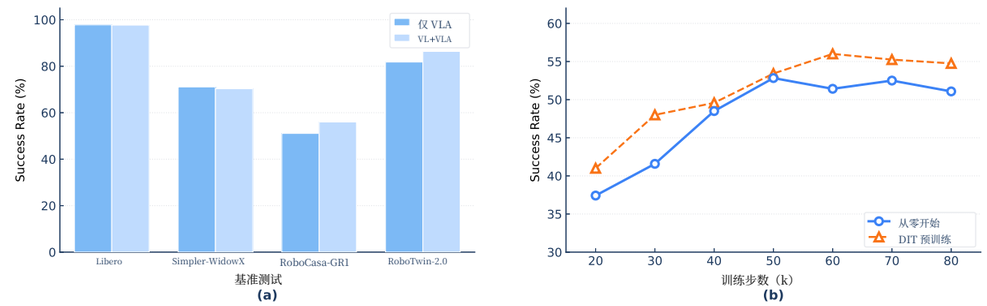

图 7：视觉-语言联合训练消融实验。（a）VL 数据对动作学习的影响。VLA-Only（仅动作数据）与 VL+VLA（动作数据 + 视觉-语言数据联合训练）在四项基准上的任务平均成功率（%）。（b）预训练 DiT 的可迁移性。将原始 Qwen3.5 VLM 主干分别与从零初始化的 DiT 解码器和预训练 DiT 配对时，RoboCasa-GR1 上的 SFT 成功率（%）与训练步数（k）的关系。预训练 DiT 收敛更快，并达到更高的峰值。

如图 7(a) 所示，将 VL 数据混入训练在操作基准上带来了明显收益。在 Libero 和 Simpler-WidowX 上，VLA-Only 与 VL+VLA 取得了几乎相同的成功率。在 RoboCasa-GR1 上，该任务需要在杂乱厨房场景中执行全身控制，且场景包含多样物体和空间布局，VL+VLA 相比 VLA-Only 提升了 **+4**.9% 的绝对值（51.1% $\rightarrow$ 56.0%）。此外，我们将预训练的 DiT 接入一个全新的 Qwen3.5-4B VLM，并与从零开始的基线进行比较，其中 DiT 采用随机初始化。如图 7(b) 所示，预训练的 DiT 在整个训练过程中始终优于从零开始的基线，在早期阶段收敛更快，并达到更高的峰值成功率。

面向异构具身的投影设计。不同机器人平台具有不同的动作维度和语义（例如 7 自由度机械臂与 29 自由度全身）。为了在异构动作格式之间学习共享表示空间，DiT 潜空间与各具身动作向量之间的投影必须协调这种不匹配。为此，我们比较了三种具有代表性的设计。

Multi-MLP。每个具身都拥有一个私有编码器 MLP，用于将其原生动作维度映射到 DiT 隐藏维度；还拥有一个私有解码器 MLP，用于映射回原维度。令 d_i 表示具身 i 的动作维度，h 表示 DiT 隐藏维度，N 表示具身数量。每个具身都拥有私有编码器 f enci： $\mathbb{R}^{d_i} \rightarrow \mathbb{R}^h$ 和解码器 f deci： $\mathbb{R}^h \rightarrow \mathbb{R}^{d_i}$，总计 $2\mathrm{h}\sum_{i=1}^{N}\mathrm{d}_i$ 个参数。

拼接。来自所有 N 个具身的动作被拼接成一个维度为 $\mathrm{D} = \sum_{i=1}^{N}\mathrm{d}_i$ 的单一向量；每个具身占据一个专用且互不重叠的片段。共享编码器 f^{enc}： $\mathbb{R}^D \rightarrow \mathbb{R}^h$ 和解码器 f^{dec}： $\mathbb{R}^h \rightarrow \mathbb{R}^D$ 处理完整向量（$2\mathrm{h}\sum_{i=1}^{N}$ d_i 个参数），每个具身从对应片段中取出其预测结果。

零填充。一个单一的共享 MLP 对所有具身进行编码；在编码前，较短的动作向量会在右侧用零填充到 $\mathrm{d}_{\max} = \max_i \mathrm{d}_i$。共享编码器 f^{enc}： $\mathbb{R}^{d_{\max}} \rightarrow \mathbb{R}^h$ 和解码器 f^{dec}： $\mathbb{R}^h \rightarrow \mathbb{R}^{d_{\max}}$ 处理填充后的向量（$2\mathrm{h}\ \mathrm{d}_{\max}$ 个参数）；每个具身只读取其 d_i 维前缀。由于 $\mathrm{d}_{\max} \leq \sum_i \mathrm{d}_i$，零填充比 Multi-MLP 和拼接使用更少的投影参数。

如表 10 所示，所有协同训练投影设计在两个基准上均取得了与单具身基线相当的性能（Bridge 单训练 62.8% 对 协同训练 ~63%；Robocasa 单训练 53.4% 对 协同训练 52–53%），这证实了多具身协同训练相对于分别在每个数据集上独立训练不会引入显著性能损失。在这三种设计中，性能差异很小（在两个基准上的绝对差异为 <1.2%），表明投影选择

Table 10: **Projection design ablation.** Results are task-average success rates (%). All co-training variants share the same training recipe.

	Single-Embodiment Training		Multi-Embodiment Co-training		
	Bridge Only	Robocasa Only	Multi-MLP	Concat.	Zero-Pad
Bridge	62.8	—	**63.3**	63.0	63.0
Robocasa	—	53.4	52.1	52.8	**53.2**

architecture has limited impact on task success once a shared latent space is established. However, Zero-Padding is architecturally the lightest: it requires only $2h\,d_{\max}$ projection parameters compared with $2h\sum_i d_i$ for Multi-MLP and Concatenation. We therefore adopt **Zero-Padding** as the default projection design.

5.2.3 Effect of RL Post-Training

We study whether RL on top of SFT provides additional gains, and whether those gains transfer to benchmarks not seen during RL training. To isolate the contribution of each post-training stage, we evaluate three checkpoints along the training pipeline: the CPT base model (**Qwen-VLA-Base**), the model after multi-task SFT (**+SFT**), and the model after RL refinement (**+RL**, i.e., Qwen-VLA-Instruct). RL rollouts are collected exclusively in SimplerEnv with sparse binary success rewards; no other benchmark data is used during the RL stage.

Table 11: **Cumulative effect of post-training stages.** Each row adds one stage on top of the previous checkpoint. RL rollouts are collected only in SimplerEnv. All numbers are success rates (%) except the DOMINO Manipulation Score (MS).

Stage	**In-Distribution**					**OOD-static**	**OOD-dynamic (DOMINO)**	
	Simpler	RoboCasa	RoboTwin-E	RoboTwin-H	LIBERO	SimplerOOD	SR	MS
CPT	64.3	40.4	64.3	66.4	90.8	25.3	21.1	37.4
+ SFT	70.8	56.0	86.3	87.1	97.8	31.6	25.7	39.1
+ RL	**73.7**	**56.7**	86.1	**87.2**	**97.9**	**32.0**	**26.6**	**39.5**

As shown in Table 11, each post-training stage provides complementary gains. SFT delivers large improvements across all benchmarks by specializing the general-purpose CPT representations to downstream task distributions. RL then provides a further uplift on top of SFT, with the largest gain on SimplerEnv (+2.9 pp, from 70.8% to 73.7%), the environment in which RL rollouts are collected. This confirms that directly optimizing closed-loop task success captures policy improvements that the imitation objective alone cannot.

More importantly, the RL gains are not confined to the training environment. On benchmarks that are *not* part of the RL rollout distribution, performance is preserved or mildly improved: RoboCasa increases by 0.7 pp, RoboTwin-Hard by 0.1 pp, LIBERO by 0.1 pp, and the OOD evaluation on SimplerEnv-OOD by 0.4 pp. RoboTwin-Easy exhibits negligible changes (<0.6 pp), confirming that RL does not cause catastrophic forgetting on held-out tasks. On the DOMINO dynamic manipulation benchmark, which tests zero-shot generalization to moving objects, both SR (25.7% $\rightarrow$ 26.6%) and MS (39.1 $\rightarrow$ 39.5) improve despite the complete absence of dynamic manipulation data during RL training. These results indicate that task-success optimization in simulation preserves performance with mild positive transfer: the policy learns to execute more decisively and recover from compounding errors, benefits that generalize across evaluation settings.

5.2.4 State Conditioning

Many robot learning methods condition the policy on explicit proprioceptive state (e.g., joint angles) in addition to visual observations, under the assumption that knowing the current robot configuration helps predict the next action. However, proprioceptive inputs also introduce embodiment-specific dependencies that may hinder cross-embodiment generalization, since different platforms have different joint spaces and kinematic structures. We examine whether explicit state conditioning benefits action learning in Qwen-VLA and, if so, where in the architecture the state should be injected. We compare three strategies on RoboTwin-2.0:

- **No State.** The model receives only visual observations and the language instruction. No proprioceptive information is provided to either the VLM backbone or the DiT action decoder.

表 10：投影设计消融实验。结果为任务平均成功率（%）。所有协同训练变体均采用相同的训练方案。

	单具身训练		多具身协同训练		
	仅 Bridge	仅 Robocasa	多 MLP	拼接。	零填充
Bridge	62.8	—	**63.3**	63.0	63.0
Robocasa	—	53.4	52.1	52.8	**53.2**

架构在建立共享潜空间后对任务成功率的影响有限。然而，Zero-Padding 在架构上最轻量：它仅需要 $2hd_{\max}$ 个投影参数，而 Multi-MLP 和 Concatenation 需要 $2h\sum_i d_i$。因此，我们采用 Zero-Padding 作为默认投影设计。

5.2.3 RL 后训练的效果

我们研究在 SFT 之上进行 RL 是否能带来额外收益，以及这些收益能否迁移到 RL 训练期间未见过的基准上。为分离每个后训练阶段的贡献，我们评估训练流程中的三个检查点：CPT 基础模型（Qwen-VLA-Base）、多任务 SFT 后的模型（+SFT），以及经过 RL 精炼后的模型（+RL，即 Qwen-VLA-Instruct）。RL 采样轨迹仅在 SimplerEnv 中采集，并使用稀疏二值成功奖励；RL 阶段不使用任何其他基准数据。

表 11：后训练阶段的累积效应。每一行都在前一个检查点之上增加一个阶段。RL rollout 仅在 SimplerEnv 中收集。除 DOMINO 操作得分（MS）外，所有数字均为成功率（%）。

阶段	**分布内**					**OOD-静态**	**OOD-动态（DOMINO）**	
	Simpler	RoboCasa	RoboTwin-E	RoboTwin-H	LIBERO	SimplerOOD	SR	MS
CPT	64.3	40.4	64.3	66.4	90.8	25.3	21.1	37.4
+ SFT	70.8	56.0	86.3	87.1	97.8	31.6	25.7	39.1
+ RL	**73.7**	**56.7**	86.1	**87.2**	**97.9**	**32.0**	**26.6**	**39.5**

如表 11 所示，每个后训练阶段都提供互补的增益。SFT 通过将通用 CPT 表征专门适配到下游任务分布，在所有基准上都带来大幅提升。随后，RL 在 SFT 基础上进一步提升，其中在 SimplerEnv 上的提升最大（+2.9 个百分点，从 70.8% 提升至 73.7%），而该环境正是收集 RL rollout 的环境。这证实，直接优化闭环任务成功率能够捕捉仅靠模仿目标无法获得的策略改进。

更重要的是，RL 带来的增益并不局限于训练环境。在不属于 RL rollout 分布的基准上，性能得以保持或略有提升：RoboCasa 提升 0.7 个百分点，RoboTwin-Hard 提升 0.1 个百分点，LIBERO 提升 0.1 个百分点，SimplerEnv-OOD 上的 OOD 评估提升 0.4 个百分点。RoboTwin-Easy 的变化可忽略（<0.6 个百分点），这证实 RL 不会在留出任务上造成灾难性遗忘。在用于测试对移动物体零样本泛化能力的 DOMINO 动态操作基准上，尽管 RL 训练期间完全缺少动态操作数据，SR（25.7% $\rightarrow$ 26.6%）和 MS（39.1 $\rightarrow$ 39.5）仍均有提升。这些结果表明，在仿真中优化任务成功率能够保持性能并带来轻微的正迁移：策略学会更果断地执行，并从累积误差中恢复，而这些收益可泛化到不同评估设置中。

5.2.4 状态条件化

许多机器人学习方法会在视觉观测之外，还将显式的本体感知状态（例如关节角度）作为策略的条件，其假设是了解当前机器人构型有助于预测下一个动作。然而，本体感知输入也会引入与具体本体相关的依赖关系，这可能阻碍跨本体泛化，因为不同平台具有不同的关节空间和运动学结构。我们考察显式状态条件化是否有助于 Qwen-VLA 中的动作学习；如果有助于，还应将状态注入架构中的哪个位置。我们在 RoboTwin-2.0 上比较了三种策略：

- **无状态** . 模型仅接收视觉观测和语言指令。无本体感知 e
  信息 tion 被提供给 VLM 主干或 DiT 动作解码器之一。

- **State in VLM Prompt.** Joint angles are discretized into 256 bins per dimension and serialized into the language prompt, e.g., `"state: [34, 128, 67, 200, 55, 12, 88] pick up the bowl"`. The VLM processes state tokens together with images and the instruction; the DiT receives no additional state input.
- **State in DiT.** Joint angles are passed as continuous-valued vectors directly to the DiT action decoder as an additional conditioning input, bypassing the VLM entirely.

Table 12: State conditioning ablation on RoboTwin-2.0 (success rates, %).

Conditioning	RoboTwin-Easy	RoboTwin-Hard
No State	88.7	87.4
State in VLM Prompt	89.3	88.7
State in DiT	89.4	88.3

As shown in Table 12, incorporating proprioceptive state yields only marginal gains regardless of the injection strategy: at most +0.7 pp on Easy and +1.3 pp on Hard relative to the vision-only baseline. The two injection strategies perform comparably ($\leq$0.4 pp difference), suggesting that neither the VLM's semantic processing nor the DiT's continuous conditioning offers a clear advantage for encoding joint-angle information. We attribute the limited impact of state conditioning to two factors. First, the multi-view visual observations already provide sufficient information about the robot's current configuration, especially for tasks where the end-effector is visible in at least one camera view. Second, the flow-matching action decoder predicts relative action displacements rather than absolute poses, reducing the need for an explicit current-state reference. Given the limited benefits and the additional complexity of maintaining embodiment-specific state interfaces across diverse platforms, we opt not to include state conditioning in our default framework, keeping the embodiment-aware text prompt as the sole platform-specific input.

6 Conclusion

We presented **Qwen-VLA**, a unified vision-language-action model that extends the Qwen vision-language backbone from perception and reasoning to embodied action generation. By formulating manipulation, navigation, egocentric action modeling, and trajectory prediction within a shared action-and-trajectory prediction framework, Qwen-VLA enables a single model to learn from heterogeneous embodied data across tasks, environments, and robot embodiments. Through embodiment-aware prompt conditioning, a DiT-based flow-matching action decoder, large-scale joint pretraining, and SFT/RL post-training, Qwen-VLA achieves strong generalist performance while retaining flexibility across different control conventions. Our results suggest that language-grounded VLA models provide a practical path toward embodied intelligence by aligning multimodal perception, instruction understanding, embodiment constraints, and executable actions. Compared with visual prediction-centric world models, Qwen-VLA emphasizes grounding multimodal understanding into actions that can be executed by embodied agents. Together, these results indicate that VLA models can serve as an actionable interface between multimodal foundation models and embodied agents, bridging high-level language reasoning with low-level physical control.

7 Limitations and Future Work

Despite its generalist capabilities, Qwen-VLA still has several limitations. First, embodied action data remains far smaller and less diverse than vision-language pretraining data, limiting robustness to long-tail objects, environments, embodiments, and contact-rich interactions. Second, joint training across vision-language understanding, navigation, and action generation introduces optimization trade-offs. While action-oriented training improves policy learning, it can modestly regress some pure vision-language and navigation evaluations, suggesting the need for better objective balancing, data curricula, and modular specialization. Third, current evaluations are still largely short-horizon and benchmark-driven, leaving long-duration, failure-prone real-world deployment as an open challenge.

These limitations motivate several future directions. Scaling real-world interaction data through autonomous collection, simulation, and sim-to-real transfer will be critical for improving robustness. Large-scale human video, from both egocentric and third-person views, can provide physical priors and temporal abstractions beyond robot demonstrations. Future models should incorporate long-horizon planning, episodic memory, and world modeling, enabling agents to track state, decompose tasks, recover from failures, and anticipate action consequences. Finally, richer physical feedback such as force, tactile,

- VLM 提示词中的状态。关节角按每个维度离散为 256 个分箱，并序列化到语言提示中，例如：“状态：`[34, 128, 67, 200, 55, 12, 88]` 拿起碗”。VLM 将状态 token 与图像和指令一起处理；DiT 不接收额外的状态输入。
- DiT 中的状态。关节角度以连续值向量的形式直接传递到 DiT 动作解码器作为额外的条件输入，完全绕过 VLM。

表 12：在 RoboTwin-2.0 上的状态条件消融（成功率，%）。

条件	RoboTwin-Easy	RoboTwin-Hard
无状态	88.7	87.4
VLM 提示中的状态	89.3	88.7
DiT 中的状态	89.4	88.3

如表 12 所示，无论采用哪种注入策略，引入本体感知状态都仅带来边际增益：相对于仅视觉基线，在 Easy 上最多为 +0.7 个百分点，在 Hard 上最多为 +1.3 个百分点。两种注入策略表现相当（相差 $\leq$0.4 个百分点），这表明无论是 VLM 的语义处理，还是 DiT 的连续条件化，在编码关节角信息方面都没有明显优势。我们将状态条件化影响有限归因于两个因素。首先，多视角视觉观测已经提供了关于机器人当前构型的充分信息，尤其是对于末端执行器在至少一个相机视角中可见的任务。其次，流匹配动作解码器预测的是相对动作位移而非绝对位姿，从而降低了对显式当前状态参考的需求。鉴于收益有限，且在不同平台之间维护特定具身形式的状态接口会带来额外复杂性，我们选择不在默认框架中加入状态条件化，仅保留具身感知的文本提示作为唯一的平台特定输入。

6 结论

我们提出了 Qwen-VLA，一种统一的视觉-语言-动作模型，它将 Qwen 视觉-语言主干从感知与推理扩展到具身动作生成。通过在共享的动作与轨迹预测框架中统一建模操作、导航、自我中心动作建模和轨迹预测，Qwen-VLA 使单一模型能够从跨任务、环境和机器人本体的异构具身数据中学习。借助本体感知的提示条件、基于 DiT 的流匹配动作解码器、大规模联合预训练以及 SFT/RL 后训练，Qwen-VLA 在保持不同控制约定下灵活性的同时，实现了强大的通用性能。我们的结果表明，基于语言的 VLA 模型通过将多模态感知、指令理解、本体约束和可执行动作对齐，为通向具身智能提供了一条切实可行的路径。与以视觉预测为中心的世界模型相比，Qwen-VLA 强调将多模态理解落地为可由具身智能体执行的动作。综上，这些结果表明 VLA 模型可以作为多模态基础模型与具身智能体之间的可操作接口，连接高层语言推理与低层物理控制。

7 局限性与未来工作

尽管 Qwen-VLA 具备通用能力，它仍存在若干局限。首先，具身动作数据在规模和多样性上仍远不及视觉-语言预训练数据，这限制了对长尾物体、环境、具身形态以及接触丰富交互的鲁棒性。其次，跨视觉-语言理解、导航和动作生成的联合训练会带来优化权衡。虽然面向动作的训练能改进策略学习，但可能使部分纯视觉-语言和导航评测出现轻微退化，这表明需要更好的目标平衡、数据课程设置和模块化专门化。第三，当前评测仍主要集中于短时程且由基准驱动，长时程、易失败的真实世界部署仍是未解决的挑战。

这些局限促使了若干未来方向。通过自主采集、仿真以及仿真到现实迁移来扩展真实世界交互数据，对于提升鲁棒性至关重要。来自第一人称和第三人称视角的大规模人类视频，能够提供超越机器人演示的物理先验和时间抽象。未来的模型应纳入长时程规划、情景记忆和世界建模，使智能体能够跟踪状态、分解任务、从失败中恢复并预判动作后果。最后，更丰富的物理反馈，例如力觉、触觉，

and proprioceptive signals, combined with large-scale reinforcement learning in simulation and the real world, may further bridge the gap between language-grounded embodied understanding and reliable control.

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以及本体感觉信号，若与仿真和真实世界中的大规模强化学习相结合，可能进一步弥合以语言为基础的具身理解与可靠控制之间的差距。

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